

4. The dimensions of 'self-inductance' are :

- (A) $[M L T^{-2} A^{-2}]$
- (B) $[M L^2 T^{-1} A^{-1}]$
- (C) $[M L^{-1} T^{-2} A^{-2}]$
- (D) $[M L^2 T^{-2} A^{-2}]$

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- (D) $[M L^2 T^{-2} A^{-2}]$

(D) $[ML^2T^{-2}A^{-2}]$

6. A metal rod of length 50 cm is held vertically and moved with a velocity of 10 m/s towards east. The horizontal component of the Earth's magnetic field at the place is 0.4 G. The emf induced across the ends of the rod is :

(A) 0.1 mV

(B) 0.2 mV

(C) 0.8 mV

(D) 1.6 mV

6. A metal rod of length 50 cm is held vertically and moved with a velocity of 10 m/s towards east. The horizontal component of the Earth's magnetic field at the place is 0.4 G. The emf induced across the ends of the rod is :

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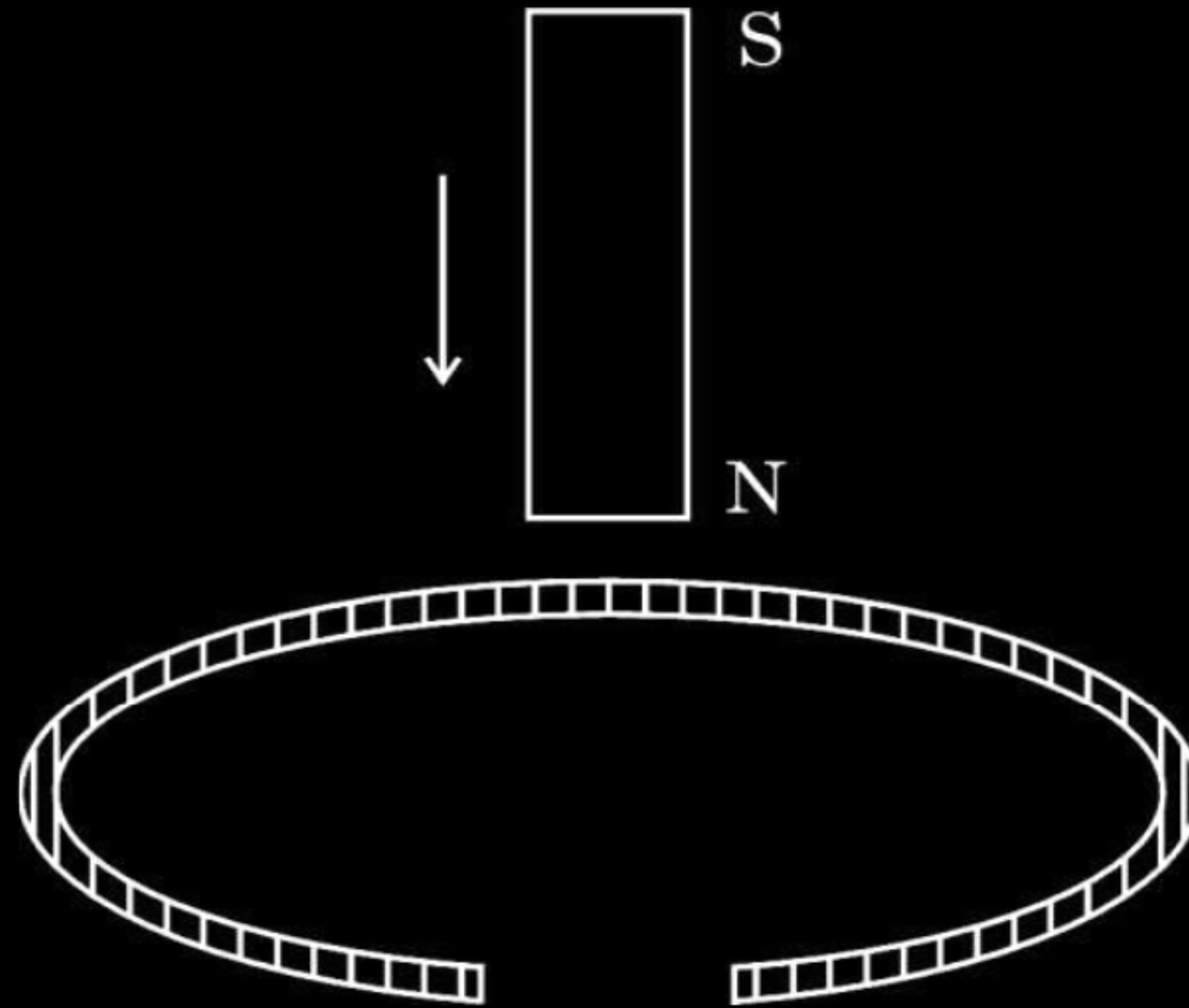
(B) 0.2 mV

(C) 0.8 mV

(D) 1.6 mV

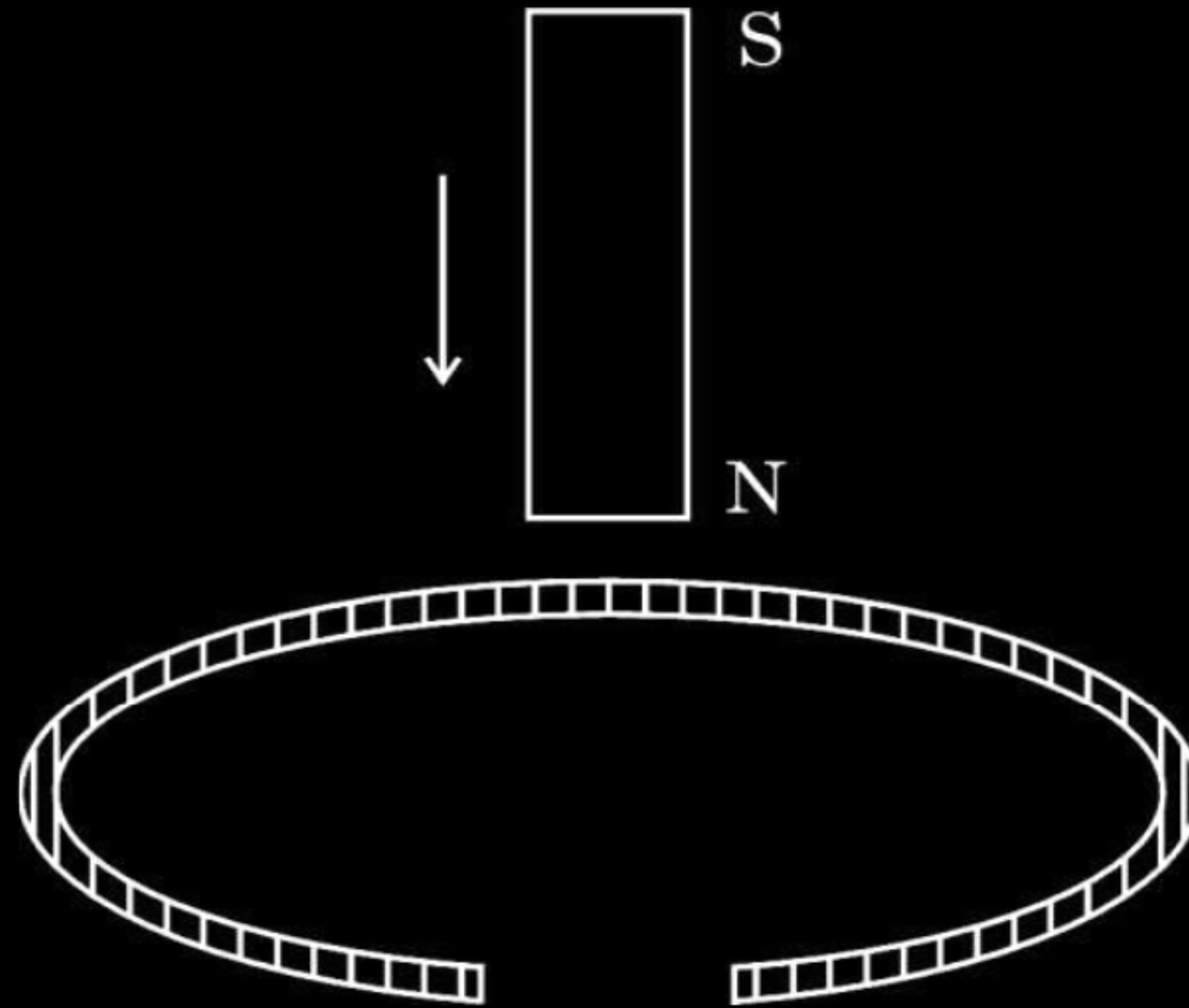
(B) 0.2 mV

4. A vertically held bar magnet is dropped along the axis of a copper ring having a cut as shown in the diagram. The acceleration of the falling magnet is :



- | | | | |
|-----|------|-----|------------------|
| (A) | zero | (B) | less than g |
| (C) | g | (D) | greater than g |

4. A vertically held bar magnet is dropped along the axis of a copper ring having a cut as shown in the diagram. The acceleration of the falling magnet is :



- | | |
|----------|----------------------|
| (A) zero | (B) less than g |
| (C) g | (D) greater than g |

(C) g

- 28.** (a) State Lenz's law. A rod MN of length L is rotated about an axis passing through its end M perpendicular to its length, with a constant angular velocity ω in a uniform magnetic field $\vec{B}$ parallel to the axis. Obtain an expression for emf induced between its ends. 3

Lenz's law

The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it.

The magnitude of the emf generated across the length dr of the rod as it moves at right angles to the magnetic field is given by

$$d\varepsilon = Bv dr$$

$$\varepsilon = \int d\varepsilon$$

$$= \int_0^L Bv dr$$

$$\varepsilon = \int_0^L B\omega r dr$$

$$\varepsilon = \frac{1}{2} BL^2 \omega$$

Alternatively:

$$\text{Area of the sector (QMN)} = \frac{1}{2} L^2 \theta$$

$$\text{Induced emf is } \varepsilon = B \times \frac{d}{dt} \left(\frac{1}{2} L^2 \theta \right)$$

$$\varepsilon = \frac{1}{2} BL^2 \frac{d\theta}{dt}$$

$$\varepsilon = \frac{1}{2} BL^2 \omega$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

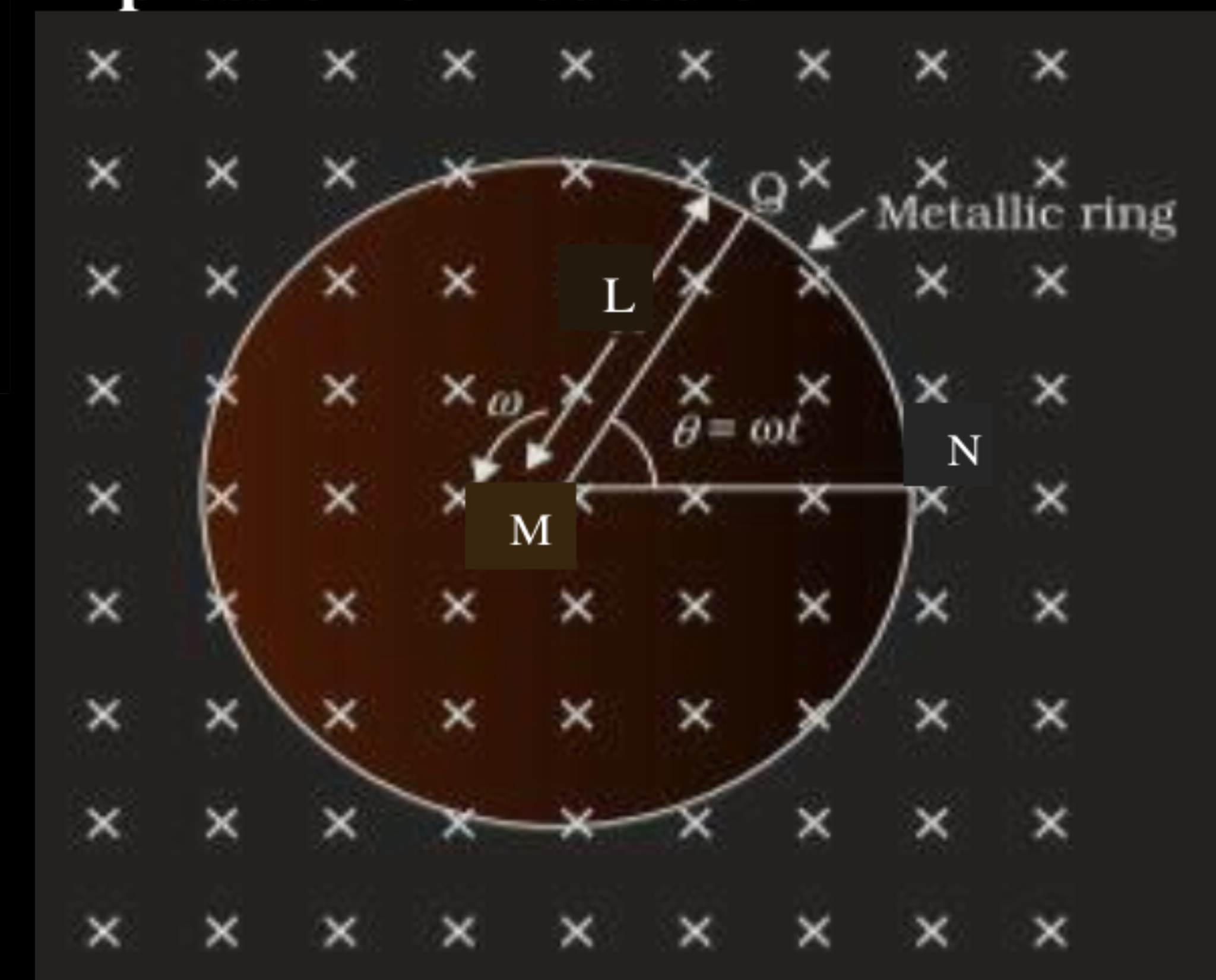
$\frac{1}{2}$

$\frac{1}{2}$

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$\frac{1}{2}$

$\frac{1}{2}$



- (b) Define 'self-inductance' of a coil. Derive an expression for self-inductance of a long solenoid of cross-sectional area A and length l , having n turns per unit length.

- (b) Define 'self-inductance' of a coil. Derive an expression for self-inductance of a long solenoid of cross-sectional area A and length l , having n turns per unit length.

Self inductance of a coil is the ratio of the flux linkage to the current flowing in the coil.

1

Alternatively:

Self inductance of a coil is defined as the flux linked with the coil when unit current flows through it.

Alternatively:

Self inductance of a coil may be defined as the magnitude of emf induced in the coil when current changes at the rate of 1 A/s in the coil.

Expression for self inductance of a long solenoid:

The magnetic field due to current flowing in the solenoid, $B = \mu_0 nI$

$\frac{1}{2}$

Total flux linked with the given solenoid

$$N\phi_B = (nl)(\mu_0 nI) A$$

$$N\phi_B = \mu_0 n^2 A l I$$

$\frac{1}{2}$

Self inductance

$$L = \frac{N\phi_B}{I}$$

$\frac{1}{2}$

$$L = \mu_0 n^2 A l$$

$\frac{1}{2}$

6. The magnetic flux through a loop changes from 3.6 Wb to 1.6 Wb in 0.4 s. The magnitude of the induced emf in the loop is — **1**
- (A) 5 V (B) 2.5 V
(C) 1.5 V (D) 0.5 V

6. The magnetic flux through a loop changes from 3.6 Wb to 1.6 Wb in 0.4 s. The magnitude of the induced emf in the loop is — **1**
- (A) 5 V (B) 2.5 V
(C) 1.5 V (D) 0.5 V

(A) 5V

13. **Assertion (A)** : Only a continuous change of magnetic flux will maintain an induced emf in a coil.

1

Reason (R) : An induced current has a direction such that the magnetic field due to the current opposes the change in the magnetic field that induces the current.

13. **Assertion (A)** : Only a continuous change of magnetic flux will maintain an induced emf in a coil. 1
- Reason (R)** : An induced current has a direction such that the magnetic field due to the current opposes the change in the magnetic field that induces the current.

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is the not the correct explanation of the Assertion (A)

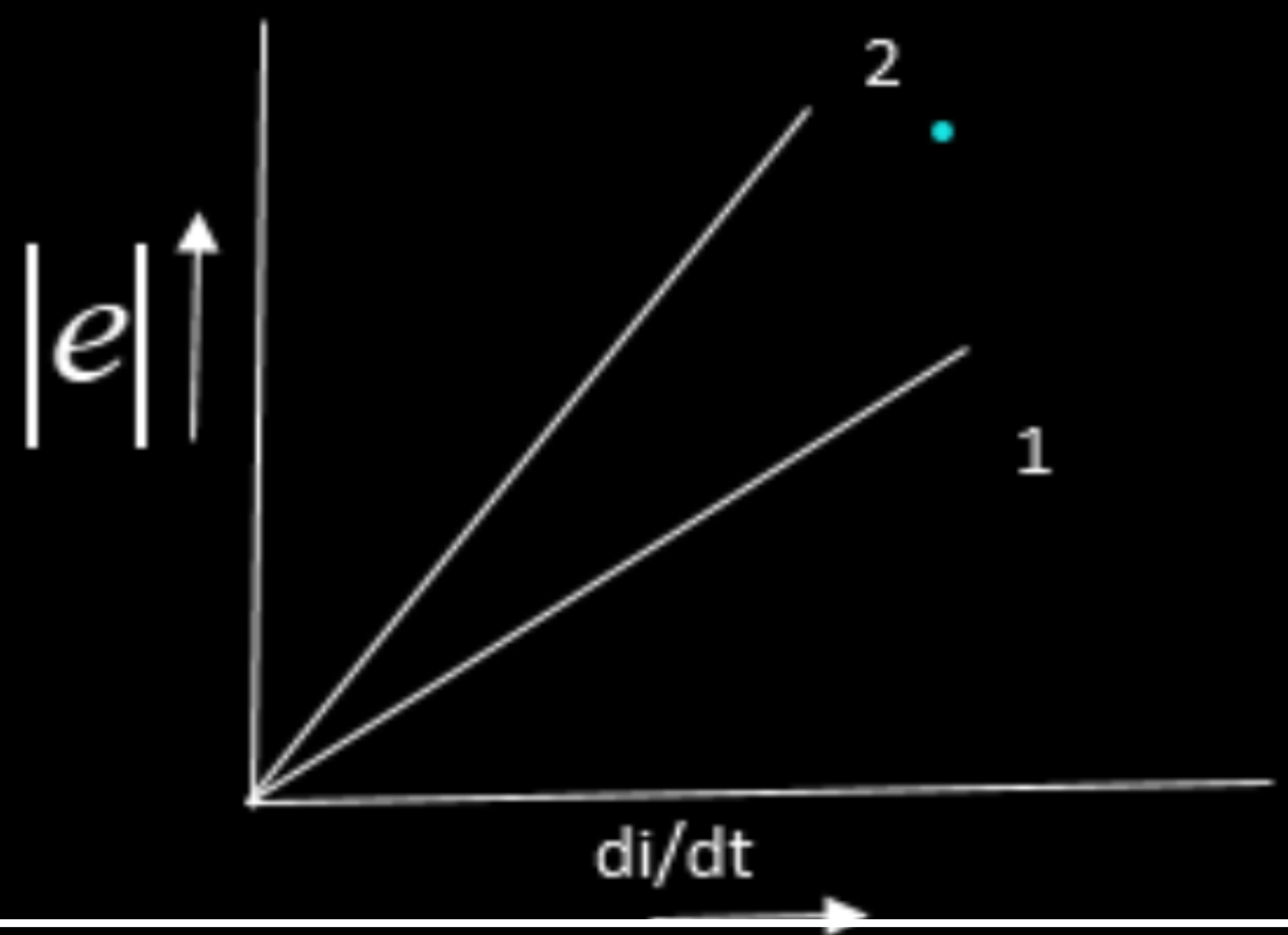
- 33.** (a) (i) Define self-inductance of a coil. Derive the expression for the energy required to build up a current I in a coil of self-inductance L .
- (ii) The currents passing through two inductors of self-inductances 10 mH and 20 mH increase with time at the same rate.
- Draw graphs showing the variation of :
- (I) the magnitude of emf induced with the rate of change of current in each inductor.
- (II) the energy stored in each inductor with the current flowing through it.

Self Inductance is magnetic flux linked with a coil when the current through the coil is unity.
Alternatively
Self Inductance is the induced emf induced in the coil when rate of change of current through the coil is unity.
To maintain growth of current, power has to be supplied from external source.

$$P = |e||I|$$
$$= \frac{dW}{dt} = LI \frac{dI}{dt}$$

$$dW = LI dI$$
$$W = \int LI dI$$
$$= \frac{1}{2} LI^2$$

(I) $E = -L \frac{dI}{dt}$



1

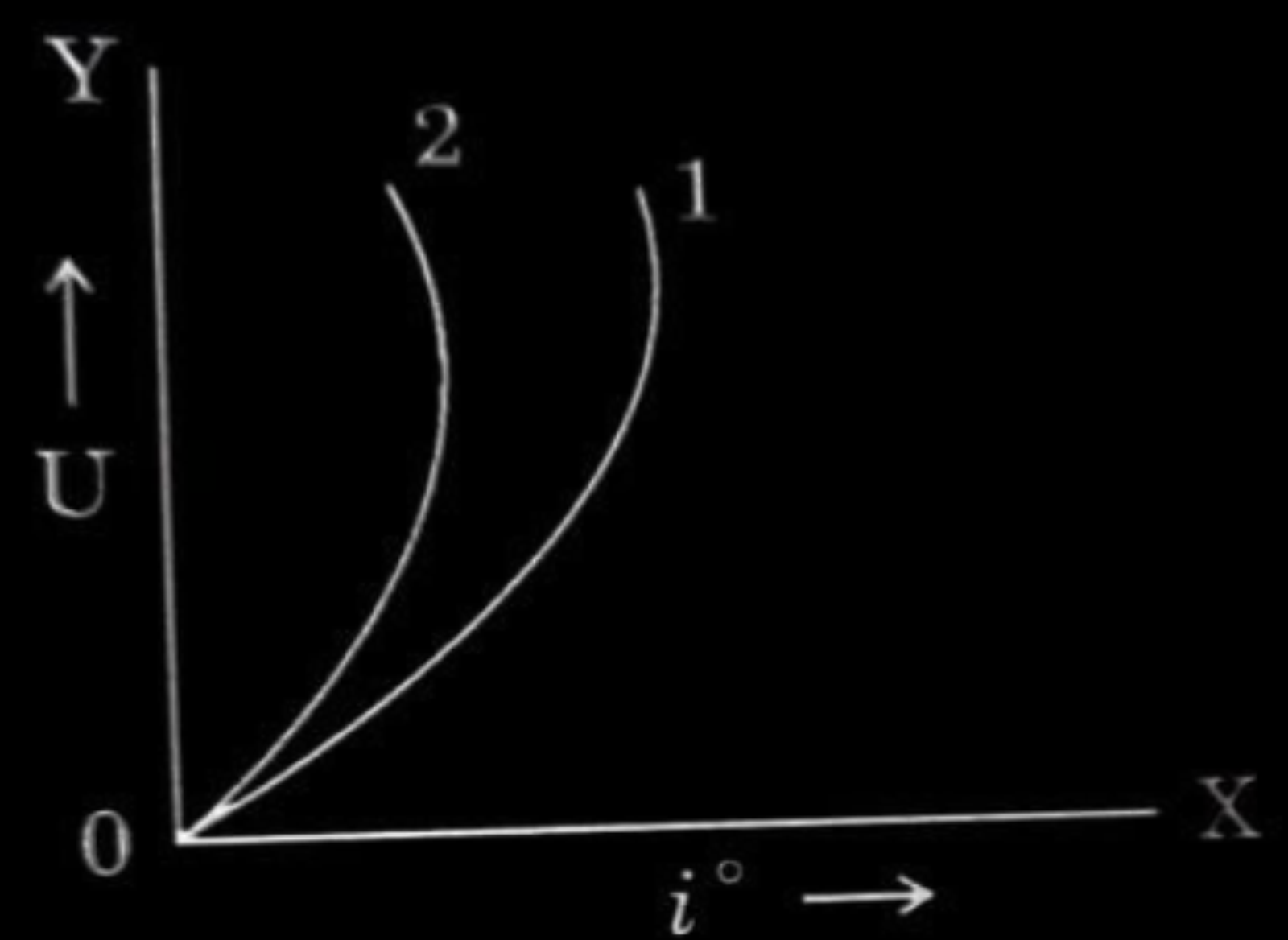
1/2

1/2

1/2

1

(II) $U = \frac{1}{2} LI^2$ Parabolic graph obtained.



(1 indicates 10mH) & (2 indicates 20mH)

1/2

1

- (b) (i) Define the term mutual inductance. Deduce the expression for the mutual inductance of two long coaxial solenoids of the same length having different radii and different number of turns.
- (ii) The current through an inductor is uniformly increased from zero to 2 A in 40 s. An emf of 5 mV is induced during this period. Find the flux linked with the inductor at $t = 10$ s.

- (b) (i) Define the term mutual inductance. Deduce the expression for the mutual inductance of two long coaxial solenoids of the same length having different radii and different number of turns.
- (ii) The current through an inductor is uniformly increased from zero to 2 A in 40 s. An emf of 5 mV is induced during this period. Find the flux linked with the inductor at $t = 10$ s.

5

(i) Mutual inductance is defined as the induced emf in primary coil when the current in secondary coil changes at the unit rate. Alternatively Mutual inductance is defined as the magnetic flux linked with the primary coil when the current in secondary coil is unity.	1
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Consider two long co-axial solenoids each of length l . Radius of inner solenoid S_1 is r_1 and number of turns per unit length is n_1 .

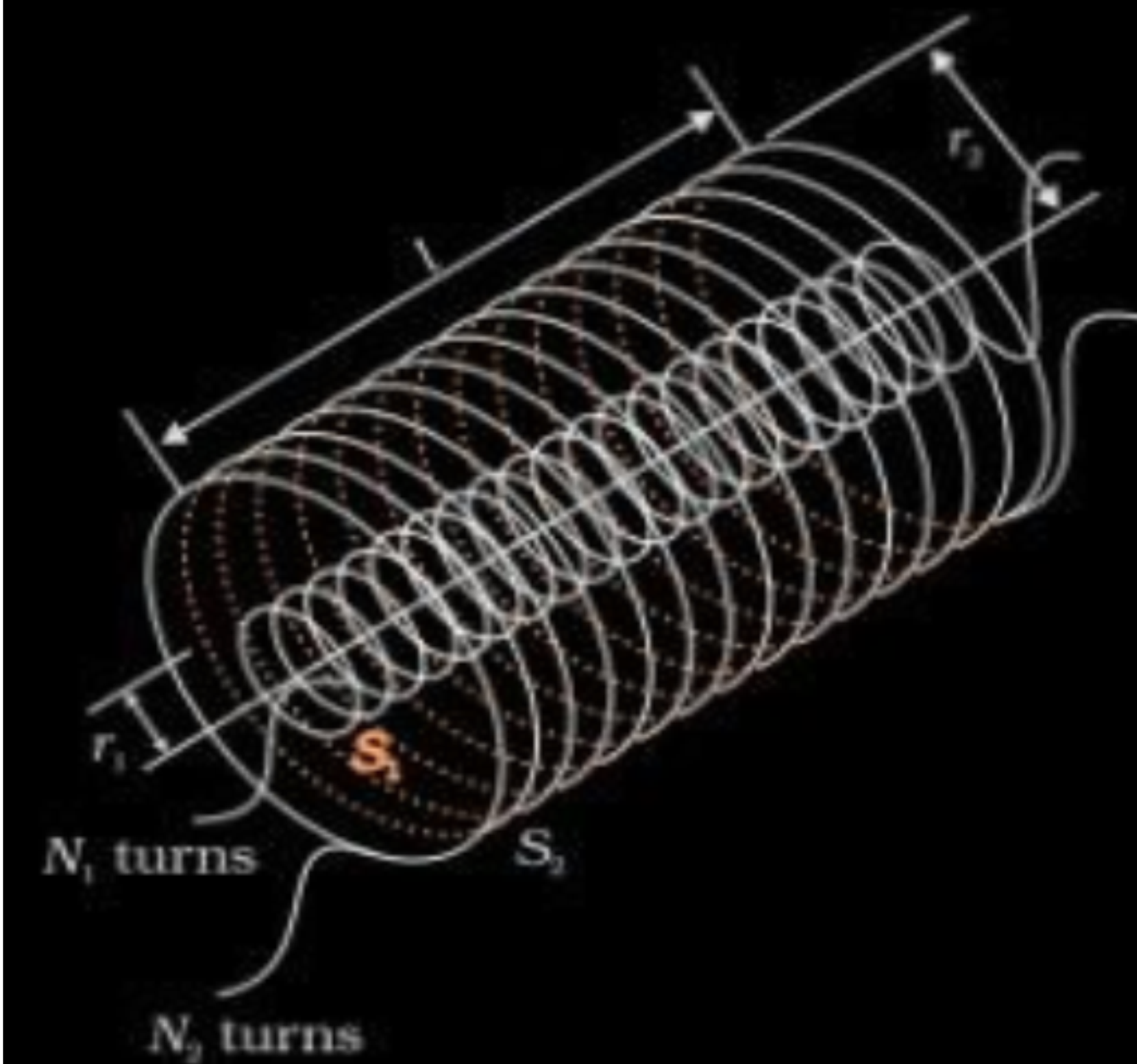
The corresponding quantities for outer solenoid S_2 are r_2 and n_2 respectively. Let N_1 and N_2 be the total number of turns of coils S_1 and S_2 respectively.

When a current I_2 is set up through S_2 , it sets up magnetic flux through S_1 .

$$\begin{aligned} N_1 \phi_1 &= M_{12} I_2 \\ &= (n_1 l) \times (\pi r_1^2) \times (\mu_0 n_2 I_2) \\ &= \mu_0 n_1 n_2 \pi r_1^2 l I_2 \\ M_{12} &= \mu_0 n_1 n_2 \pi r_1^2 l = M_{21} \end{aligned}$$

(ii)

$$\begin{aligned} |e| &= L \frac{dI}{dt} \\ L &= \frac{e}{dI/dt} \\ &= \frac{5 \times 10^{-3}}{2/40} \\ &= 0.1 \text{ H} \\ \phi &= LI \\ &= 0.1 \times \frac{2}{40} \times 10 \\ &= 0.05 \text{ Wb} \end{aligned}$$



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

6. The magnetic flux linked with a coil changes with time t as $\phi = (8t^2 + 5t + 7)$, where t is in seconds and ϕ is in Wb. The value of emf induced in the coil at $t = 4$ s is :

(A) 32 V

(B) 37 V

(C) 64 V

(D) 69 V

6. The magnetic flux linked with a coil changes with time t as $\phi = (8t^2 + 5t + 7)$, where t is in seconds and ϕ is in Wb. The value of emf induced in the coil at $t = 4$ s is :

(A) 32 V

(B) 37 V

(C) 64 V

(D) 69 V

(D) 69 V

6. The magnetic flux linked with a closed coil (in Wb) varies with time t (in s) as $\phi = 5t^2 + 4t - 2$. If the resistance of the circuit is $14\ \Omega$, the magnitude of induced current in the coil at $t = 1\text{ s}$ will be :

(A) 0.5 A

(B) 1.0 A

(C) 1.5 A

(D) 2.0 A

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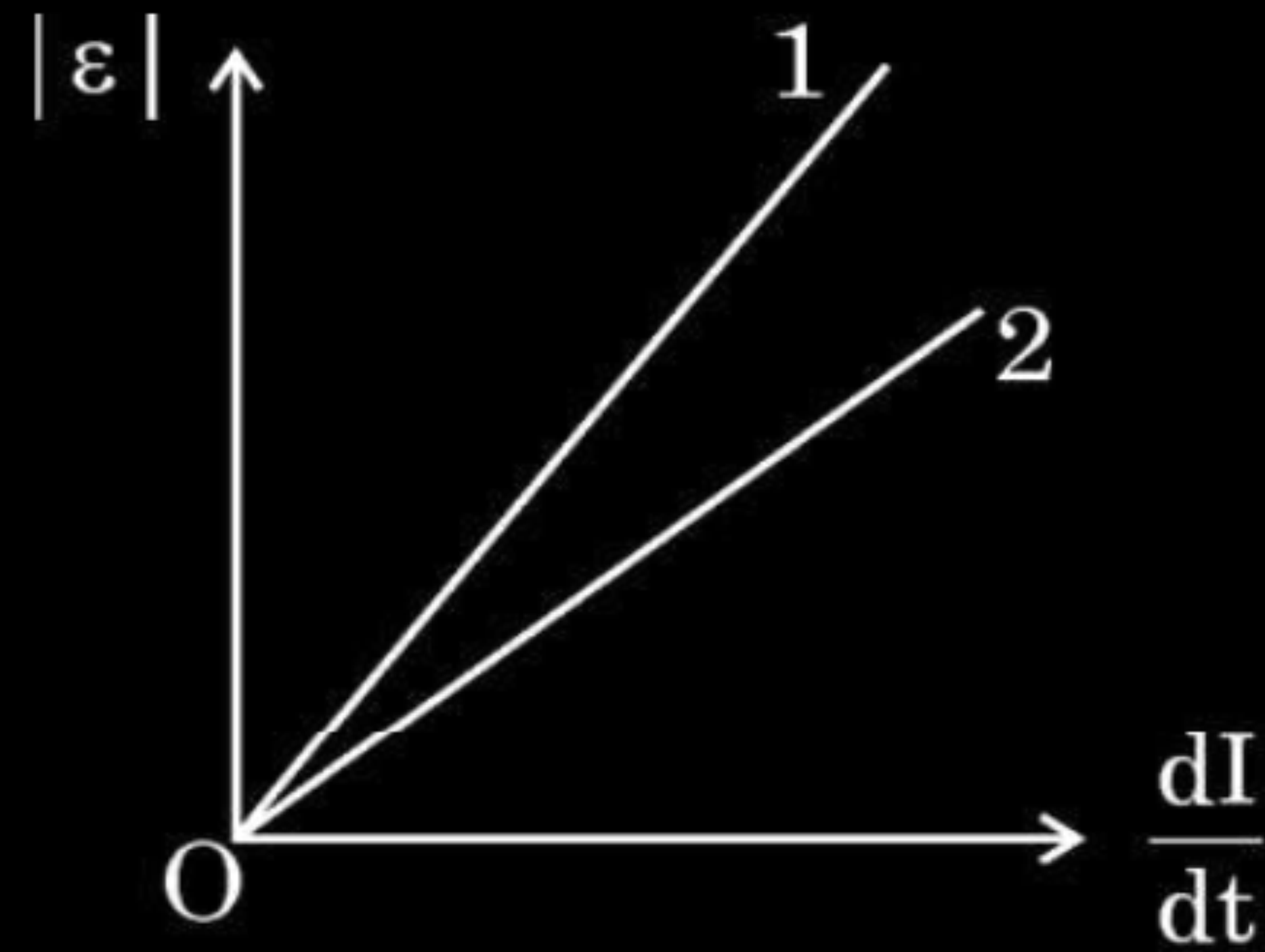
(B) 1.0 A

(C) 1.5 A

(D) 2.0 A

(B) 1.0 A

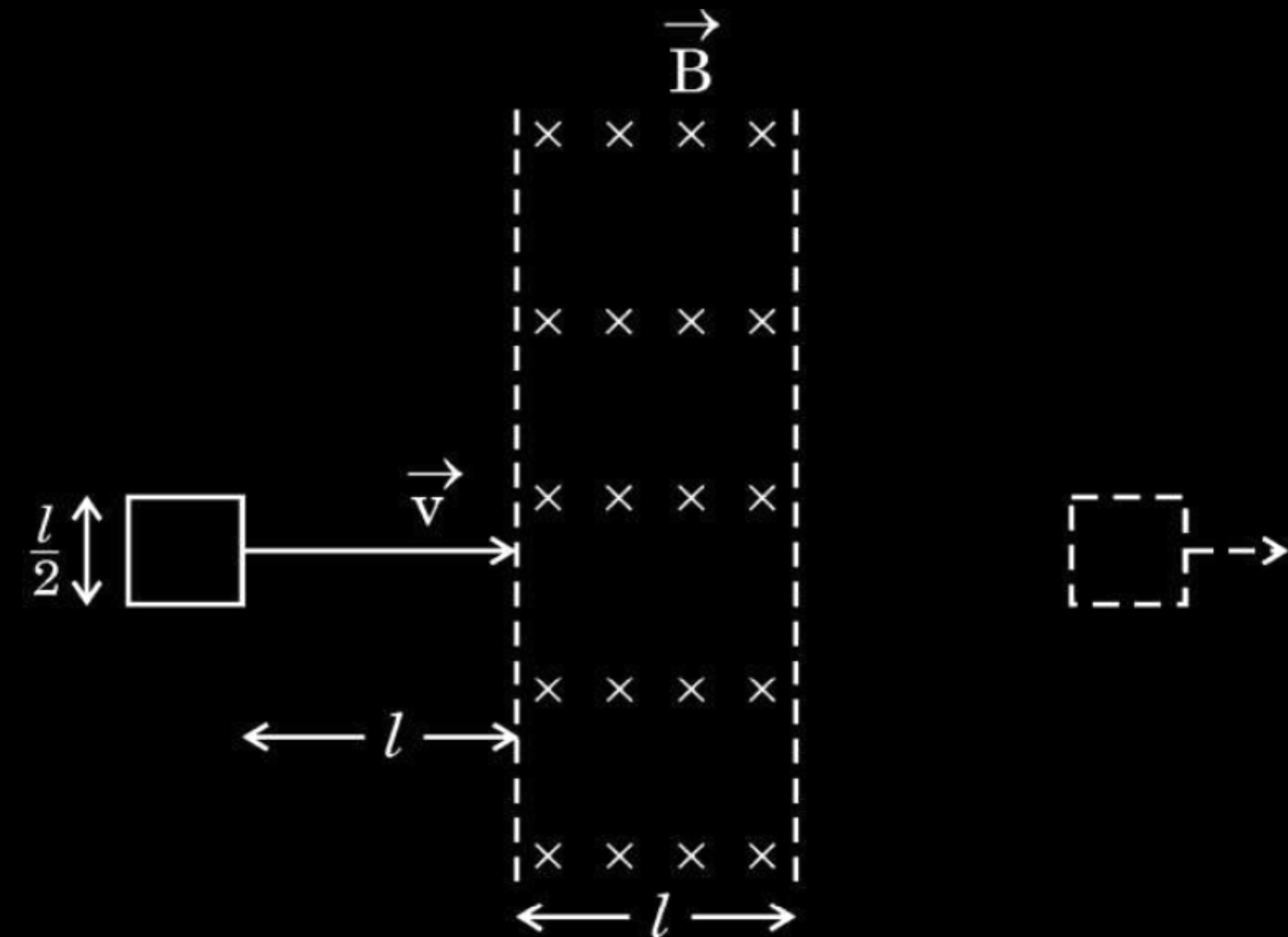
- 24.** (a) Obtain an expression for the self-inductance of a long solenoid of length ' l ', cross-sectional area ' A ' and having ' N ' turns.
- (b) The figure shows the plot of magnitude of induced emf (ε) versus the rate of change of current in two coils '1' and '2'. Which coil has greater value of self-inductance and why ?



Magnetic field due to a current I flowing in the solenoid $B = \mu_0 n I$	$\frac{1}{2}$
Total flux linked with solenoid is	
$N\phi_B = (nl)(\mu_0 n I)A$	$\frac{1}{2}$
$= \mu_0 n^2 A l$	
$L = \frac{N\phi_B}{I}$	$\frac{1}{2}$
$L = \mu_0 n^2 A l$	$\frac{1}{2}$
(b) Coil 1	
Slope of line 1, $\frac{ \varepsilon }{dI/dt} = L$ is more for coil 1 than coil 2	1

33. (a) (i) State Lenz's law and explain how this law is a consequence of conservation of energy principle.

(ii) A square shaped loop of side $\frac{l}{2}$ is initially lying outside a region of uniform magnetic field $\vec{B}$ as shown in the figure. The loop is moved towards right with a constant velocity $\vec{v}$ till it goes out of the region of magnetic field.

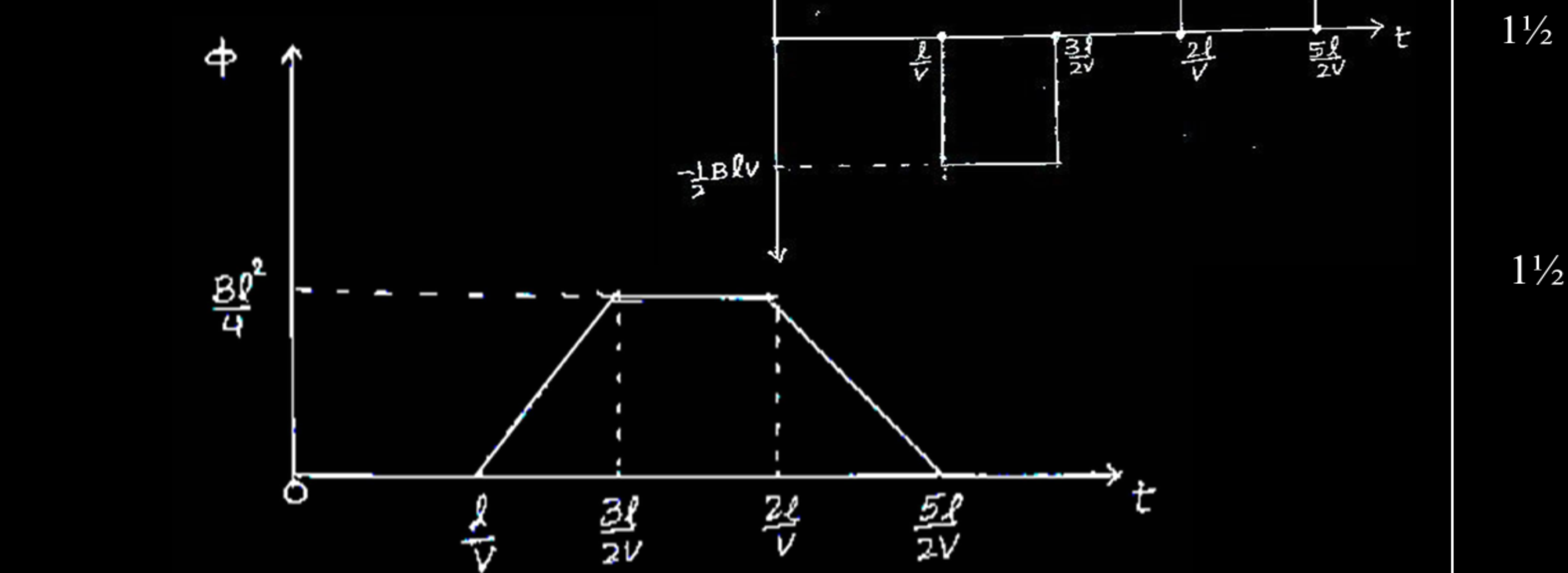


- (I) What will be the directions of induced current when the loop enters the field and when it leaves the field ?
- (II) Draw the plots showing the variation of magnetic flux ϕ linked with the loop with time t and variation of induced emf E with time t . Mark the relevant values of E , ϕ and t on the graphs.

Lenz's law – Polarity of the induced emf is such that it tends to produce a current, which opposes the change in magnetic flux that produces it.
 1/2

When magnet is moved closer/ away from the loop, same/ opposite pole is developed on the approaching face of the loop. So mechanical work is required to move a magnet which gets converted into electrical energy which is consistent with the law of conservation of energy.
 1/2

(ii)
 1/2
 (I) Anticlockwise
 1/2
 (II) Clockwise



6. When current in a coil changes at a steady rate from 8 A to 6 A in 4 ms, an emf of 1.5 V is induced in it. The value of self-inductance of the coil is :

(A) 6 mH

(B) 12 mH

(C) 3 mH

(D) 9 mH

6. When current in a coil changes at a steady rate from 8 A to 6 A in 4 ms, an emf of 1.5 V is induced in it. The value of self-inductance of the coil is :

(A) 6 mH

(B) 12 mH

(C) 3 mH

(D) 9 mH

(C) 3mH

- 24.** (a) Differentiate between magnetic flux through an area and magnetic field at a point.
- (b) A bar magnet is held with its length along the axis of a closed coil. Initially the south pole of the magnet faces the coil. If the magnet is moved towards the coil, explain how a current is induced in the coil and in what direction.

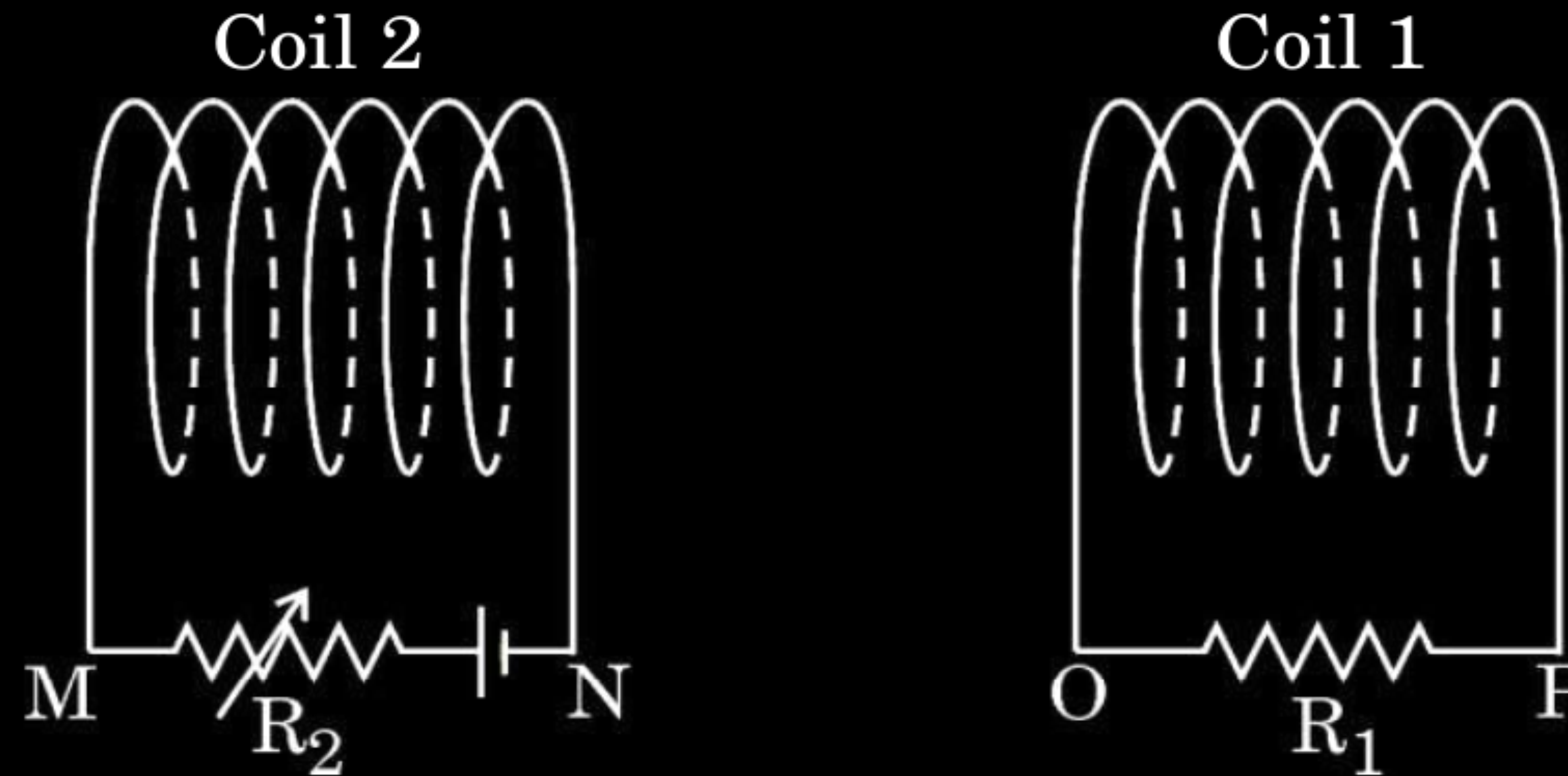
- 24.** (a) Differentiate between magnetic flux through an area and magnetic field at a point.
- (b) A bar magnet is held with its length along the axis of a closed coil. Initially the south pole of the magnet faces the coil. If the magnet is moved towards the coil, explain how a current is induced in the coil and in what direction.

3

(a) Magnetic flux is equal to total number of magnetic field lines passing normal to given area.	$\frac{1}{2}$
Magnetic field at a point in the space around the magnet or moving charge where magnetic force can be experienced.	$\frac{1}{2}$
(b) When south pole of the bar magnet moves closer to coil, the magnetic flux through the coil increases. Hence according to Faraday's law induced emf/current generate in the coil.	1
According to Lenz's law near end of the coil become south pole and as a consequence, current flows in the coil is clockwise direction.	1

- 24.** Two coils '1' and '2' are placed close to each other as shown in the figure. Find the direction of induced current in coil '1' in each of the following situations, justifying your answers :

3



- Coil '2' is moving towards coil '1'.
- Coil '2' is moving away from coil '1'.
- The resistance connected with coil '2' is increased keeping both the coils stationary.

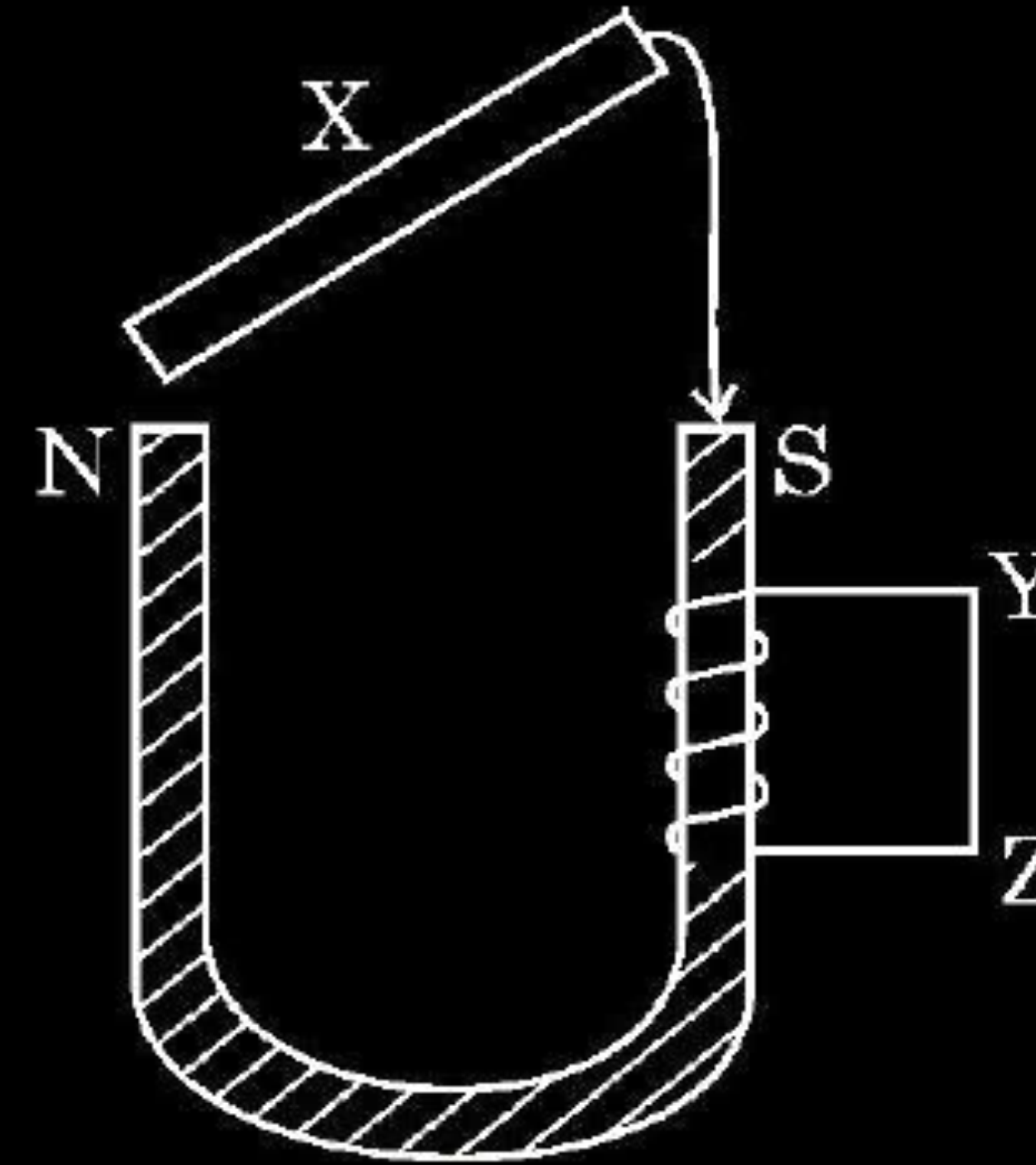
Finding direction of induced current in coil and justification

- | | |
|---|-----------------------------|
| (a) coil '2' is moving toward coil '1' | $\frac{1}{2} + \frac{1}{2}$ |
| (b) coil '2' is moving away from coil '1' | $\frac{1}{2} + \frac{1}{2}$ |
| (c) Resistance connected with coil '2' is increased keeping both the coils stationary | $\frac{1}{2} + \frac{1}{2}$ |

(a) Induced current will be clockwise. Due to the direction flow of current in coil 2, the face approaching to coil 1 behaves like south pole. Induced polarity on coil 1 viewed from the side of coil 2 will be south pole	$\frac{1}{2}$ $\frac{1}{2}$
(b) Induced current will be Anticlockwise Due to direction of flow of current in coil 2, face going away from coil 1 behaves like south pole. Induced polarity on coil 1 viewed from the side of coil 2, will be north pole.	$\frac{1}{2}$ $\frac{1}{2}$
(c) Induced current will be Anticlockwise Till the resistance in coil 2 is increasing, the magnetic flux associated with coil 1 is decreasing. Hence according to Lenz's law current will induce in coil 1, momentarily and after that no induced current in coil	$\frac{1}{2}$ $\frac{1}{2}$

5. A soft iron rod X is allowed to fall on the two poles of a U shaped permanent magnet as shown in figure. A coil is wrapped over one arm of the U shaped magnet.

1

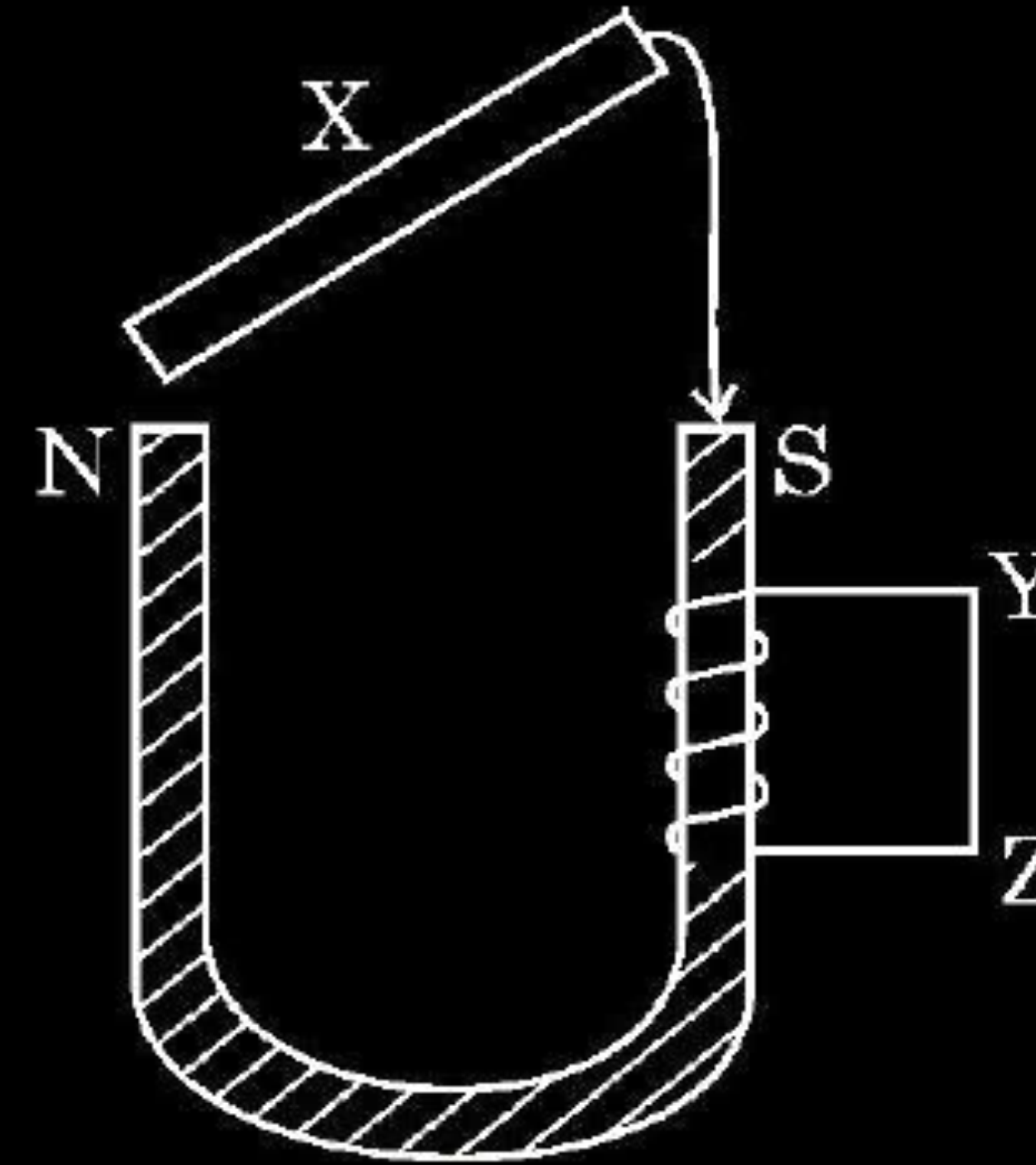


During fall of the rod, the current in the coil will be

- | | |
|-------------------------|---------------------------|
| (A) clockwise current | (B) anticlockwise current |
| (C) alternating current | (D) zero |

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1



During fall of the rod, the current in the coil will be

- | | |
|-------------------------|---------------------------|
| (A) clockwise current | (B) anticlockwise current |
| (C) alternating current | (D) zero |

(B) anticlockwise current

5. Two long solenoids of radii r_1 and r_2 ($>r_1$) and number of turns per unit length n_1 and n_2 respectively are co-axially wrapped one over the other. The ratio of self-inductance of inner solenoid to their mutual inductance is –

(A) $\frac{n_1}{n_2}$

(B) $\frac{n_2}{n_1}$

(C) $\frac{n_1 r_1^2}{n_2 r_2^2}$

(D) $\frac{n_2 r_1^2}{n_1 r_2^2}$

5. Two long solenoids of radii r_1 and r_2 ($>r_1$) and number of turns per unit length n_1 and n_2 respectively are co-axially wrapped one over the other. The ratio of self-inductance of inner solenoid to their mutual inductance is –

1

(A) $\frac{n_1}{n_2}$

(B) $\frac{n_2}{n_1}$

(C) $\frac{n_1 r_1^2}{n_2 r_2^2}$

(D) $\frac{n_2 r_1^2}{n_1 r_2^2}$

(A) $\frac{n_1}{n_2}$

7. A coil of an ac generator, having 100 turns and area 0.1 m^2 each, rotates at half a rotation per second in a magnetic field of 0.02 T . The maximum emf generated in the coil is

(A) 0.31 V

(B) 0.20 V

(C) 0.63 V

(D) 0.10 V

7. A coil of an ac generator, having 100 turns and area 0.1 m^2 each, rotates at half a rotation per second in a magnetic field of 0.02 T . The maximum emf generated in the coil is

(A) 0.31 V

(B) 0.20 V

(C) 0.63 V

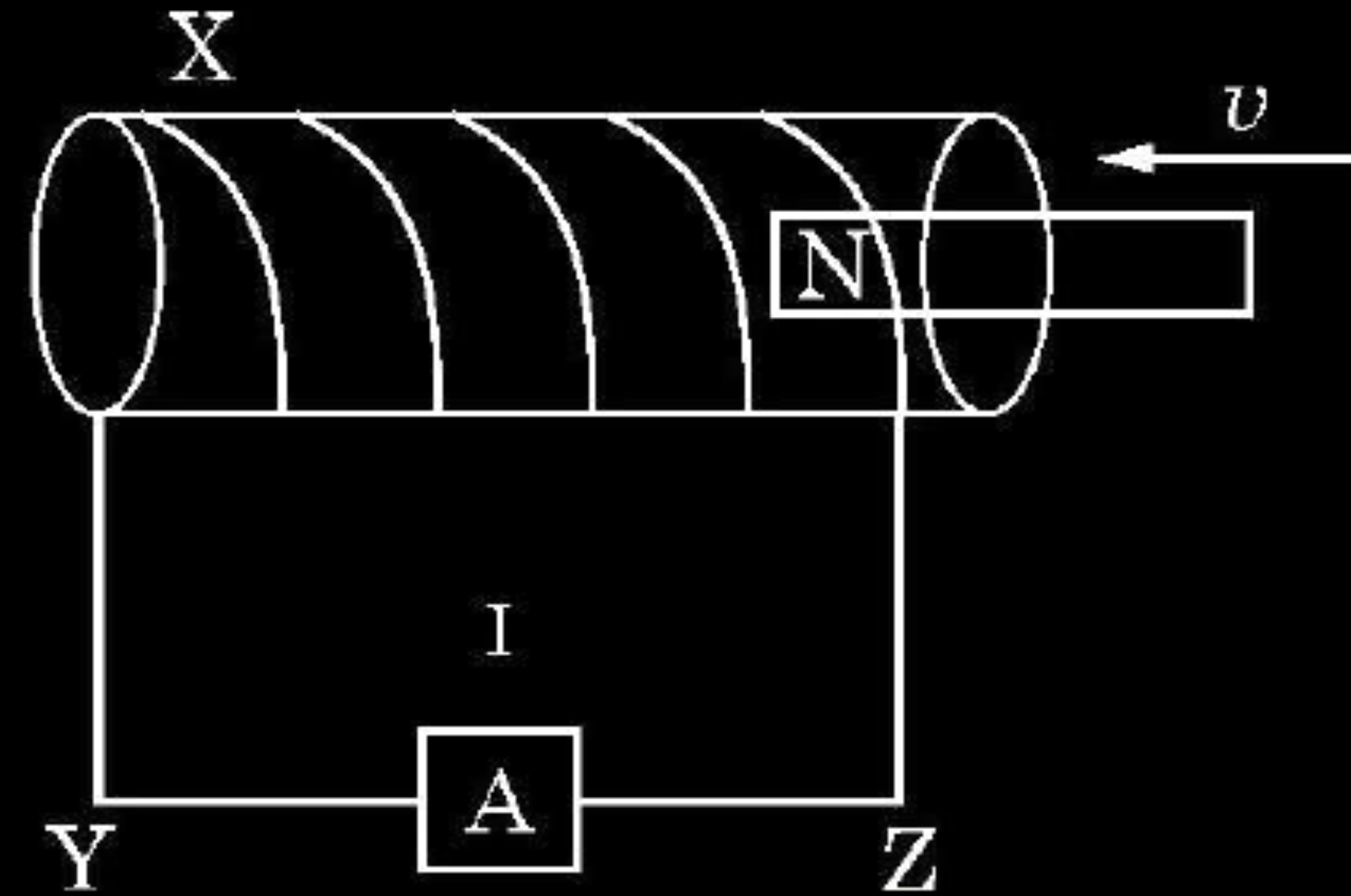
(D) 0.10 V

1

(C) 0.63 V

5. In the figure X is a coil wound over a hollow wooden pipe.

1

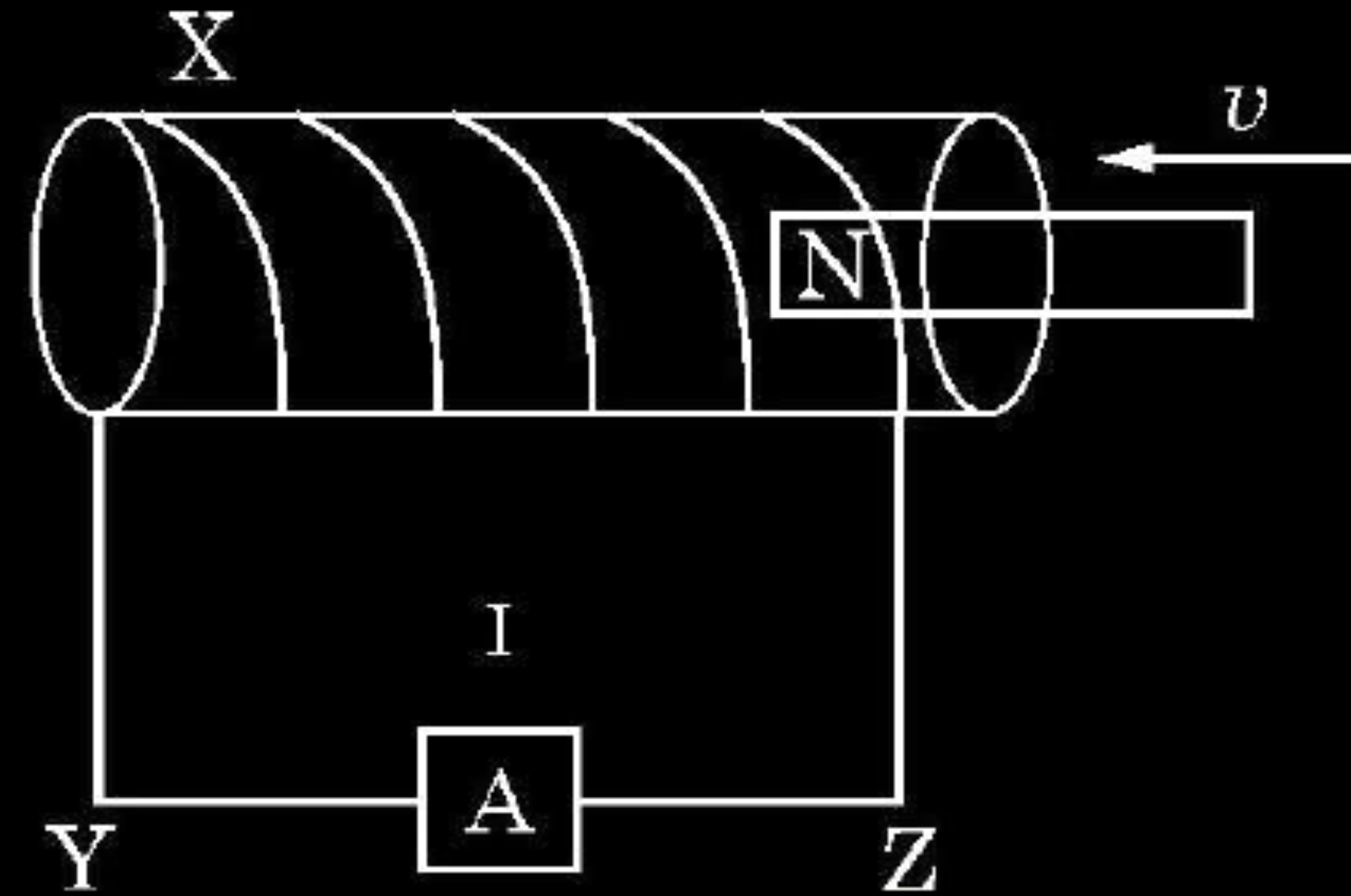


A permanent magnet is pushed at a constant speed v from the right into the pipe and it comes out at the left end of the pipe. During the entry and the exit of the magnet, the current in the wire YZ will be from

- | | |
|----------------------------|----------------------------|
| (A) Y to Z and then Y to Z | (B) Z to Y and then Y to Z |
| (C) Y to Z and then Z to Y | (D) Z to Y and then Z to Y |

5. In the figure X is a coil wound over a hollow wooden pipe.

1

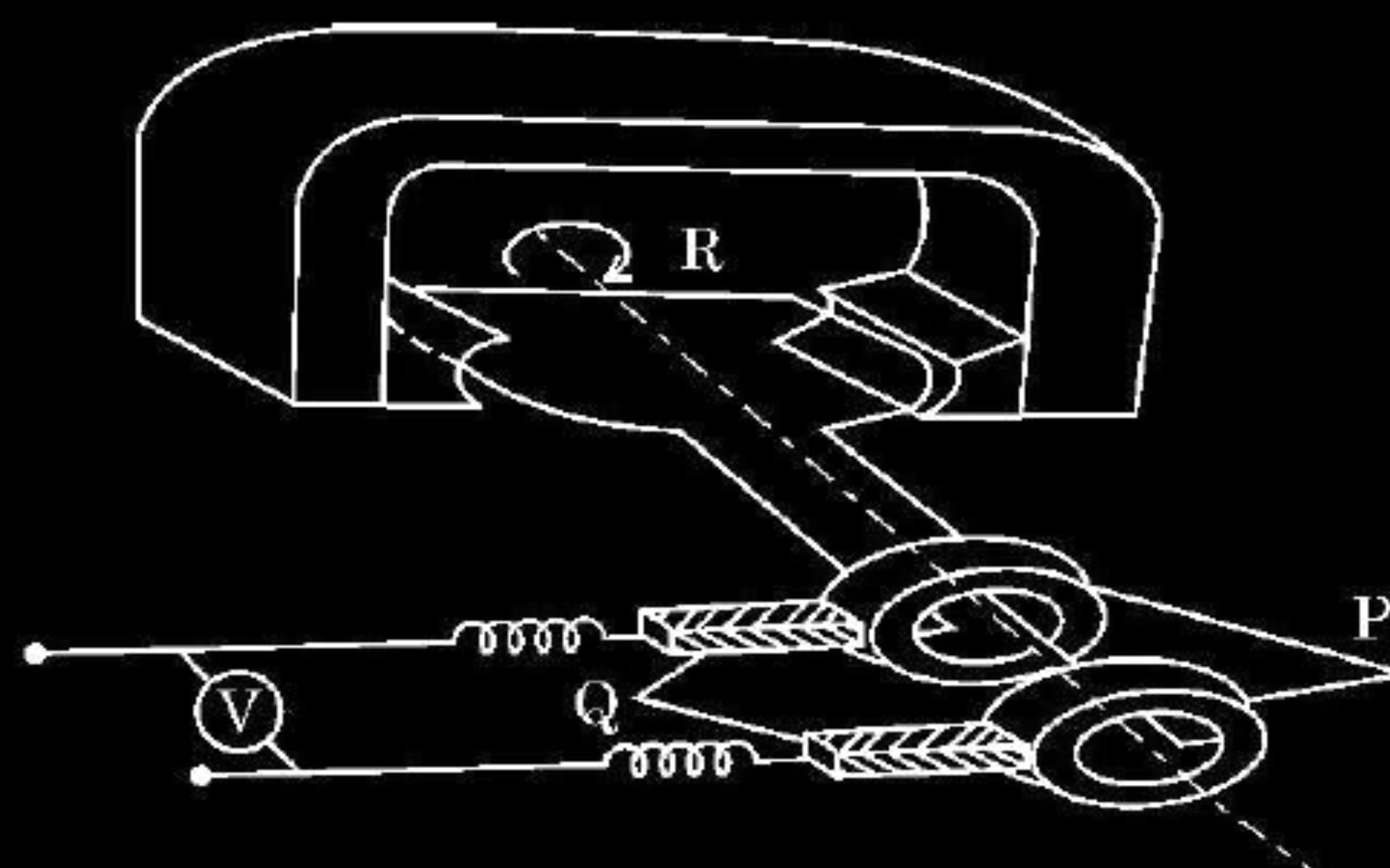


A permanent magnet is pushed at a constant speed v from the right into the pipe and it comes out at the left end of the pipe. During the entry and the exit of the magnet, the current in the wire YZ will be from

- | | |
|----------------------------|----------------------------|
| (A) Y to Z and then Y to Z | (B) Z to Y and then Y to Z |
| (C) Y to Z and then Z to Y | (D) Z to Y and then Z to Y |

(B) Z to Y and then Y to Z

24. (a) State Lenz's law.
(b) In the given figure :



- (i) Identify the machine.
- (ii) Name the parts P and Q and R of the machine.
- (iii) Give the polarities of the magnetic poles.
- (iv) Write the two ways of increasing the output voltage.

(a) Lenz's law- The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it.	$\frac{1}{2}$
(b) (i) AC generator	$\frac{1}{2}$
(ii) P – Slip rings Q – Carbon brushes R- Armature coil	$\frac{1}{2}$
(iii) Left side of the magnet is North & right side is South or vice-versa.	$\frac{1}{2}$
(iv) (Any two)	
-By increasing the number of turns in the armature coil. -By increasing the speed of rotation of the armature coil. -By increasing the strength of the magnetic field B.	$\frac{1}{2} + \frac{1}{2}$

5. You are required to design an air-filled solenoid of inductance 0.016 H having a length 0.81 m and radius 0.02 m . The number of turns in the solenoid should be

(A) 2592

(B) 2866

(C) 2976

(D) 3140

5. You are required to design an air-filled solenoid of inductance 0.016 H having a length 0.81 m and radius 0.02 m . The number of turns in the solenoid should be

(A) 2592

(B) 2866

(C) 2976

(D) 3140

1

(B) 2866

15. **Assertion (A)** : It is difficult to move a magnet into a coil of large number of turns when the circuit of the coil is closed. **1**
- Reason (R)** : The direction of induced current in a coil with its circuit closed, due to motion of a magnet, is such that it opposes the cause.

15. **Assertion (A)** : It is difficult to move a magnet into a coil of large number of turns when the circuit of the coil is closed. 1
- Reason (R)** : The direction of induced current in a coil with its circuit closed, due to motion of a magnet, is such that it opposes the cause.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

4. A circular coil of diameter 15 mm having 300 turns is placed in a magnetic field of 30 mT such that the plane of the coil is perpendicular to the direction of magnetic field. The magnetic field is reduced uniformly to zero in 20 ms and again increased uniformly to 30 mT in 40 ms. If the emfs induced in the two time intervals are e_1 and e_2 respectively, then the value of e_1/e_2 is

(A) $\frac{1}{2}$

(B) 1

(C) 2

(D) 4

4. A circular coil of diameter 15 mm having 300 turns is placed in a magnetic field of 30 mT such that the plane of the coil is perpendicular to the direction of magnetic field. The magnetic field is reduced uniformly to zero in 20 ms and again increased uniformly to 30 mT in 40 ms. If the emfs induced in the two time intervals are e_1 and e_2 respectively, then the value of e_1/e_2 is

(A) $\frac{1}{2}$

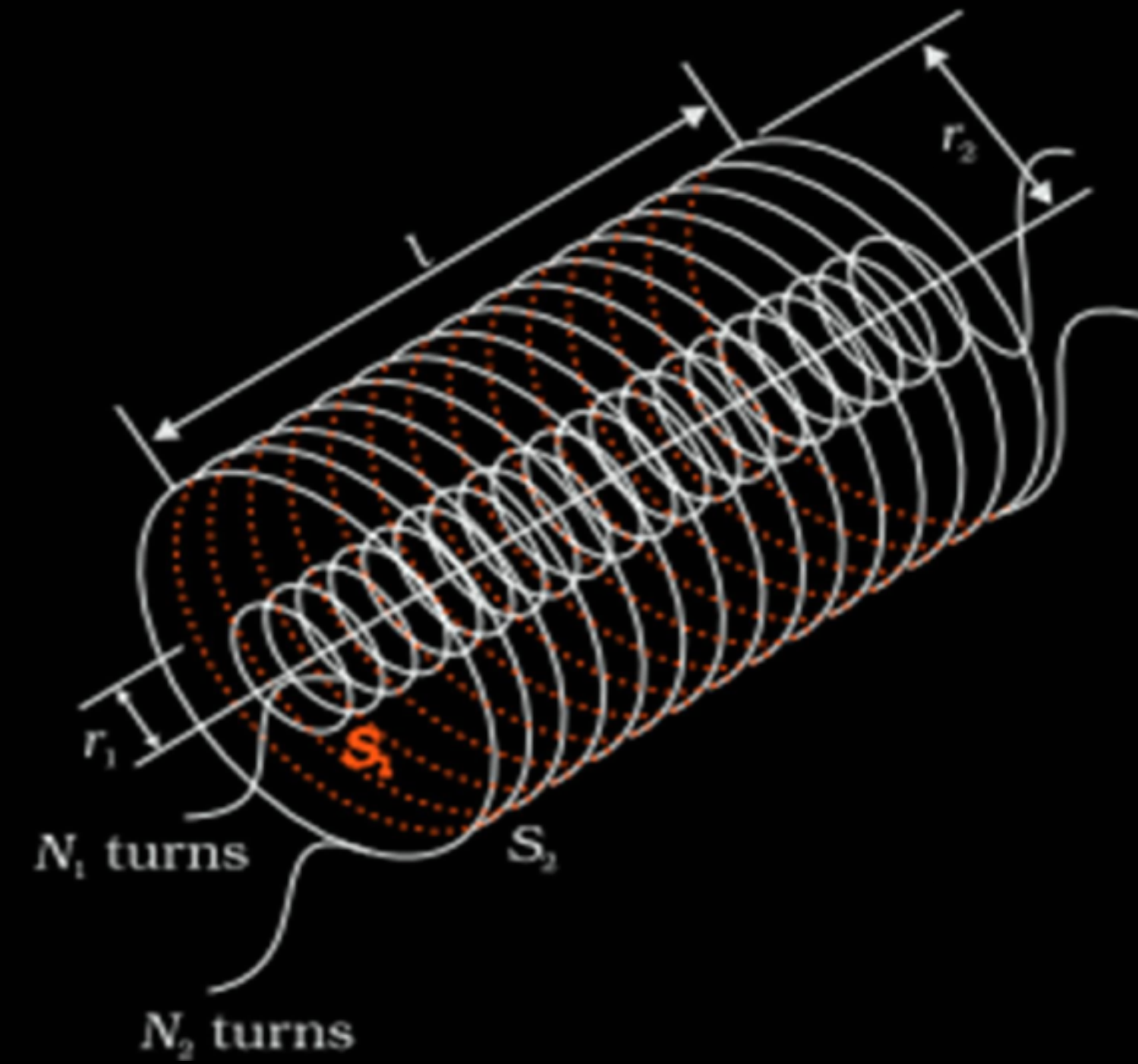
(B) 1

(C) 2

(D) 4

(C) 2

24. Consider two long co-axial solenoids S_1 and S_2 , each of length l ($\gg r_2$) and of radius r_1 and r_2 ($r_2 > r_1$). The number of turns per unit length are n_1 and n_2 respectively. Derive an expression for mutual inductance M_{12} of solenoid S_1 with respect to solenoid S_2 . Show that $M_{21} = M_{12}$.



Let N_1 and N_2 be the total number of turns of coils S_1 and S_2 respectively.
When current I_2 is set up in S_2 , flux linkage with solenoid S_1 is –

$$N_1\phi_1 = M_{12}I_2 \quad \text{--(i)}$$

$$N_1\phi_1 = (n_1l)(\pi r_1^2)(\mu_0 n_2 I_2)$$

$$N_1\phi_1 = \mu_0 n_1 n_2 \pi r_1^2 l I_2 \quad \text{--(ii)}$$

From (i) and (ii)

$$M_{12} = \mu_0 n_1 n_2 \pi r_1^2 l$$

Considering Reverse case, when I_1 current is set up in S_1 , flux linkage with S_2 is –

$$N_2\phi_2 = M_{21}I_1 \quad \text{--(iii)}$$

$$N_2\phi_2 = (n_2l)(\pi r_1^2)(\mu_0 n_1 I_1) \quad \text{--(iv)}$$

From (iii) and (iv)

$$M_{21} = n_1 n_2 \pi r_1^2 l$$

$$\therefore M_{12} = M_{21}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

4. A coil has 100 turns, each of area 0.05 m^2 and total resistance 1.5Ω . It is inserted at an instant in a magnetic field of 90 mT , with its axis parallel to the field. The charge induced in the coil at that instant is :

(A) 3.0 mC

(B) 0.30 C

(C) 0.45 C

(D) 1.5 C

4. A coil has 100 turns, each of area 0.05 m^2 and total resistance 1.5Ω . It is inserted at an instant in a magnetic field of 90 mT , with its axis parallel to the field. The charge induced in the coil at that instant is : **1**
- (A) 3.0 mC (B) 0.30 C
(C) 0.45 C (D) 1.5 C

(B) 0.30 C

24. (a) Show that the energy required to build up the current I in a coil of inductance L is $\frac{1}{2} LI^2$.
- (b) Considering the case of magnetic field produced by air-filled current carrying solenoid, show that the magnetic energy density of a magnetic field B is $\frac{B^2}{2\mu_0}$.

a) Induced emf in an inductor

$$|\varepsilon| = L \frac{dI}{dt}$$

1/2

Rate of work done at any instant

$$\frac{dW}{dt} = |\varepsilon| I$$

1/2

Total Amount of work done in establishing current I

$$W = \int dW = \int_0^I LI dI$$

Energy required to build up current I is

$$W = \frac{1}{2} L I^2$$

1/2

b) The Magnetic Energy is

$$W = U_B = \frac{1}{2} L I^2$$

$$= \frac{1}{2} L \left(\frac{B}{n \mu_0} \right)^2 \quad \text{as } B = n \mu_0 I$$

1/2

Using $L = \mu_0 n^2 A l$

$$U_B = \frac{1}{2} (\mu_0 n^2 A l) \left(\frac{B^2}{\mu_0^2 n^2} \right)$$

1/2

Energy density = $\frac{U_B}{\text{volume}}$

$$\begin{aligned} \frac{U_B}{\text{volume}} &= \frac{1}{2} \times \mu_0 n^2 A l \times \frac{B^2}{\mu_0^2 n^2} \times \frac{1}{A l} \\ &= \frac{1}{2} \frac{B^2}{\mu_0} \end{aligned}$$

1/2

5. A rectangular loop of area vector $\vec{A} = [\hat{i} + \sqrt{3} \hat{j}] \times 10^{-2} \text{ m}^2$ is placed in a magnetic field $\vec{B} = [4.0t^2 + 2.0] \hat{i}$, where B is in tesla and t in seconds. At $t = 1.0 \text{ s}$, the magnitude of induced emf in the loop is :

(A) 80 mV

(B) 0.11 V

(C) 0.22 V

(D) 0.8 V

5. A rectangular loop of area vector $\vec{A} = [\hat{i} + \sqrt{3} \hat{j}] \times 10^{-2} \text{ m}^2$ is placed in a magnetic field $\vec{B} = [4.0t^2 + 2.0] \hat{i}$, where B is in tesla and t in seconds. At $t = 1.0 \text{ s}$, the magnitude of induced emf in the loop is :

(A) 80 mV

(B) 0.11 V

(C) 0.22 V

(D) 0.8 V

(A) 80 mV

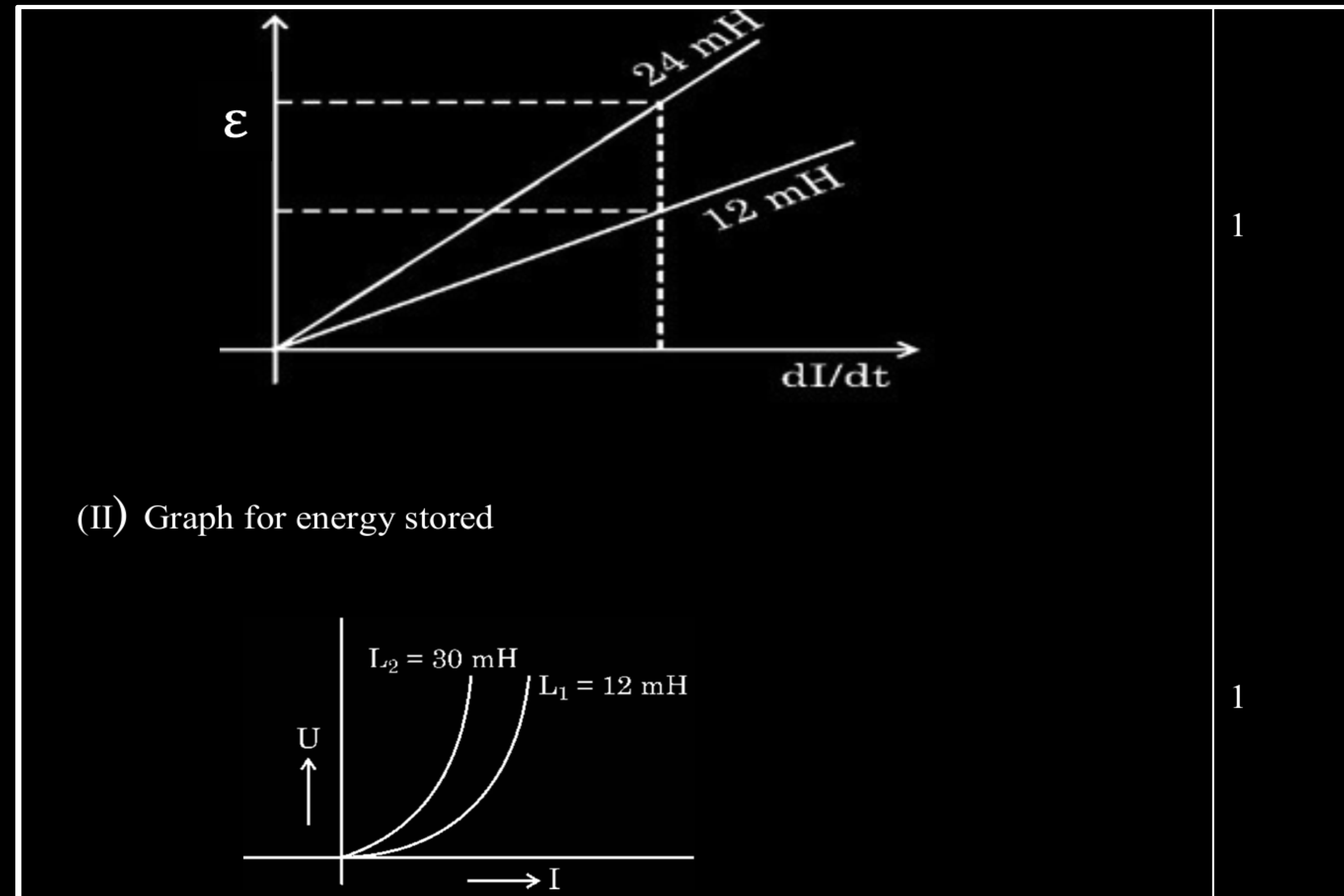
- 12.** Whenever a magnet is moved either towards or away from a conducting coil, an emf is induced whose magnitude is independent of the :
- (A) number of turns in the coil
 - (B) resistance of the coil
 - (C) speed with which the magnet is moved
 - (D) area of the coil

- 12.** Whenever a magnet is moved either towards or away from a conducting coil, an emf is induced whose magnitude is independent of the :
- (A) number of turns in the coil
 - (B) resistance of the coil
 - (C) speed with which the magnet is moved
 - (D) area of the coil

(B) the resistance of the coil

- (ii) The currents through two inductors of self-inductance 12 mH and 24 mH are increasing with time at the same rate. Draw graphs showing the variation of the :
- (I) magnitude of emf induced with the rate of change of current in each inductor.
 - (II) energy stored in each inductor with the current flowing through it.

- (ii) The currents through two inductors of self-inductance 12 mH and 24 mH are increasing with time at the same rate. Draw graphs showing the variation of the :
- (I) magnitude of emf induced with the rate of change of current in each inductor.
- (II) energy stored in each inductor with the current flowing through it.



4. A current of 5.0 A in one coil is reduced to zero in 10 ms inducing an emf of 9.0 V in the neighbouring coil. The mutual inductance of the two coils is :

(A) 18 mH

(B) 1.8 mH

(C) 0.18 mH

(D) 0.18 H

4. A current of 5.0 A in one coil is reduced to zero in 10 ms inducing an emf of 9.0 V in the neighbouring coil. The mutual inductance of the two coils is :

(A) 18 mH

(B) 1.8 mH

(C) 0.18 mH

(D) 0.18 H

(A) 18 mH

6. Which of the following will *not* increase the output voltage in a generator ?
- (A) Rotating the coil rapidly
 - (B) Changing the material of the coil
 - (C) Increasing the area of coil
 - (D) Increasing the magnetic field through the coil

6. Which of the following will *not* increase the output voltage in a generator ?
- (A) Rotating the coil rapidly
 - (B) Changing the material of the coil
 - (C) Increasing the area of coil
 - (D) Increasing the magnetic field through the coil

(B) Changing the material of the coil.

- 23.** (a) State Faraday's Law of electromagnetic induction.
- (b) A rectangular coil of area A having number of turns N is rotated with angular velocity ω in a uniform magnetic field $\vec{B}$ directly perpendicular to the plane of the coil. Obtain an expression for the instantaneous value of emf induced in the coil. When will the induced emf be maximum ?

- 23.** (a) State Faraday's Law of electromagnetic induction.
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3

(a) The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

1

(award $\frac{1}{2}$ mark if the student only writes $\varepsilon = -\frac{d\phi_B}{dt}$)

(b) $\phi = BA \cos \theta = BA \cos \omega t$

For N turns $\phi = NBA \cos \omega t$

$\frac{1}{2}$

$$\varepsilon = -\frac{d\phi_B}{dt}$$

$\frac{1}{2}$

$$\varepsilon = +NBA \omega \sin \omega t$$

$\frac{1}{2}$

For maximum ε , $\omega t = \frac{\pi}{2}$

$\frac{1}{2}$

$$\therefore \varepsilon_0 = NBA\omega$$

5. The mutual inductance of two coils, in a given orientation is 50 mH. If the current in one of the coils changes as $i = 1.0 \sin\left(100 \pi t + \frac{\pi}{3}\right)$ A, the peak value of emf (in volt) induced in the other coil will be

(A) $\frac{\pi}{5}$

(B) 5π

(C) 0.5π

(D) 0.05π

5. The mutual inductance of two coils, in a given orientation is 50 mH. If the current in one of the coils changes as $i = 1.0 \sin\left(100 \pi t + \frac{\pi}{3}\right)$ A, the peak value of emf (in volt) induced in the other coil will be

(A) $\frac{\pi}{5}$

(B) 5π

(C) 0.5π

(D) 0.05π

(B) 5π

22. (a) (i) State Lenz's Law. In a closed circuit, the induced current opposes the change in magnetic flux that produced it as per the law of conservation of energy. Justify.
- (ii) A metal rod of length 2 m is rotated with a frequency 60 rev/s about an axis passing through its centre and perpendicular to its length. A uniform magnetic field of 2T perpendicular to its plane of rotation is switched-on in the region. Calculate the e.m.f. induced between the centre and the end of the rod.

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(a) (i) The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it. In a closed loop, when the polarity of induced emf is such that, the induced current favours the change in magnetic flux then the magnetic flux and consequently the current will go on increasing without any external source of energy. This violates law of conservation of energy.

$$\varepsilon = \frac{1}{2} B l^2 \omega$$

$$= \frac{1}{2} \times 2 \times (2)^2 \times (2\pi \times 60)$$

$$= 480\pi \text{ V}$$

$$= 1.51 \times 10^3 \text{ V}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

5. A coil of area of cross-section 0.5 m^2 is placed in a magnetic field acting normally to its plane. The field varies as $B = 0.5t^2 + 2t$, where B is in tesla and t in seconds. The emf induced in the coil at $t = 1\text{s}$ is
- | | |
|---------------------|---------------------|
| (A) 0.5 V | (B) 1.0 V |
| (C) 1.5 V | (D) 3.0 V |

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- (A) 0.5 V (B) 1.0 V
(C) 1.5 V (D) 3.0 V

(C) 1.5V

5. Two coils are placed near each other. When the current in one coil is changed at the rate of 5 A/s , an emf of 2 mV is induced in the other. The mutual inductance of the two coils is

(A) 0.4 mH

(B) 2.5 mH

(C) 10 mH

(D) 2.5 H

5. Two coils are placed near each other. When the current in one coil is changed at the rate of 5 A/s , an emf of 2 mV is induced in the other. The mutual inductance of the two coils is

(A) 0.4 mH

(B) 2.5 mH

(C) 10 mH

(D) 2.5 H

(A) 0.4 mH

6. A coil of N turns is placed in a magnetic field $\vec{B}$ such that $\vec{B}$ is perpendicular to the plane of the coil. $\vec{B}$ changes with time as $B = B_0 \cos\left(\frac{2\pi}{T}t\right)$ where T is time period. The magnitude of emf induced in the coil will be maximum at

(A) $t = \frac{nT}{8}$

(B) $t = \frac{nT}{4}$

(C) $t = \frac{nT}{2}$

(D) $t = nT$

Here, $n = 1, 2, 3, 4, \dots$

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(B) $t = \frac{nT}{4}$

(C) $t = \frac{nT}{2}$

(D) $t = nT$

Here, $n = 1, 2, 3, 4, \dots$

1

No option is correct, award 1 mark.

7. A circular loop A of radius R carries a current I . Another circular loop B of radius $r\left(=\frac{R}{20}\right)$ is placed concentrically in the plane of A. The magnetic flux linked with loop B is proportional to

1

- | | |
|-----------------------|----------------|
| (A) R | (B) $\sqrt{R}$ |
| (C) $R^{\frac{3}{2}}$ | (D) R^2 |

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- | | |
|-----------------------|----------------|
| (A) R | (B) $\sqrt{R}$ |
| (C) $R^{\frac{3}{2}}$ | (D) R^2 |

(A) R

22. (a) (i) Define mutual inductance. Write its SI unit. **3**
- (ii) Derive an expression for the mutual inductance of a system of two long coaxial solenoids of same length l , having turns N_1 and N_2 and of radii r_1 and r_2 ($> r_1$).

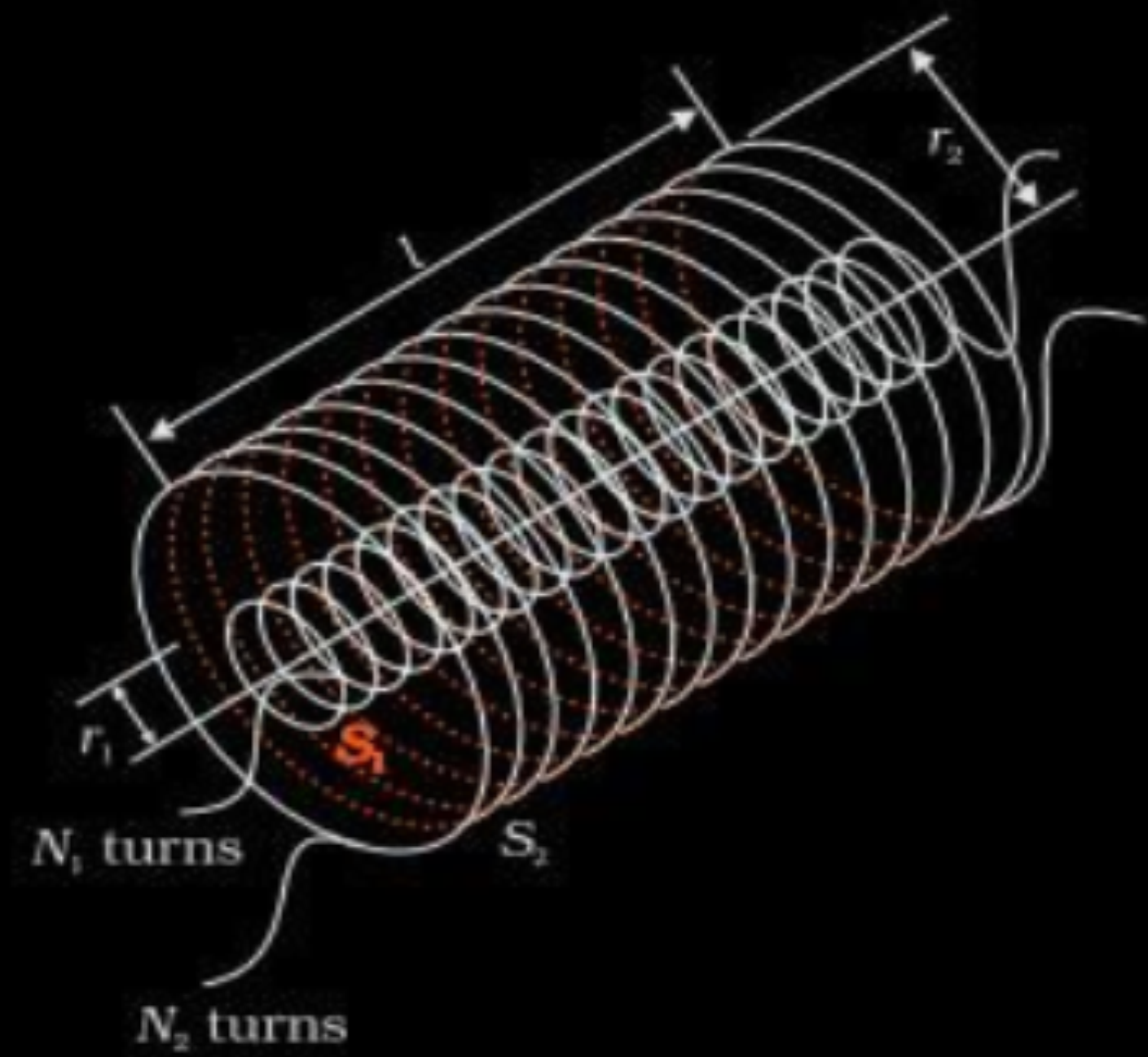
(i) Mutual inductance between two coils is defined as the magnetic flux associated with a coil when unit current flows through neighbouring coil.

Alternatively

Mutual inductance between two coils is defined as the magnitude of induced emf in a coil when the rate of change of current in neighbouring coil is unity.

SI unit of mutual inductance is henry(H).

(ii)



When current I_2 flows in outer solenoid, the resulting flux linkage with inner solenoid.

$$N_1\phi_1 = N_1B_2A_1$$

$$N_1\phi_1 = N_1\left(\frac{\mu_0N_2I_2}{l}\right)\pi r_1^2$$

$$N_1\phi_1 = \frac{\mu_0N_1N_2\pi r_1^2I_2}{l} \dots\dots\dots(1)$$

$$N_1\phi_1 = M_{12}I_2 \dots\dots\dots (2)$$

From equations (1) and (2)

$$M_{12} = \frac{\mu_0N_1N_2\pi r_1^2}{l}$$

1/2

1/2

1/2

1/2

1/2

1/2

7. Consider a solenoid of length l and area of cross-section A with fixed number of turns. The self-inductance of the solenoid will increase if :
- (A) both l and A are increased
 - (B) l is decreased and A is increased
 - (C) l is increased and A is decreased
 - (D) both l and A are decreased

7. Consider a solenoid of length l and area of cross-section A with fixed number of turns. The self-inductance of the solenoid will increase if :
- (A) both l and A are increased
 - (B) l is decreased and A is increased
 - (C) l is increased and A is decreased
 - (D) both l and A are decreased

(B) l is decreased and A is increased

24. Prove that induced charge depends on the net change in the magnetic flux and not on the time interval of the flux change.

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3

Using Faraday's law of electromagnetic induction	
$ \mathcal{E} = \frac{\Delta\phi}{\Delta t}$	$\frac{1}{2}$
$I = \frac{ \mathcal{E} }{R}$	$\frac{1}{2}$
$I = \frac{1}{R} \left(\frac{\Delta\phi}{\Delta t} \right)$	$\frac{1}{2}$
$\frac{\Delta Q}{\Delta t} = \frac{1}{R} \left(\frac{\Delta\phi}{\Delta t} \right)$	$\frac{1}{2}$
$\Delta Q = \frac{\Delta\phi}{R}$	$\frac{1}{2}$
Hence induced charge depends on change in magnetic flux, not on the time interval of flux change.	$\frac{1}{2}$

- (ii) The magnetic flux through a coil of resistance $5\ \Omega$ increases with time as :

$$\phi = (2.0\ t^3 + 5.0\ t^2 + 6.0\ t)\ \text{mWb}$$

Find the magnitude of induced current through the coil at $t = 2\ \text{s}$.

- (ii) The magnetic flux through a coil of resistance $5\ \Omega$ increases with time as :

$$\phi = (2.0\ t^3 + 5.0\ t^2 + 6.0\ t)\ \text{mWb}$$

Find the magnitude of induced current through the coil at $t = 2\ \text{s}$.

$$(ii)\ \phi = (2.0t^3 + 5.0t^2 + 6.0t)\ \text{mWb}$$

$$|\varepsilon| = \frac{d\phi}{dt} = 50 \times 10^{-3}\ \text{V}$$

$$I = \frac{|\varepsilon|}{R}$$

$$I = \frac{50 \times 10^{-3}}{5}\ \text{A} = 10\ \text{mA}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) (i) A rectangular coil of N turns and area of cross-section A is rotated at a steady angular speed ω in a uniform magnetic field. Obtain an expression for the emf induced in the coil at any instant of time.
- (ii) Two coplanar and concentric circular loops L_1 and L_2 are placed coaxially with their centres coinciding. The radii of L_1 and L_2 are 1 cm and 100 cm respectively. Calculate the mutual inductance of the loops. (Take $\pi^2 = 10$)

(i) The flux at any instant t is

$$\phi = NBA \cos\theta = NBA \cos\omega t$$

From Faraday's law

$$\varepsilon = -\frac{d\phi_B}{dt}$$

$$= -NBA \frac{d}{dt} (\cos\omega t)$$

$$\varepsilon = -NBA \omega \sin\omega t$$

$$(ii) \quad M = \frac{\mu_0 \pi r_1^2}{2r_2} = \frac{4\pi \times 10^{-7} \times \pi r_1^2}{2r_2}$$

$$= \frac{2 \times 10 \times 10^{-7} \times (10^{-2})^2}{100 \times 10^{-7}}$$

$$= 2 \times 10^{-10} \text{ H}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

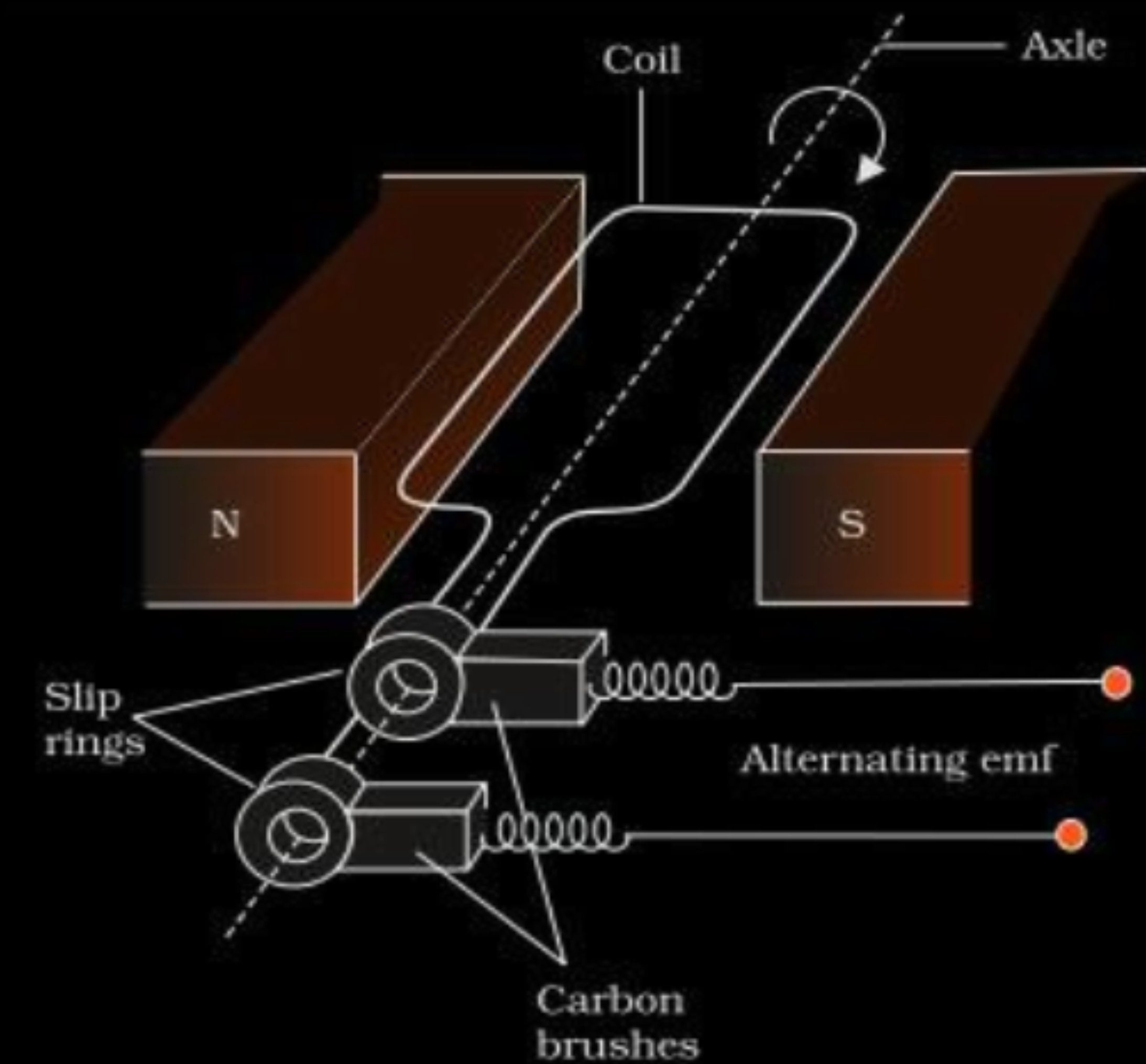
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$



6. The mutual inductance of two coils C_1 and C_2 is 20 mH. In coil C_1 , the current changes from 4 A to zero in 0.2 s. If the resistance of coil C_2 is $4\ \Omega$, then the charge that flows through it per second will be :

(A) 4.0 C

(B) 1.5 C

(C) 0.05 C

(D) 0.1 C

6. The mutual inductance of two coils C_1 and C_2 is 20 mH. In coil C_1 , the current changes from 4 A to zero in 0.2 s. If the resistance of coil C_2 is 4 Ω , then the charge that flows through it per second will be :

(A) 4.0 C

(B) 1.5 C

(C) 0.05 C

(D) 0.1 C

(D) 0.1 C

6. The current in a coil of 15 mH increases uniformly from zero to 4 A in 0.004 s. The emf induced in the coil will be :

(A) 22.5 V

(B) 17.5 V

(C) 15.0 V

(D) 12.5 V

6. The current in a coil of 15 mH increases uniformly from zero to 4 A in 0.004 s. The emf induced in the coil will be :

(A) 22.5 V

(B) 17.5 V

(C) 15.0 V

(D) 12.5 V

(C) 15.0 V

3. The emf induced in a coil rotating in a magnetic field does ***not*** depend upon the following :

- (A) Area of the coil
- (B) Resistance of the coil
- (C) Number of turns in the coil
- (D) Angular speed of rotation of the coil

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- (A) Area of the coil
- (B) Resistance of the coil
- (C) Number of turns in the coil
- (D) Angular speed of rotation of the coil

(B) Resistance of the coil

15. *Assertion (A) :* Lenz's law is a consequence of the law of conservation of energy.

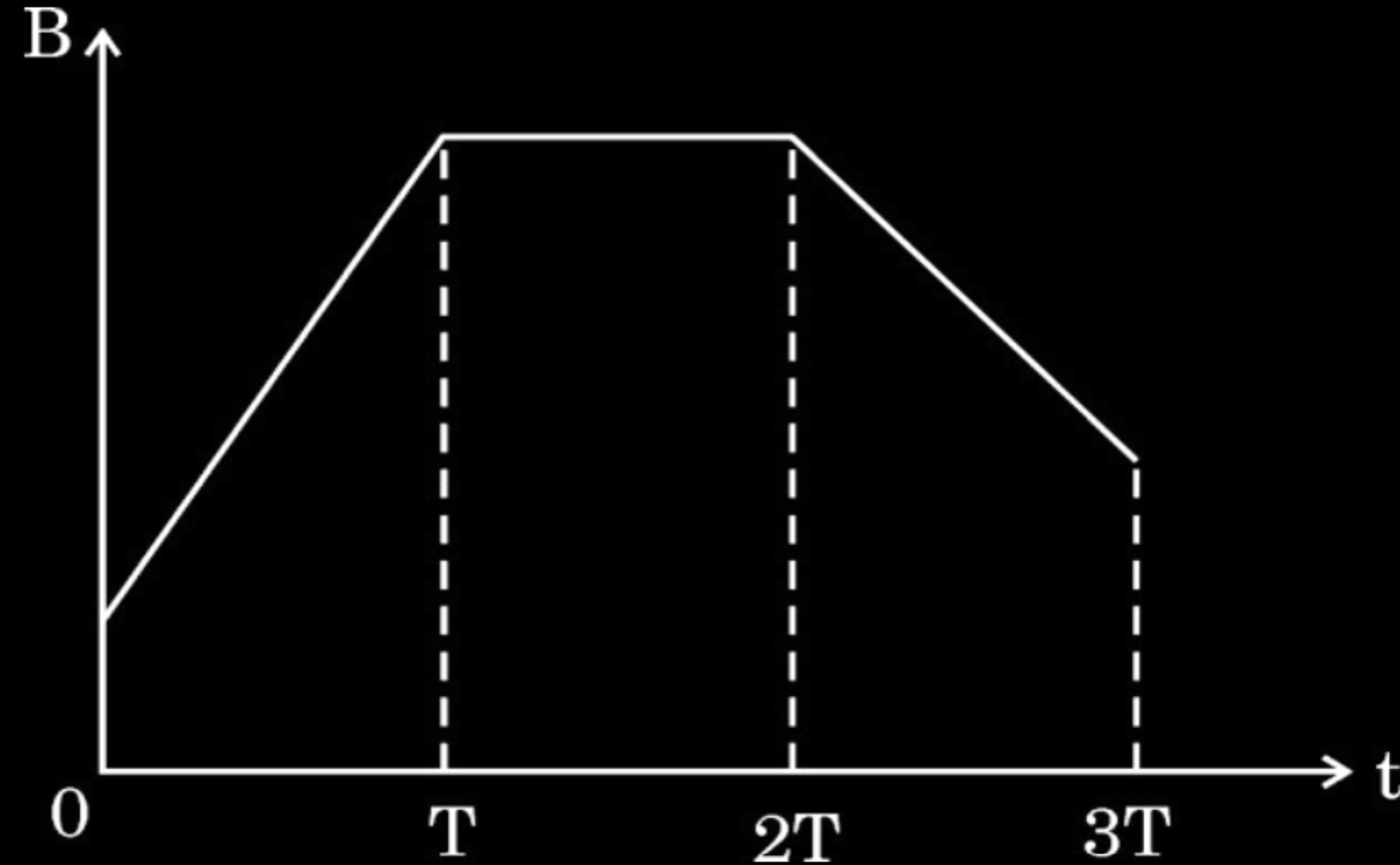
Reason (R) : There is no power loss in an ideal inductor.

15. *Assertion (A) :* Lenz's law is a consequence of the law of conservation of energy.

Reason (R) : There is no power loss in an ideal inductor.

(B) Both Assertion(A) and Reason (R) are true but Reason(R) is not the correct explanation of the Assertion (A)

3. A conducting loop is placed in a magnetic field, normal to its plane. The magnitude of the magnetic field varies with time as shown in the figure. If ϵ_1 , ϵ_2 and ϵ_3 are magnitudes of induced emfs during periods $0 \leq t \leq T$, $T \leq t \leq 2T$ and $2T \leq t \leq 3T$, then :



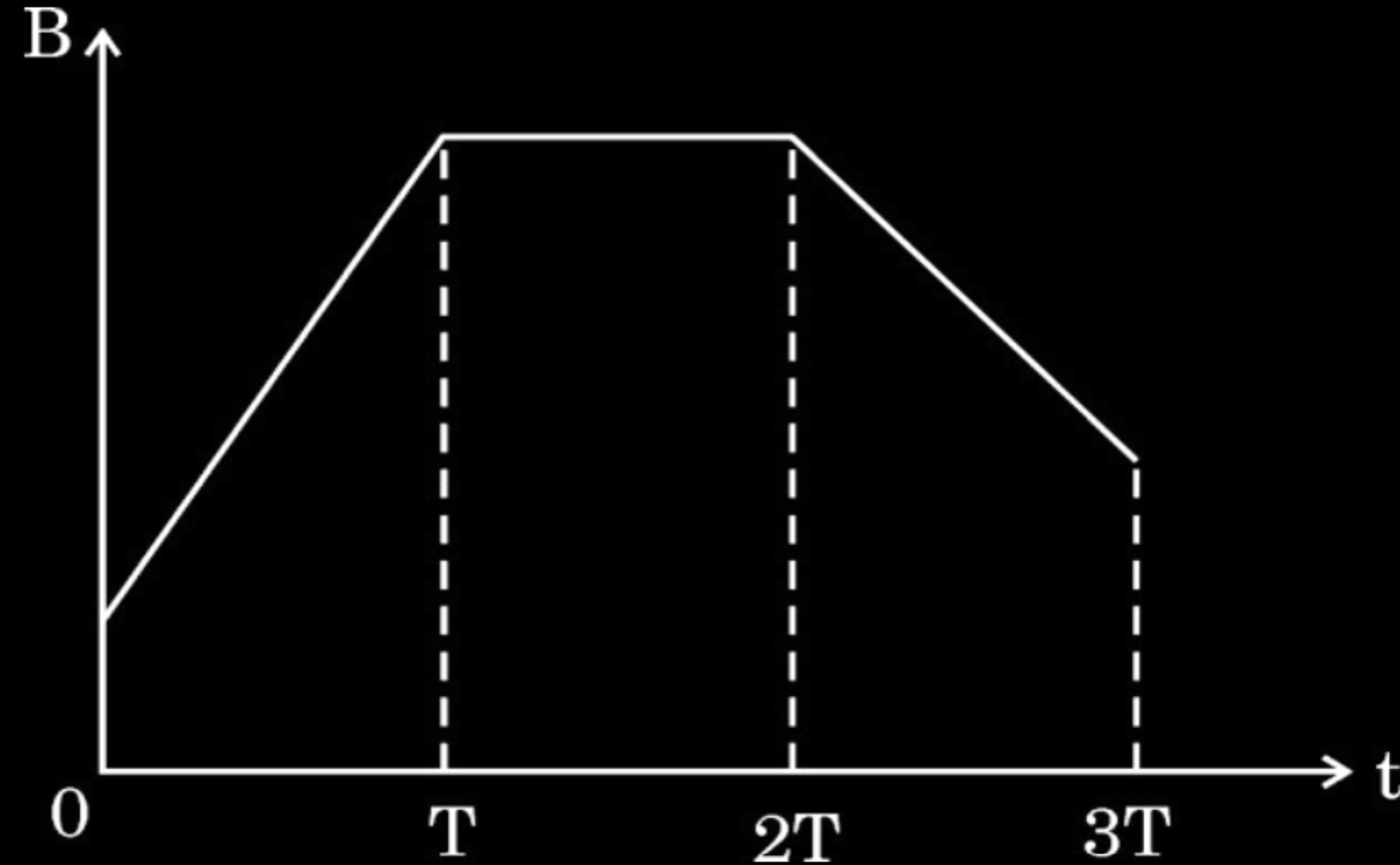
(A) $\epsilon_1 > \epsilon_2 > \epsilon_3$

(C) $\epsilon_3 > \epsilon_1 > \epsilon_2$

(B) $\epsilon_2 > \epsilon_3 > \epsilon_1$

(D) $\epsilon_1 > \epsilon_3 > \epsilon_2$

3. A conducting loop is placed in a magnetic field, normal to its plane. The magnitude of the magnetic field varies with time as shown in the figure. If ε_1 , ε_2 and ε_3 are magnitudes of induced emfs during periods $0 \leq t \leq T$, $T \leq t \leq 2T$ and $2T \leq t \leq 3T$, then :



(A) $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$

(C) $\varepsilon_3 > \varepsilon_1 > \varepsilon_2$

(B) $\varepsilon_2 > \varepsilon_3 > \varepsilon_1$

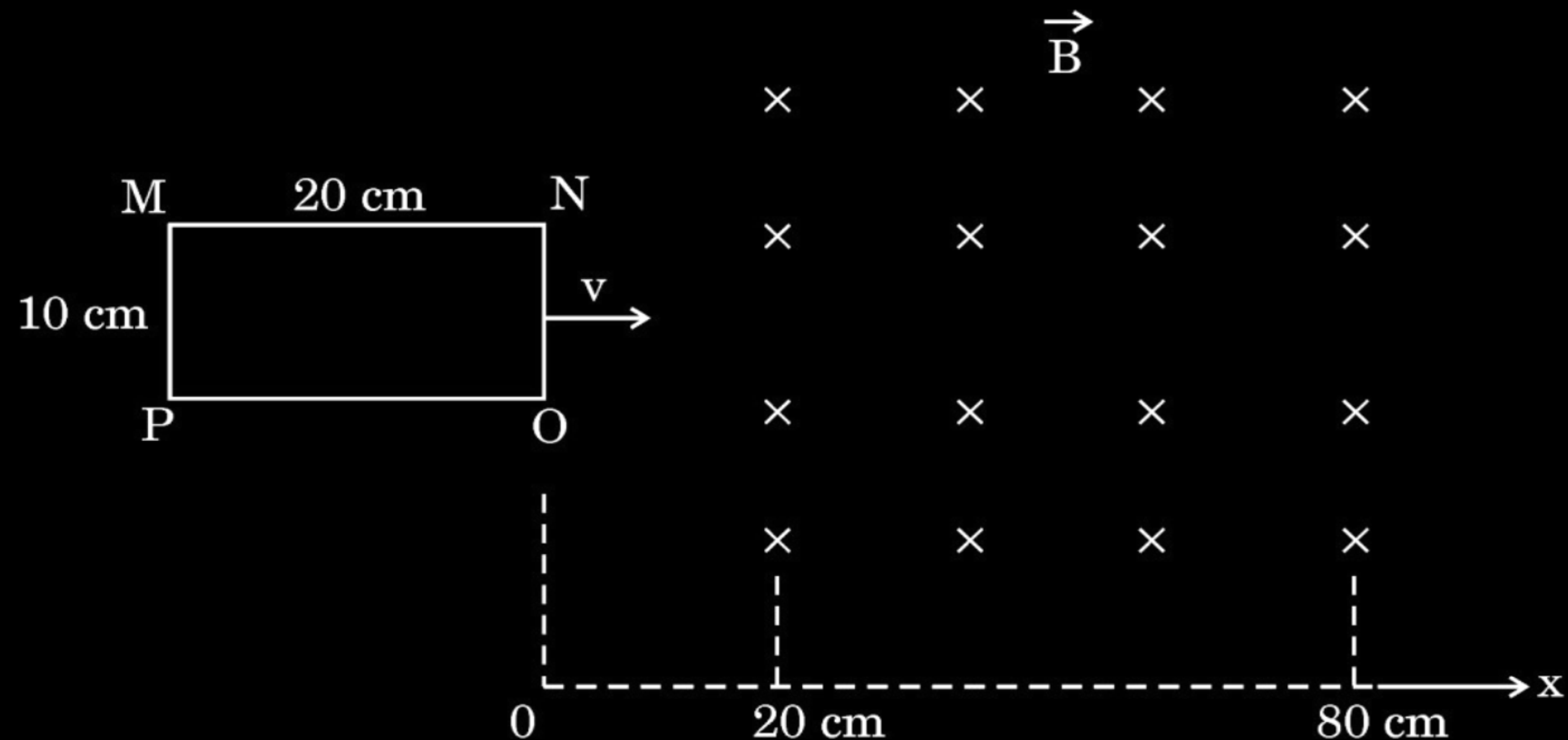
(D) $\varepsilon_1 > \varepsilon_3 > \varepsilon_2$

(D) $\varepsilon_1 > \varepsilon_3 > \varepsilon_2$

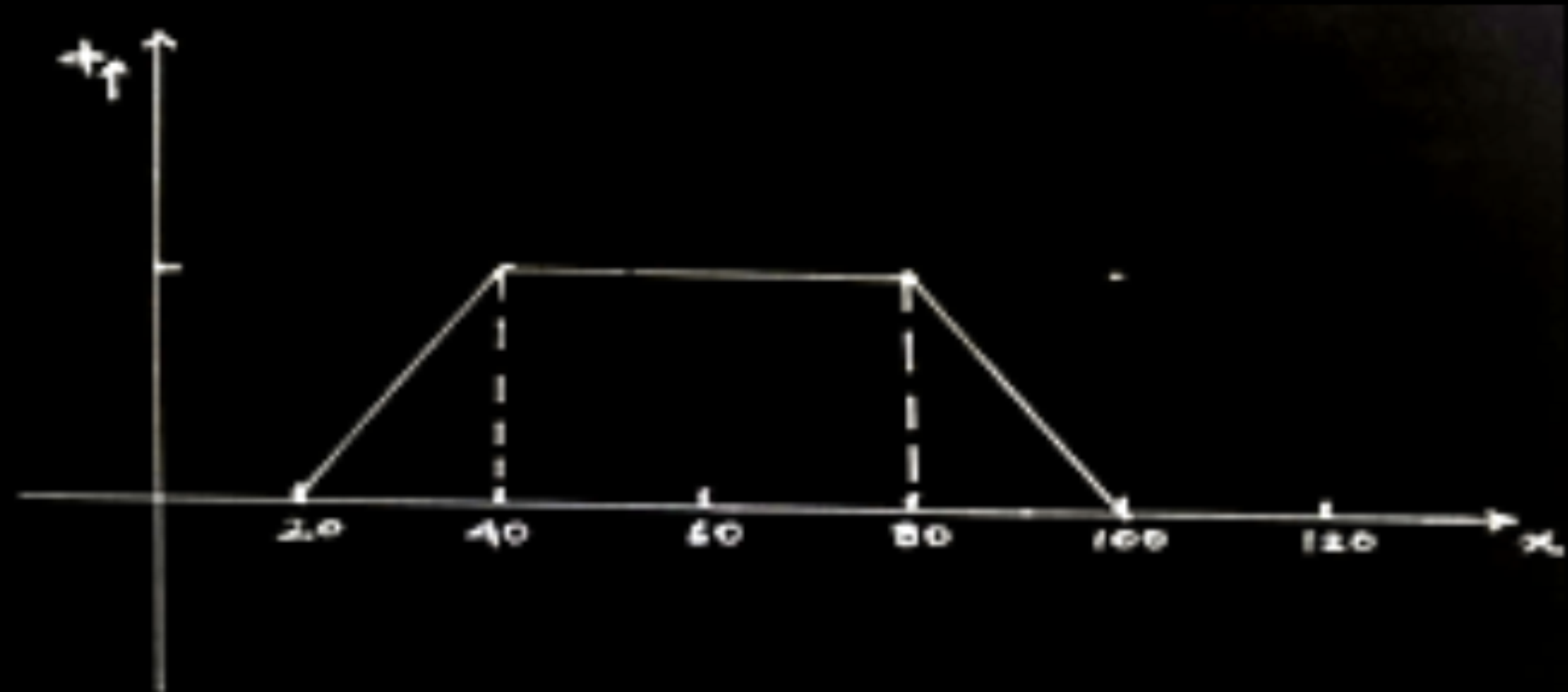
- 24.** A rectangular loop of sides $10\text{ cm} \times 20\text{ cm}$ is kept outside a region of uniform magnetic field $|\vec{B}| = 5\text{ mT}$ as shown in the figure. The loop is moved with the velocity of 5 cm/s till it goes completely out of the magnetic field.

- (a) Plot a graph showing variation of the magnetic flux ϕ with x ($0 \leq x \leq 100\text{ cm}$).
- (b) Find the maximum value of magnetic flux linked with the loop.
- (c) Will an external work be required to be done to move the loop through the magnetic field ?

3



(a)



1 1/2

(b) $\phi = B.A$

$$= 5 \times 10^{-3} \times 20 \times 10^{-2} \times 10 \times 10^{-2}$$

$$= 10^{-4} \text{ Wb}$$

1/2

1/2

1/2

(c) Yes, external work is required.

- 24.** A 100-turn coil of radius 1.6 cm and resistance $5.0\ \Omega$ is co-axial with a solenoid of 250 turns/cm and radius 1.8 cm. The solenoid current drops from 1.5 A to zero in 25 ms. Calculate the current induced in the coil in this duration. (Take $\pi^2 = 10$)

24. A 100-turn coil of radius 1.6 cm and resistance 5.0 Ω is co-axial with a solenoid of 250 turns/cm and radius 1.8 cm. The solenoid current drops from 1.5 A to zero in 25 ms. Calculate the current induced in the coil in this duration. (Take $\pi^2 = 10$)

3

$$\text{Induced emf } (\varepsilon) = \frac{-Nd\phi}{dt}$$

$$= \frac{-NAdB}{dt}$$

$$= -NA \frac{d}{dt} (\mu_0 nI)$$

$$= -N \mu_0 n (\pi r^2) \frac{dI}{dt}$$

$$\varepsilon = \frac{100 \times 4\pi \times 10^{-7} \times 250 \times 10^2 \times \pi \times (1.6 \times 10^{-2})^2 \times 1.5}{25 \times 10^{-3}}$$

$$= 0.1536 \text{ V}$$

$$I = \frac{\varepsilon}{R}$$

$$= 0.03 \text{ A}$$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

15. *Assertion (A) :* The mutual inductance between two coils is maximum when the coils are wound on each other.

Reason (R) : The flux linkage between two coils is maximum when they are wound on each other.

15. *Assertion (A) :* The mutual inductance between two coils is maximum when the coils are wound on each other.

Reason (R) : The flux linkage between two coils is maximum when they are wound on each other.

(A) Assertion(A) is true , Reason (R) is true and Reason (R) is correct explanation of the Assertion (A)

7. A conducting circular loop is placed in a uniform magnetic field $B = 50 \text{ mT}$ with its plane perpendicular to the magnetic field. The radius of the loop is made to shrink at a constant rate of 1 mm s^{-1} . At the instant the radius of the loop is 4 cm , the induced emf in the loop is :

(A) $\pi \mu\text{V}$

(B) $2\pi \mu\text{V}$

(C) $4\pi \mu\text{V}$

(D) $8\pi \mu\text{V}$

7. A conducting circular loop is placed in a uniform magnetic field $B = 50 \text{ mT}$ with its plane perpendicular to the magnetic field. The radius of the loop is made to shrink at a constant rate of 1 mm s^{-1} . At the instant the radius of the loop is 4 cm , the induced emf in the loop is :

(A) $\pi \mu\text{V}$

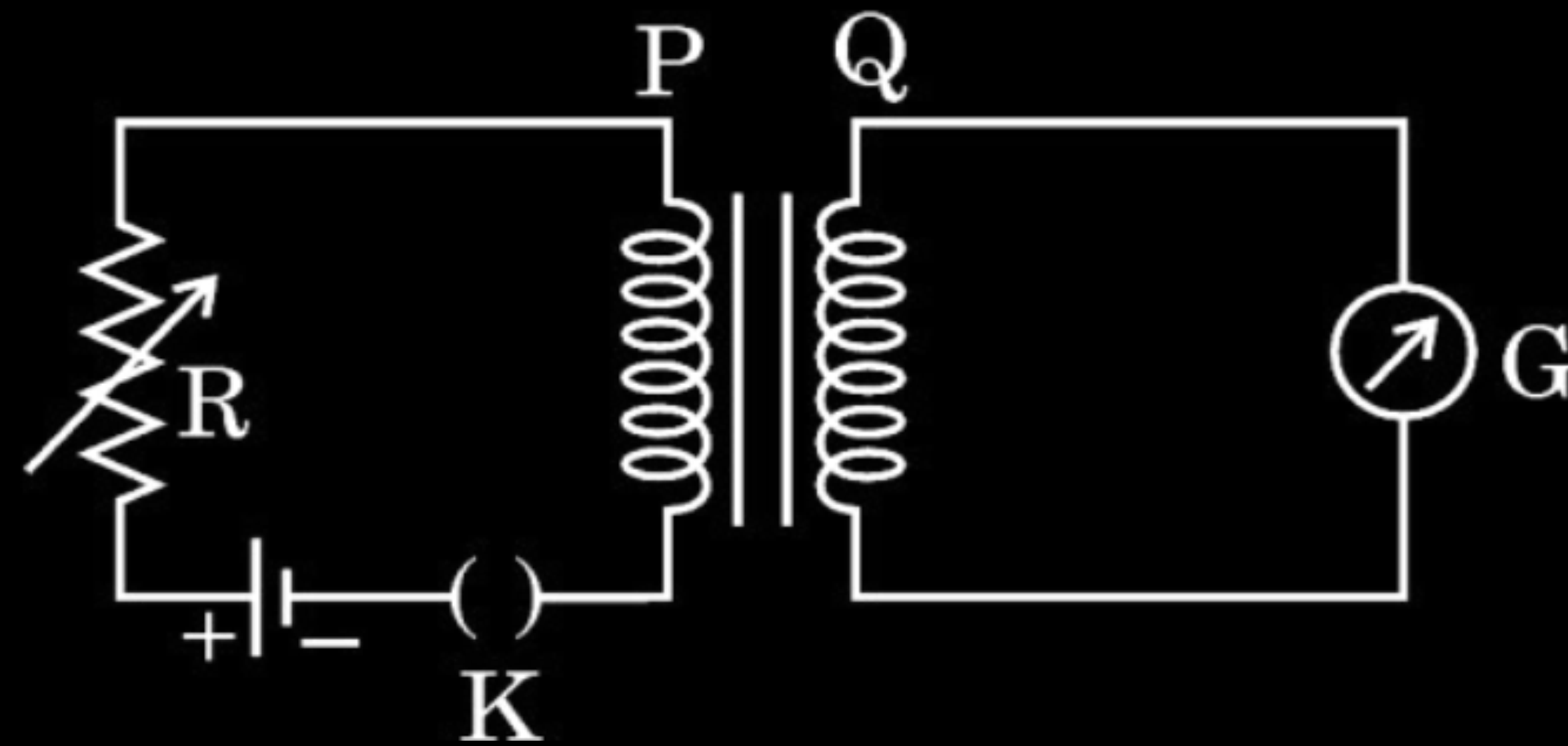
(B) $2\pi \mu\text{V}$

(C) $4\pi \mu\text{V}$

(D) $8\pi \mu\text{V}$

(C) $4\pi\mu\text{V}$

- 25.** Consider the arrangement of two coils P and Q shown in the figure. When current in coil P is switched on or switched off, a current flows in coil Q.
- (a) Explain the phenomenon involved in it.
 - (b) Mention two factors on which the current produced in coil Q depends.
 - (c) Give the direction of current in coil Q when there is a current in the coil P and (i) R is increased, and (ii) R is decreased.



(a) Mutual Induction When an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf in it.	1
(b) Factors on which the current produced in coil Q depends will be: (Any two) (i) Number of turns in coil P and Q (ii) Current flowing through coil P. (iii) Resistance of coil Q. (iv) Mutual Induction between the two coils.	$\frac{1}{2} + \frac{1}{2}$
(c) The direction of current through coil Q: (i) Clockwise when R is increased. (ii) Anticlockwise when R is decreased.	$\frac{1}{2}$ $\frac{1}{2}$

- 32.** (a) (i) Define the term 'self-inductance' of a coil. Obtain an expression for self-inductance of a solenoid of area of cross-section A , length L and having n turns per unit length.
- (ii) The current flowing in a coil of inductance 50 mH is reduced from 10 A to 0 in 20 ms . Calculate the change in flux linked with the coil.

(i) Self-Inductance is the flux linked with a coil when unit current flows through it.

Alternatively:

Self -Inductance is the emf induced in a coil when the rate of change of current through the coil is unity.

Magnetic field due to current I flowing through the solenoid

$$B = \mu_0 n I$$

Flux linked with the solenoid $\phi = \vec{B} \cdot \vec{A}$

$$= \mu_0 n I A$$

Total flux linked with the solenoid $= (nL)(\mu_0 n I) A$

$$= \mu_0 n^2 A L I$$

$$\therefore \text{Self -Inductance} = \mu_0 n^2 A L$$

(ii)

$$\Delta \phi = L \Delta I$$

$$= 50 \times 10^{-3} (10)$$

$$= 0.5 \text{ Wb}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

- (b) (i) Briefly explain the working of an ac generator. Obtain the expression for the instantaneous value of the emf induced in the generator.

(i) The armature coil is mechanically rotated in the uniform magnetic field by some external means. The rotation of the coil causes the magnetic flux through it to change so an emf is induced in it.

Magnetic flux linked with the coil in orientation θ .

$$\phi = NBA \cos \theta$$

$$= NBA \cos \omega t$$

$$e = - \frac{d\phi}{dt}$$

$$= NBA \omega \sin \omega t$$

$$e = e_0 \sin \omega t$$

1

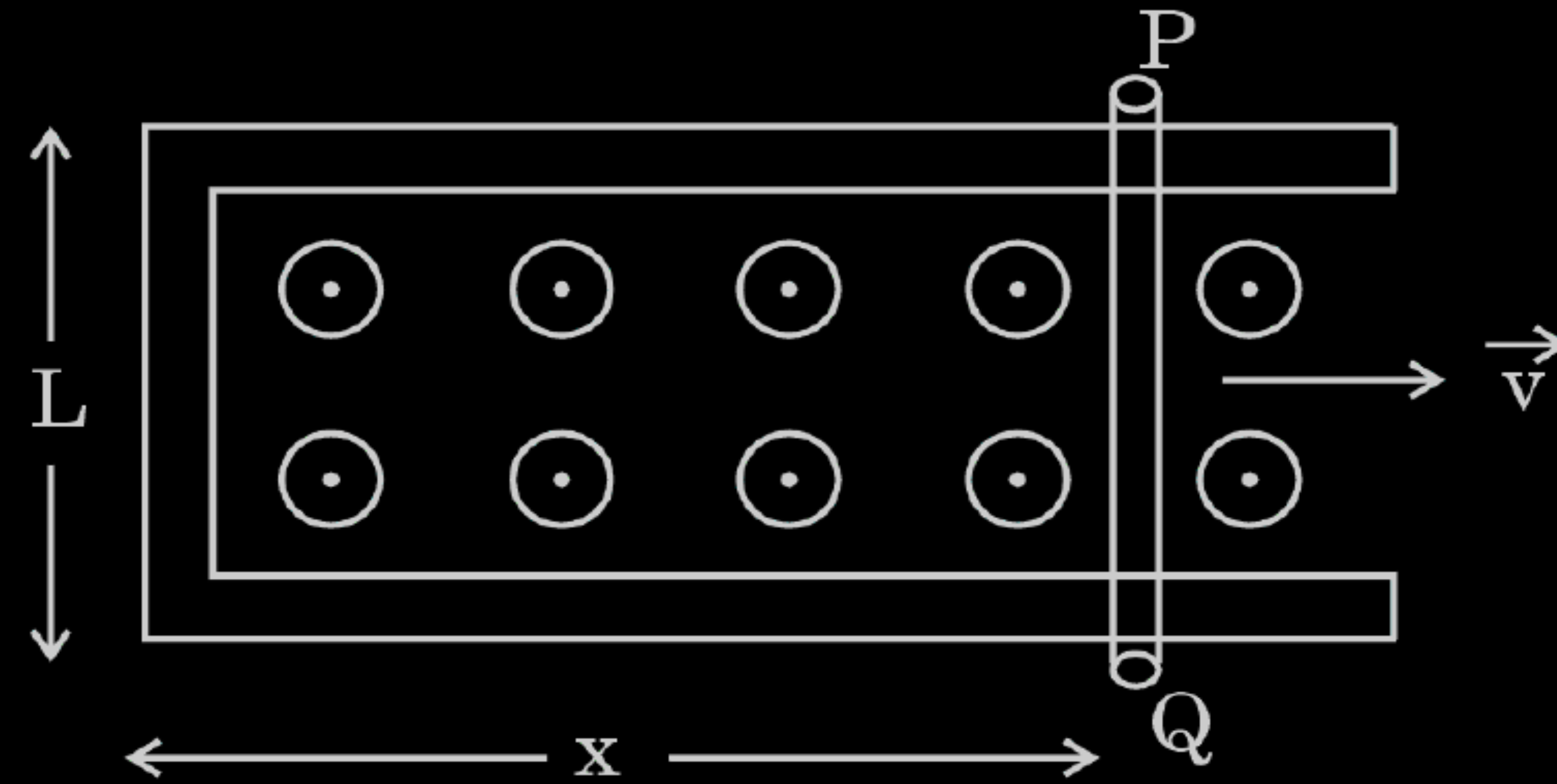
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4. A rod PQ of resistance R lies across frictionless conducting rails in a constant uniform magnetic field $\vec{B}$, as shown in the figure. Assuming that the rails have negligible resistance, the force required to pull the rod to the right at constant velocity $\vec{v}$ is :



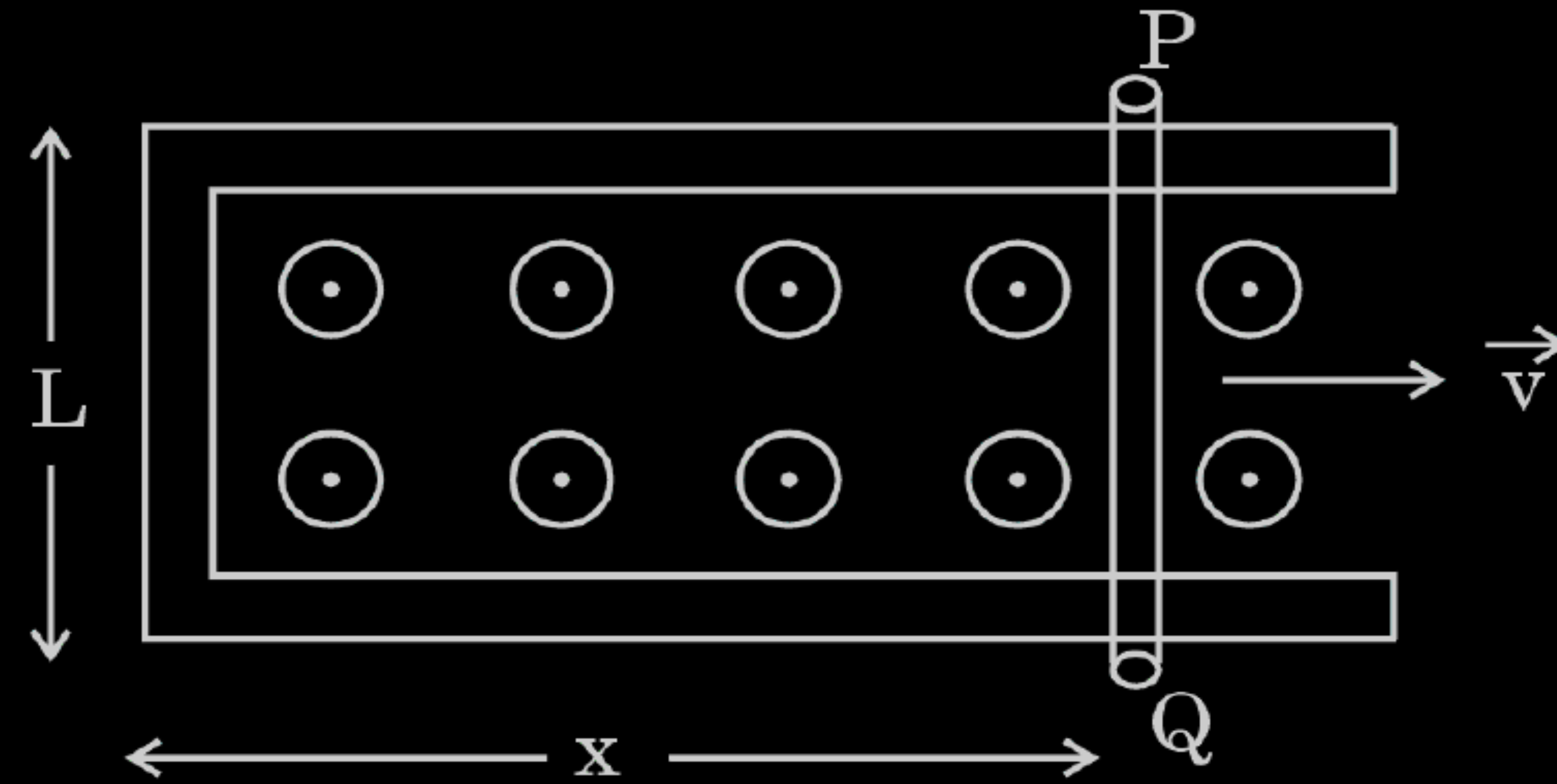
(A) 0

(B) Blv

(C) $\frac{Blv}{R}$

(D) $\frac{B^2 l^2 v}{R}$

4. A rod PQ of resistance R lies across frictionless conducting rails in a constant uniform magnetic field $\vec{B}$, as shown in the figure. Assuming that the rails have negligible resistance, the force required to pull the rod to the right at constant velocity $\vec{v}$ is :



(A) 0

(B) Blv

(C) $\frac{Blv}{R}$

(D) $\frac{B^2 l^2 v}{R}$

(D) $\frac{B^2 l^2 v}{R}$

26. A small circular loop of area $\frac{6}{\pi} \text{ cm}^2$ is placed inside a long solenoid at its centre such that its axis makes an angle of 60° with the axis of the solenoid. The number of turns per cm is 10 in the solenoid. The current in the solenoid changes uniformly from 5 A to zero in 10 ms. Calculate the emf induced in the loop.

26. A small circular loop of area $\frac{6}{\pi} \text{ cm}^2$ is placed inside a long solenoid at its centre such that its axis makes an angle of 60° with the axis of the solenoid. The number of turns per cm is 10 in the solenoid. The current in the solenoid changes uniformly from 5 A to zero in 10 ms. Calculate the emf induced in the loop.

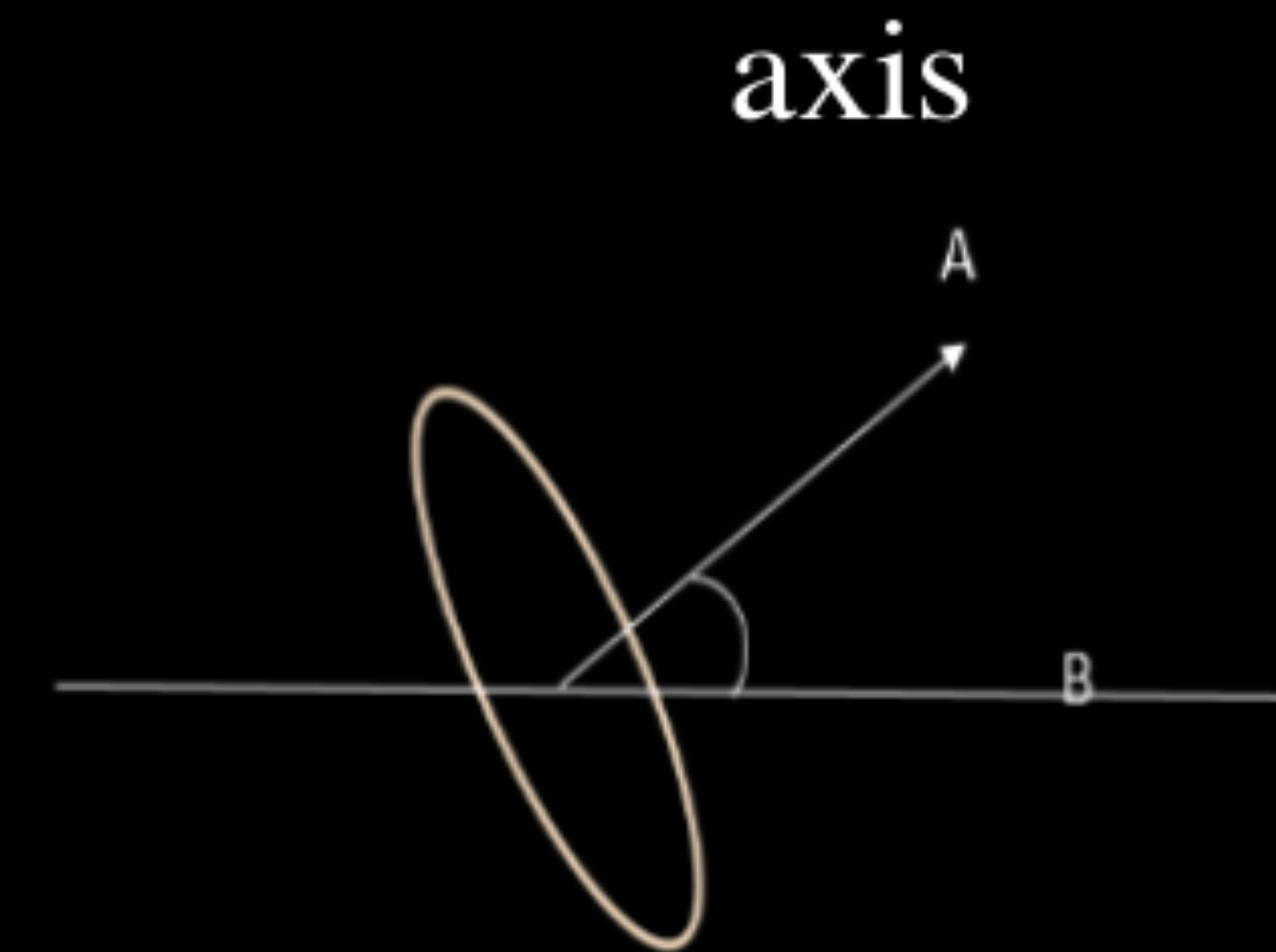
3

E.m.f induced in the circular loop:

$$\begin{aligned}
 |\mathcal{E}| &= \frac{d\phi}{dt} \\
 &= \frac{d}{dt}(BA\cos\theta) \\
 &= \frac{d}{dt}(\mu_0 n I A \cos\theta)
 \end{aligned}$$

$$|\mathcal{E}| = \left(\mu_0 n A \cos\theta \cdot \frac{dI}{dt} \right)$$

$$\begin{aligned}
 |\mathcal{E}| &= 4\pi \times 10^{-7} \times 1000 \times \frac{6}{\pi} \times 10^{-4} \times \cos 60^\circ \times \frac{5}{10^{-2}} \\
 &= 6 \times 10^{-5} \text{ V}
 \end{aligned}$$



$\frac{1}{2}$

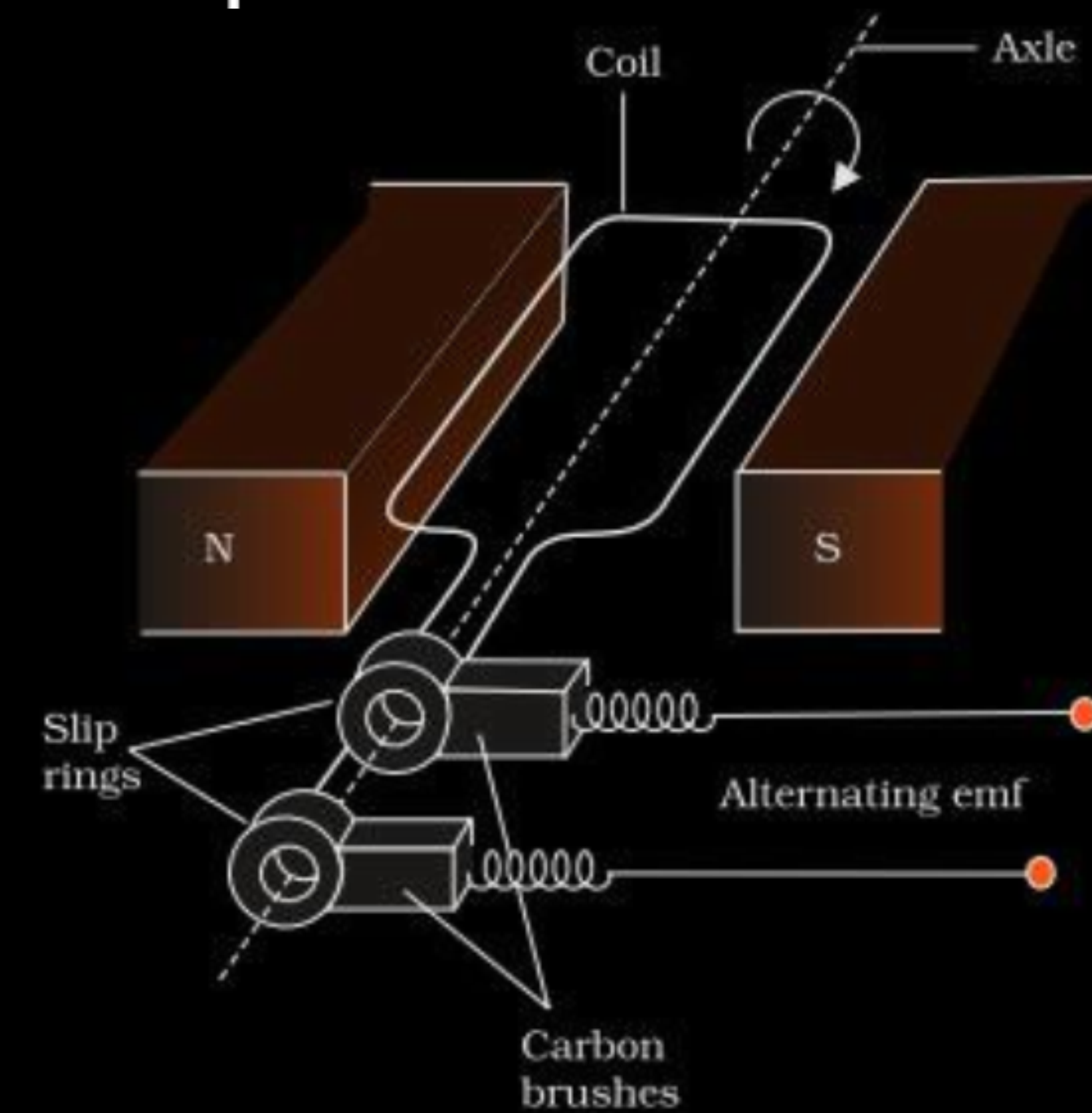
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- (i) With the help of a labelled diagram, explain the working of an ac generator. Obtain the expression for the emf induced at an instant 't'.

- (i) With the help of a labelled diagram, explain the working of an ac generator. Obtain the expression for the emf induced at an instant 't'.



Working of ac generator

When coil is rotated in a uniform magnetic field with a constant angular speed ω , magnetic flux through it changes. As a result, an e.m.f is induced in the coil.

Flux linked with the coil at any instant 't' is

$$\varphi_B = BA \cos \omega t$$

$$\varepsilon = -N \frac{d\varphi_B}{dt}$$

$$\varepsilon = NBA\omega \sin \omega t$$

- (i) State Faraday's law of electromagnetic induction and mention the utility of Lenz's law. Obtain an expression for self-inductance of a coil in terms of its geometry and permeability of the medium.

- (i) The magnitude of induced e.m.f in a circuit is equal to the time rate of change of magnetic flux through the circuit

$\frac{1}{2}$

Utility of Lenz's law

It give polarity of the induced e.m.f .

$\frac{1}{2}$

Expression for self inductance

Consider a long solenoid of cross-sectional area A and length l , having n turns per unit length. If I is the current flowing in the solenoid, magnetic field inside the solenoids is

$$B = \mu_0 n I$$

$\frac{1}{2}$

Total magnetic flux linked with the solenoid is

$$N\phi_B = (nl)(\mu_0 n I)(A)$$

$$N\phi_B = \mu_0 n^2 A l I$$

$\frac{1}{2}$

Self inductance

$$L = \frac{N\phi_B}{I}$$

$\frac{1}{2}$

$$L = \mu_0 n^2 A l$$

If solenoid is filled with a material of relative permeability μ_r , then

$$L = \mu_r \mu_0 n^2 A l$$

$\frac{1}{2}$

6. A bar magnet is thrown from a distance into a closed loop with its north pole towards the loop and passes through it. The induced current in the loop is :
- (A) first clockwise and then anticlockwise
 - (B) first anticlockwise and then clockwise
 - (C) clockwise only
 - (D) anticlockwise only

6. A bar magnet is thrown from a distance into a closed loop with its north pole towards the loop and passes through it. The induced current in the loop is :
- (A) first clockwise and then anticlockwise
 - (B) first anticlockwise and then clockwise
 - (C) clockwise only
 - (D) anticlockwise only

(B) first anticlockwise and then clockwise

23. (a) Define coefficient of self-inductance.
- (b) A solenoid has length 12 cm, area of cross-section 24 cm^2 and 1500 turns. Calculate the self-inductance of the solenoid.

23. (a) Define coefficient of self-inductance.
- (b) A solenoid has length 12 cm, area of cross-section 24 cm^2 and 1500 turns. Calculate the self-inductance of the solenoid.

3

(a) Coefficient of self – induction – It is the flux linked with a coil when the current flowing through the coil is unity. Alternatively- Coefficient of self – induction – It is the self induced emf in a coil when the current in the coil changes at the rate of 1 A/S	1
(b) Coefficient of self induction (L) $L = \frac{N^2 \mu_0 A}{l}$ $L = \frac{(1500)^2 \times 4\pi \times 10^{-7} \times 24 \times 10^{-4}}{12 \times 10^{-2}}$ $L = 18\pi \times 10^{-3} \text{ H}$ $= 5.65 \times 10^{-2} \text{ H}$	$\frac{1}{2}$ $\frac{1}{2}$ 1

24. A coil of area of cross-section 'A' has N number of turns. It is rotating with an angular speed ω inside a uniform magnetic field B. If the total resistance of circuit of coil is R, find :

3

- (a) the maximum current in the circuit.
- (b) the flux through the coil when the current is zero.
- (c) the flux through the coil when the current is maximum.

(a) Induced $Emf(\varepsilon) = N \frac{-d\phi_B}{dt} = -NBA \frac{d}{dt}(\cos \omega t)$

Induced $emf(\varepsilon) = NBA \omega \sin \omega t$

$Current(I) = \frac{\varepsilon}{R} = \frac{NBA \omega \sin \omega t}{R}$

Max current $(I_{\max}) = \frac{NBA \omega}{R}$

(b) Flux through the coil

$\phi_B = NBA \cos \omega t$

When current is zero, $\omega t = 0^\circ$

$\phi_B = NBA$

(c) When current is maximum, $\omega t = 90^\circ$

$\phi_B = 0$

1

1

1

4. The magnetic flux (in SI units) through a coil varies with time as $\phi = 3t^2 + 4t + 7$. The ratio of emf induced in the coil at $t = 2$ s to that at $t = 1$ s will be :

(a) 2

(b) 0.8

(c) 1.6

(d) 4

4. The magnetic flux (in SI units) through a coil varies with time as $\phi = 3t^2 + 4t + 7$. The ratio of emf induced in the coil at $t = 2$ s to that at $t = 1$ s will be :

(a) 2

(b) 0.8

(c) 1.6

(d) 4

(c) 1.6

8. Which of the following statements is *not* correct ?
- (a) Lenz's law is the consequence of the law of conservation of energy.
 - (b) The magnitude of induced emf in a coil is directly proportional to the rate of change of magnetic flux.
 - (c) Inserting an iron core in a coil decreases its self-inductance.
 - (d) An emf can be induced in a straight conductor by moving it perpendicularly through a uniform magnetic field.

8. Which of the following statements is *not* correct ?
- (a) Lenz's law is the consequence of the law of conservation of energy.
 - (b) The magnitude of induced emf in a coil is directly proportional to the rate of change of magnetic flux.
 - (c) Inserting an iron core in a coil decreases its self-inductance.
 - (d) An emf can be induced in a straight conductor by moving it perpendicularly through a uniform magnetic field.

(c) Inserting an iron core in a coil decrease its self inductance

28. Define the term 'self-inductance' of a coil. Obtain the expression for the self-inductance of a long solenoid of length l , cross-sectional area A and having N turns.

- 28.** Define the term ‘self-inductance’ of a coil. Obtain the expression for the self-inductance of a long solenoid of length l , cross-sectional area A and having N turns.

3

It is defined as magnetic flux through a solenoid when unit current passes through the solenoid.

$$L = \frac{\phi}{I}$$

Flux linkage through coil of N turns

$$N\phi_B \propto I \quad \text{_____ (i)}$$

$$N\phi_B = LI \quad \text{_____ (ii)}$$

Induced emf $\varepsilon = -N \frac{d\phi_B}{dt} = -NL \frac{dI}{dt} \quad \text{_____ (iii)}$

$$N\phi_B = N \frac{A\mu_o NI}{l} \quad \text{_____ (iv)}$$

From eq. (ii) and (iv)

$$L = \mu_o \frac{N^2 A}{l} \quad \text{_____ (v)}$$

1

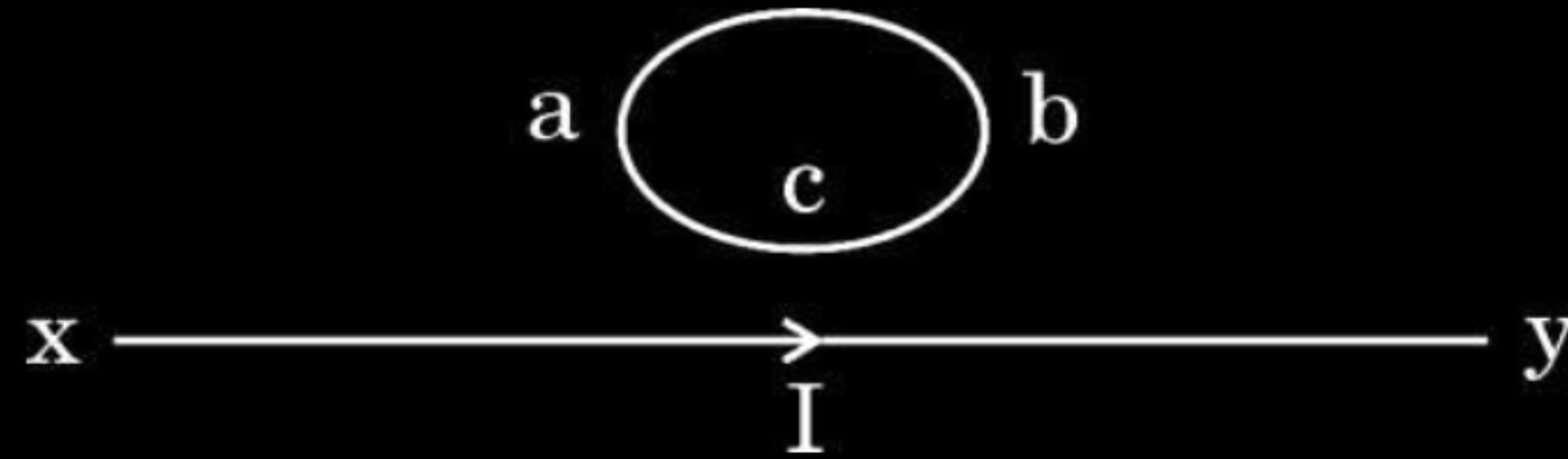
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$\frac{1}{2}$

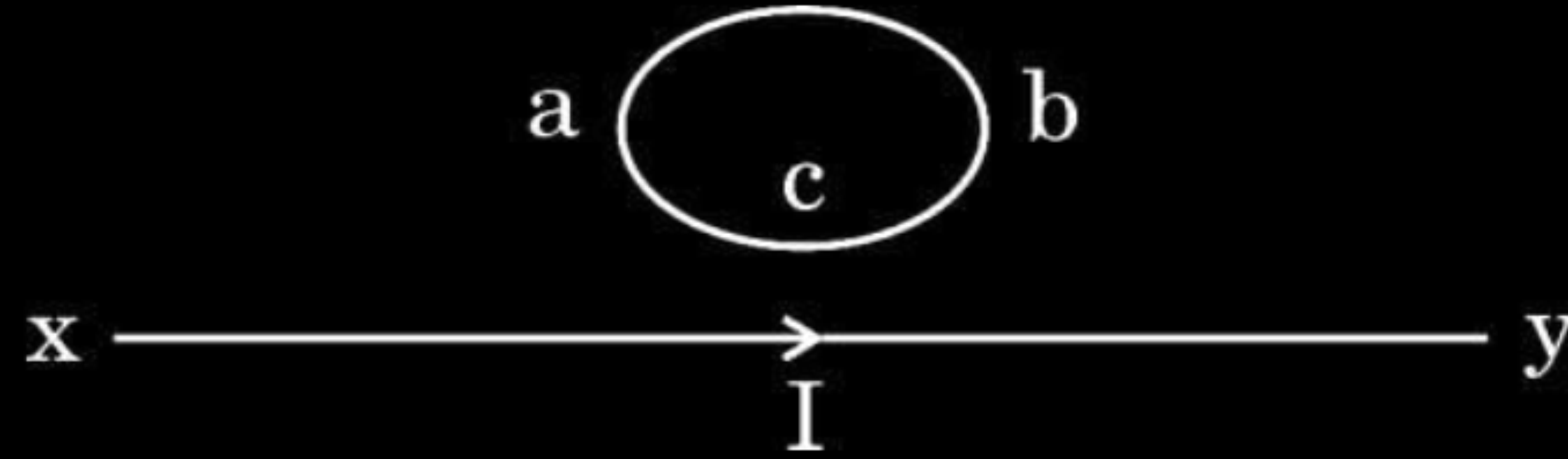
$\frac{1}{2}$

11. The direction of induced current in the loop abc is :



- (a) along abc if I decreases
- (b) along acb if I increases
- (c) along abc if I is constant
- (d) along abc if I increases

11. The direction of induced current in the loop abc is :



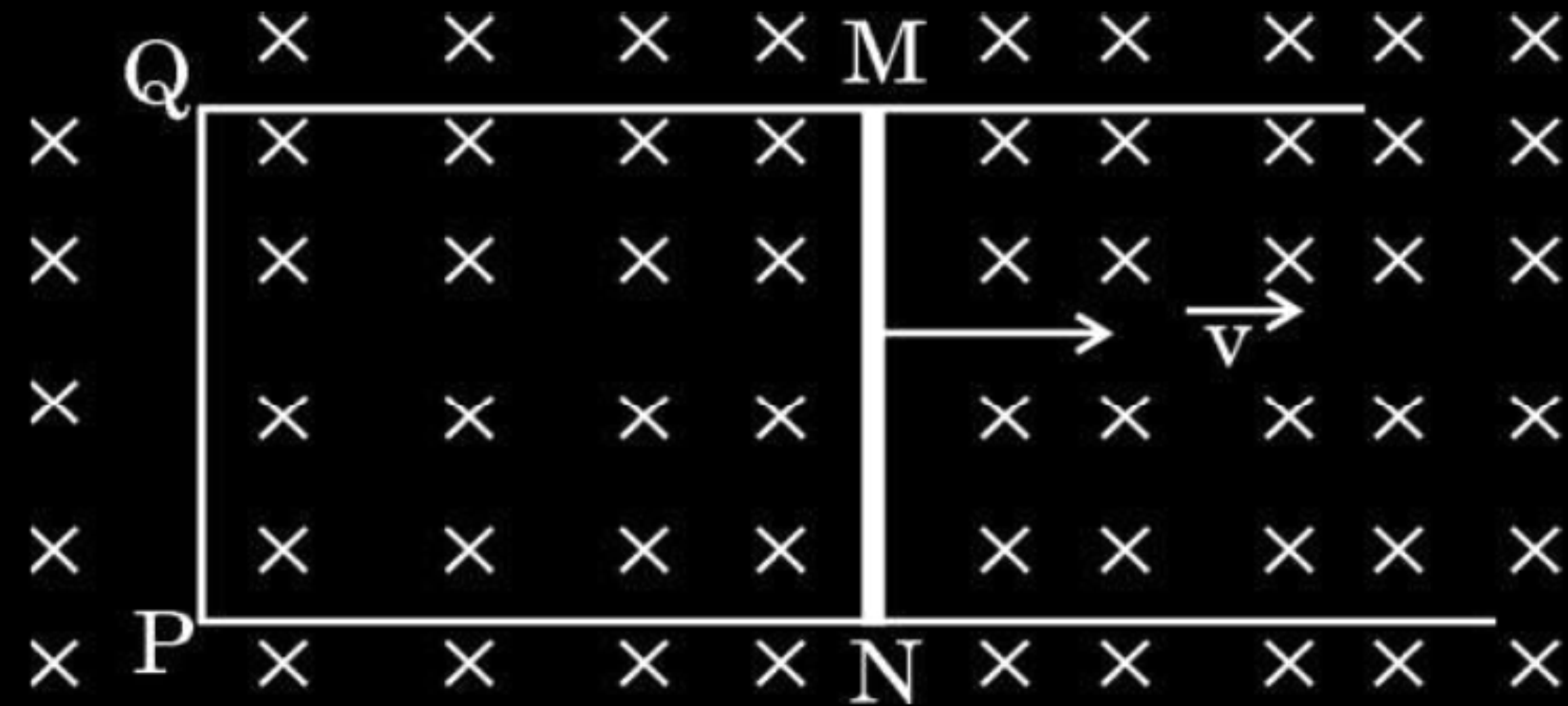
- (a) along abc if I decreases
- (b) along acb if I increases
- (c) along abc if I is constant
- (d) along abc if I increases

(d) along abc if I increases

- 30.** A rectangular conductor MNPQ with a movable arm MN (resistance r) is kept in a uniform magnetic field as shown in the figure. Resistance of arms MQ, QP and PN are negligible. Obtain the expression for the :

3

- (a) current induced in the loop specifying its direction, and
- (b) power required to move the arm.



- 30.** (a) Two coplanar concentric circular loops of radii R and r ($\ll R$) are arranged coaxially. Obtain the expression for their mutual inductance.

- 30.** (a) Two coplanar concentric circular loops of radii R and r ($\ll R$) are arranged coaxially. Obtain the expression for their mutual inductance.

(a) Magnetic field due to current I flowing in the loop with radius R , at its centre.

$$B = \frac{\mu_0 I}{2R}$$

Magnetic flux linked with the smaller coil

$$\phi = \frac{\mu_0 I}{2R} \cdot \pi r^2$$

which is equal to MI

hence
$$M = \frac{\mu_0 \pi r^2}{2R}$$

$\frac{1}{2}$

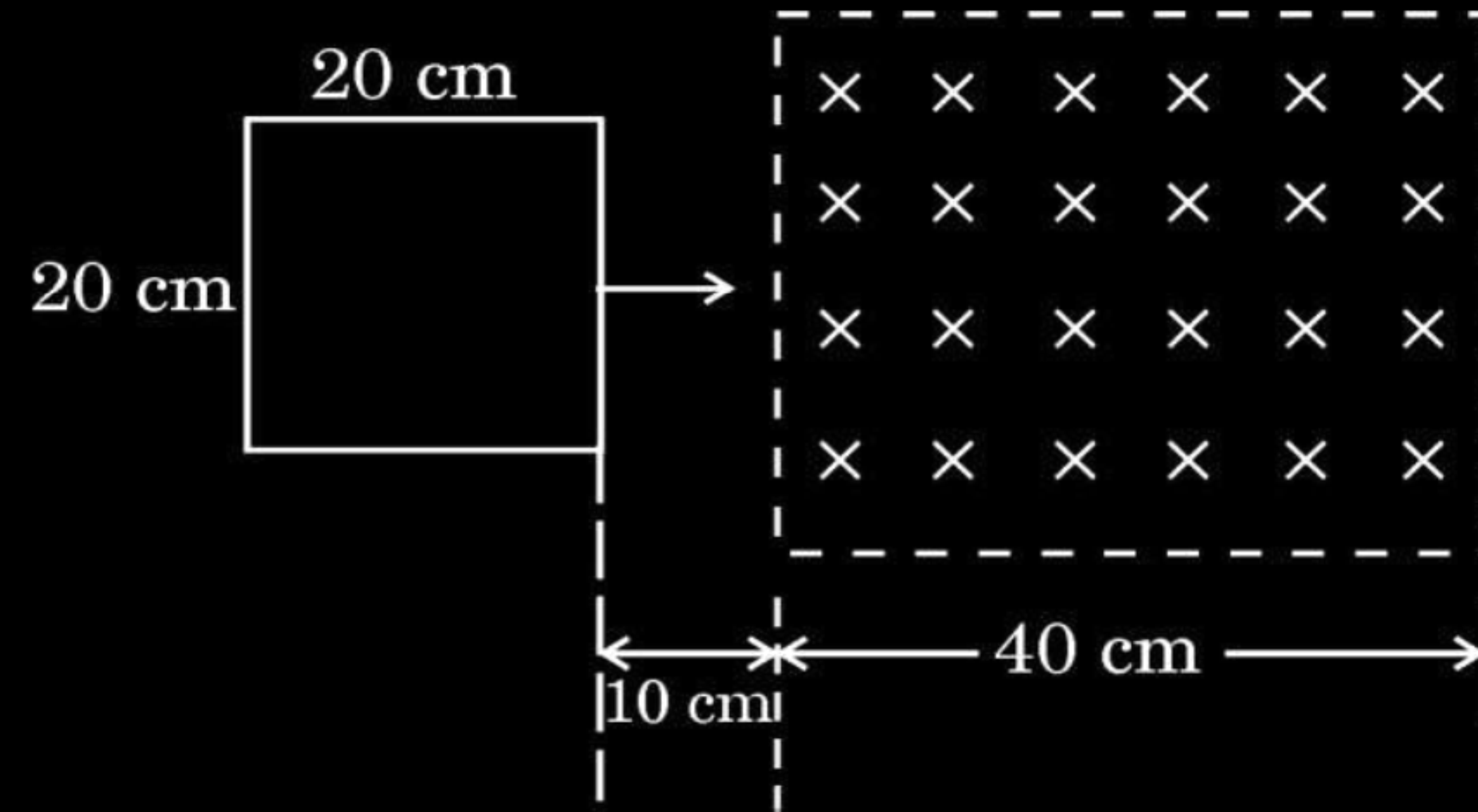
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

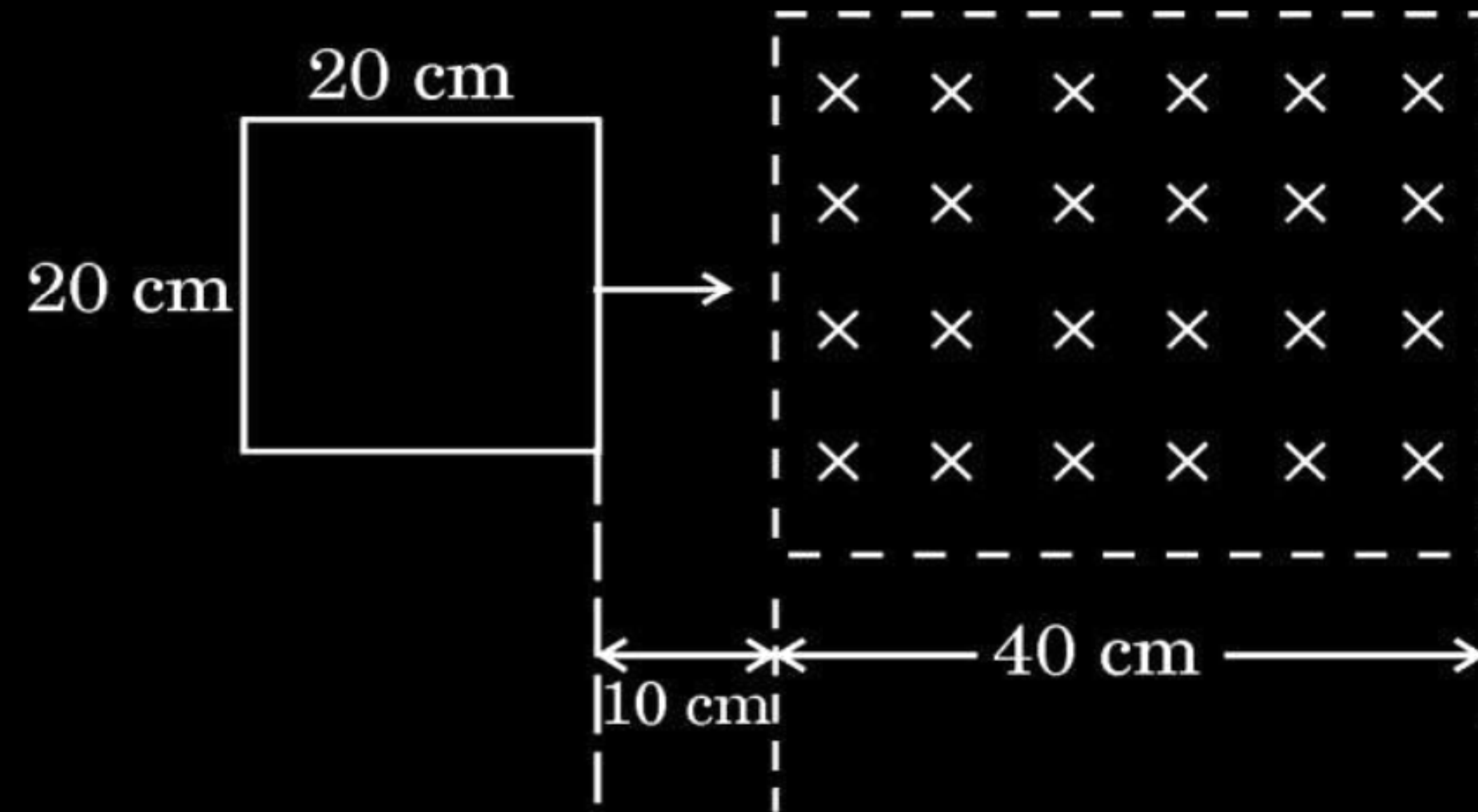
- (b) A square loop of side 20 cm starts moving at $t = 0$ with a velocity of 5 cm/s towards a region of uniform magnetic field as shown in the figure. Specify the time interval(s) during which induced emf is produced in the loop.

3



- (b) A square loop of side 20 cm starts moving at $t = 0$ with a velocity of 5 cm/s towards a region of uniform magnetic field as shown in the figure. Specify the time interval(s) during which induced emf is produced in the loop.

3



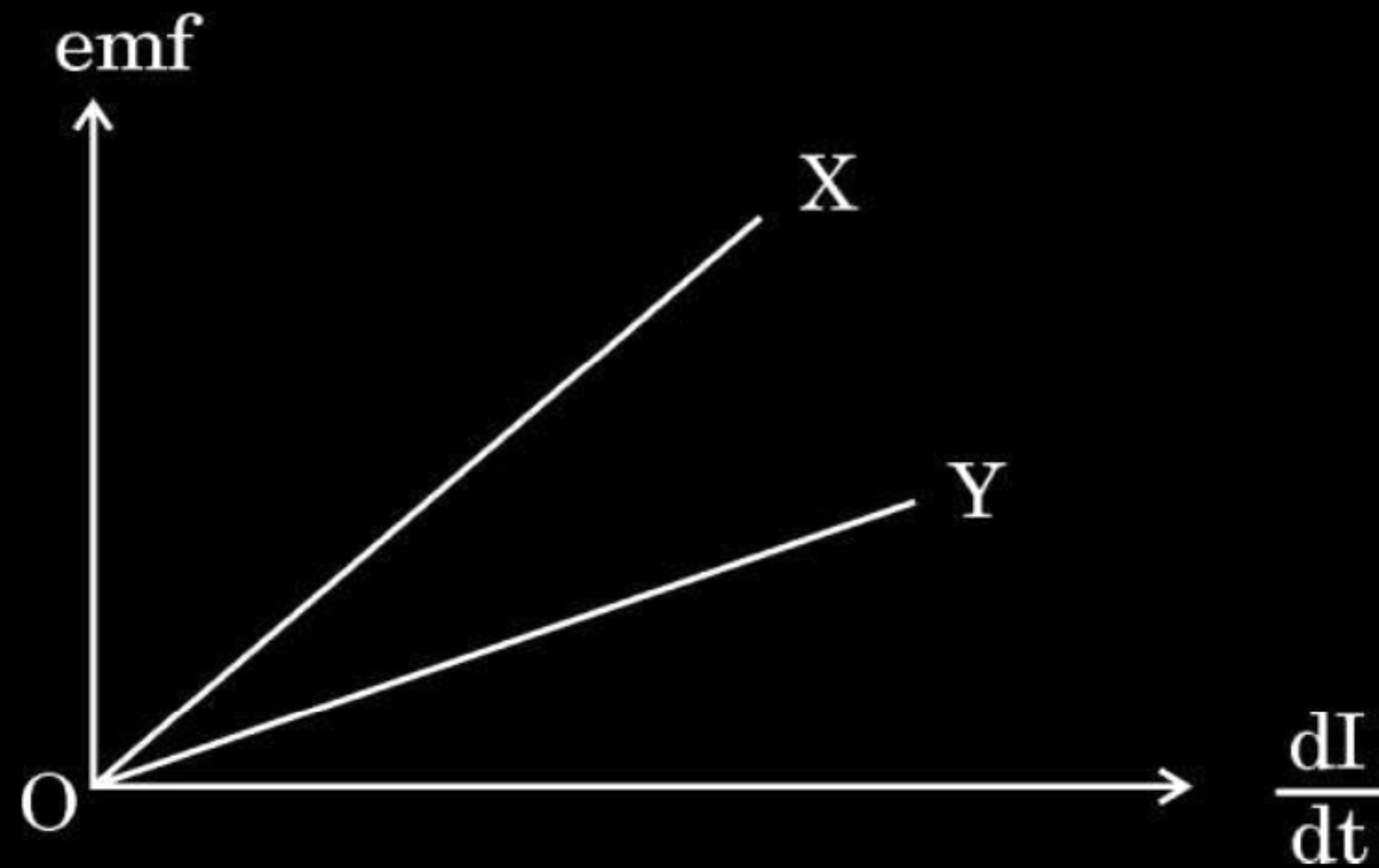
(b) $t = \frac{l}{v}$

Induced e.m.f. during time interval

$2\text{ s} < t \leq 6\text{ s} \quad \& \quad 10\text{ s} \leq t \leq 14\text{ s}$

$\frac{1}{2} + \frac{1}{2}$

- 27.** (a) The figure shows the variation of induced emf as a function of rate of change of current for two identical solenoids X and Y. One is air cored and the other is iron cored. Which one of them is iron cored ? Why ?



- (b) Obtain an expression for self-inductance of a long solenoid of length L and cross-sectional area A having N turns.

(a) X is iron cored.

Reason- From the graph-

$$\text{Slope of the graph} = \frac{\varepsilon}{\left| \frac{dI}{dt} \right|} = L$$

Slope of X is more than that of Y. Hence X is iron cored because Inductance of iron cored coil is more than that of air cored coil.

(b) Magnetic field due to solenoid (axis)

$$B = \frac{\mu_0 N I}{l}$$

Magnetic flux through the solenoid

$$\phi_B = N \phi = \frac{\mu_0 N^2 A}{L} I$$

$$\text{Since self-inductance} = \frac{\phi_B}{I}$$

$$= \frac{\mu_0 N^2 A}{L}$$

1/2

1/2

1/2

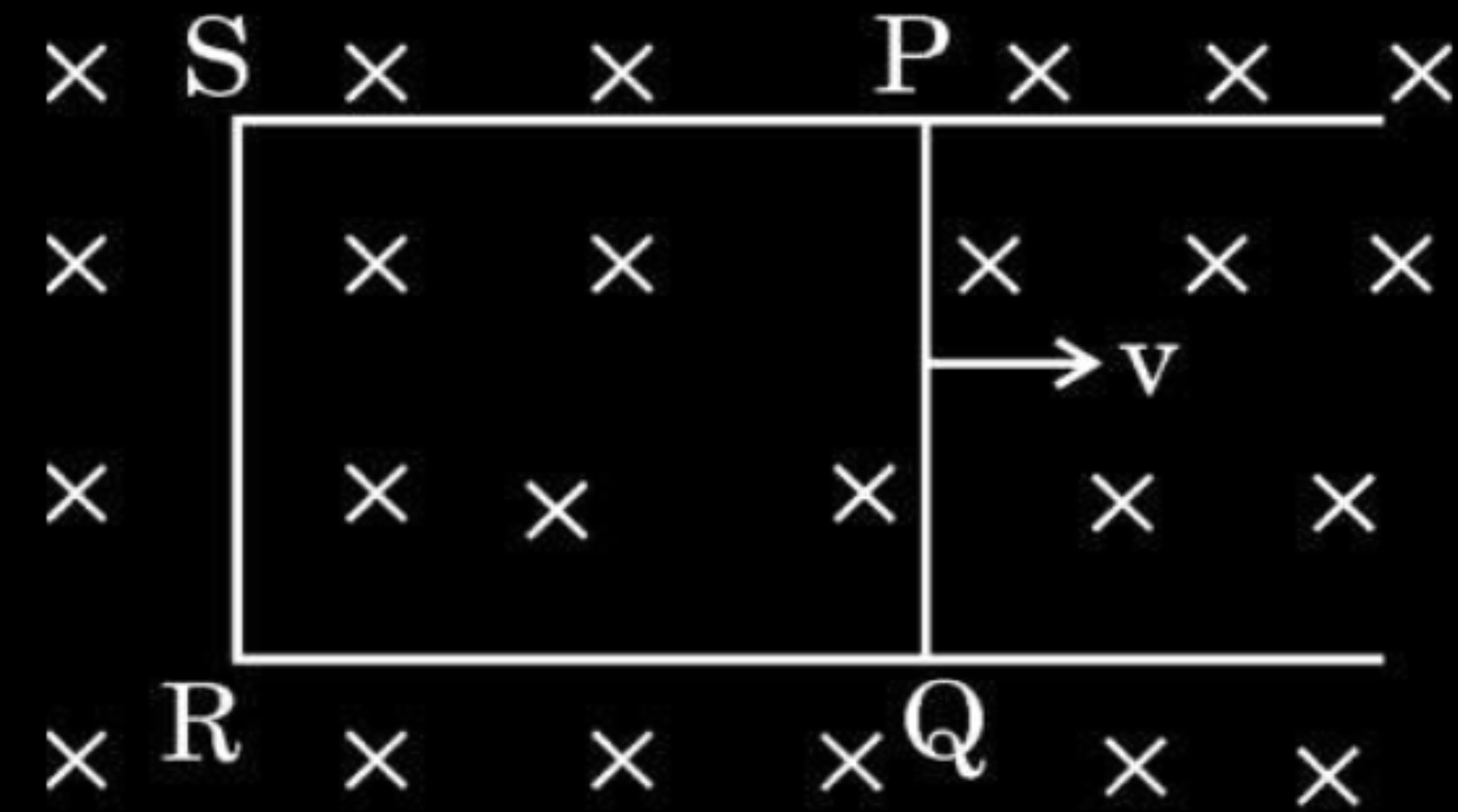
1/2

1/2

1/2

28. The figure shows a rectangular conductor PQRS in which the arm PQ of length 10 cm and resistance $0.4 \, \Omega$ is free to move. It is kept in a uniform magnetic field $B = 0.2 \, \text{T}$ acting perpendicular into the plane of PQRS. If arm PQ is moved with a velocity v of 5 cm/s as shown, find :

3



- the current induced in the loop, and
- the power required to move the arm.

(Resistances of arms PS, SR and RQ are negligible.)

(a)

$$\begin{aligned}i &= \frac{Blv}{R} \\&= \frac{0.2 \times (10 \times 10^{-2}) \times (5 \times 10^{-2})}{0.4} \\i &= 2.5 \times 10^{-3} \text{ A}\end{aligned}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

(b)

$$\begin{aligned}P &= i^2 R \\&= (2.5 \times 10^{-3})^2 \times 0.4 \\P &= 2.5 \times 10^{-6} \text{ W}\end{aligned}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- 29.** A rectangular loop of sides 25 cm and 20 cm is lying in x-y plane. It is subjected to a magnetic field $\vec{B} = (5t^2 + 2t + 10)\hat{k}$, where B is in Tesla and t is in seconds. If the resistance of the loop is $4\ \Omega$, find the emf induced and the induced current in the loop at $t = 5\text{ s}$.

$\phi_B = \vec{B} \cdot \vec{A} = BA \cos 0^\circ$ $= (5t^2 + 2t + 10) \times (25 \times 10^{-2} \times 20 \times 10^{-2})$ $ \mathcal{E} = \left -\frac{d\phi_B}{dt} \right $	1/2
$= 5 \times 10^{-2} \times (10t + 2)$	1/2
At t = 5 sec;	
$\mathcal{E} = (5 \times 10^{-2}) \times (52)$	1/2
$\mathcal{E} = 2.6 \text{ V}$	1/2
$I = \frac{\mathcal{E}}{R}$	1/2
$= \frac{2.6}{4}$	
$I = 0.65 \text{ A}$	1/2

- 27.** What is meant by the term ‘mutual inductance’ of a pair of coils ? Obtain an expression for the mutual inductance of two long coaxial solenoids, each of length l but having different number of turns N_1 and N_2 and radii r_1 and r_2 ($r_2 > r_1$).

Mutual Inductance-It is the magnetic flux in the secondary coil due to the flow of unit current in the primary coil.

Alternatively:

It is the emf induced in the secondary coil when rate of change of current in the primary coil is unity.

Expression for Mutual Inductance

Let N_1 and N_2 be the total number of turns of coils S_1 & S_2 , respectively.

When a current I_2 is set up through S_2 , it in turn sets up a magnetic flux through $S_1 \Rightarrow N_1 \phi_1 = M_{12} I_2$ -----(1)

M_{12} is the mutual inductance of solenoid S_1 w.r.t. S_2 .

Magnetic field due to I_2 in $S_2 = \mu_o \left(\frac{N_2}{l} \right) I_2$ -----(2)

$N_1 \phi_1 = (N_1)(\pi r_1^2) \left(\mu_o \frac{N_2}{l} I_2 \right)$ -----(2)

From equation (1) & equation (2)

$$M_{12} = \frac{\mu_o N_1 N_2 \pi r_1^2}{l}$$

1. Two identical circular coaxial coils A and B, arranged in vertical planes parallel to each other, carry currents in the same direction. If the distance between the coils is decreased at a constant rate, the current :
- (a) increases in A and decreases in B.
 - (b) decreases in both A and B.
 - (c) increases in both A and B.
 - (d) remains same in both A and B.

1. Two identical circular coaxial coils A and B, arranged in vertical planes parallel to each other, carry currents in the same direction. If the distance between the coils is decreased at a constant rate, the current :
- (a) increases in A and decreases in B.
 - (b) decreases in both A and B.
 - (c) increases in both A and B.
 - (d) remains same in both A and B.

(b) Decreases in both A and B.

- 9.** A circular coil of radius 8.0 cm and 40 turns is rotated about its vertical diameter with an angular speed of $\frac{25}{\pi}$ rad s⁻¹ in a uniform horizontal magnetic field of magnitude 3.0×10^{-2} T. The maximum emf induced in the coil is :

(a) 0.12 V

(b) 0.15 V

(c) 0.19 V

(d) 0.22 V

- 9.** A circular coil of radius 8.0 cm and 40 turns is rotated about its vertical diameter with an angular speed of $\frac{25}{\pi}$ rad s⁻¹ in a uniform horizontal magnetic field of magnitude 3.0×10^{-2} T. The maximum emf induced in the coil is :

(a) 0.12 V

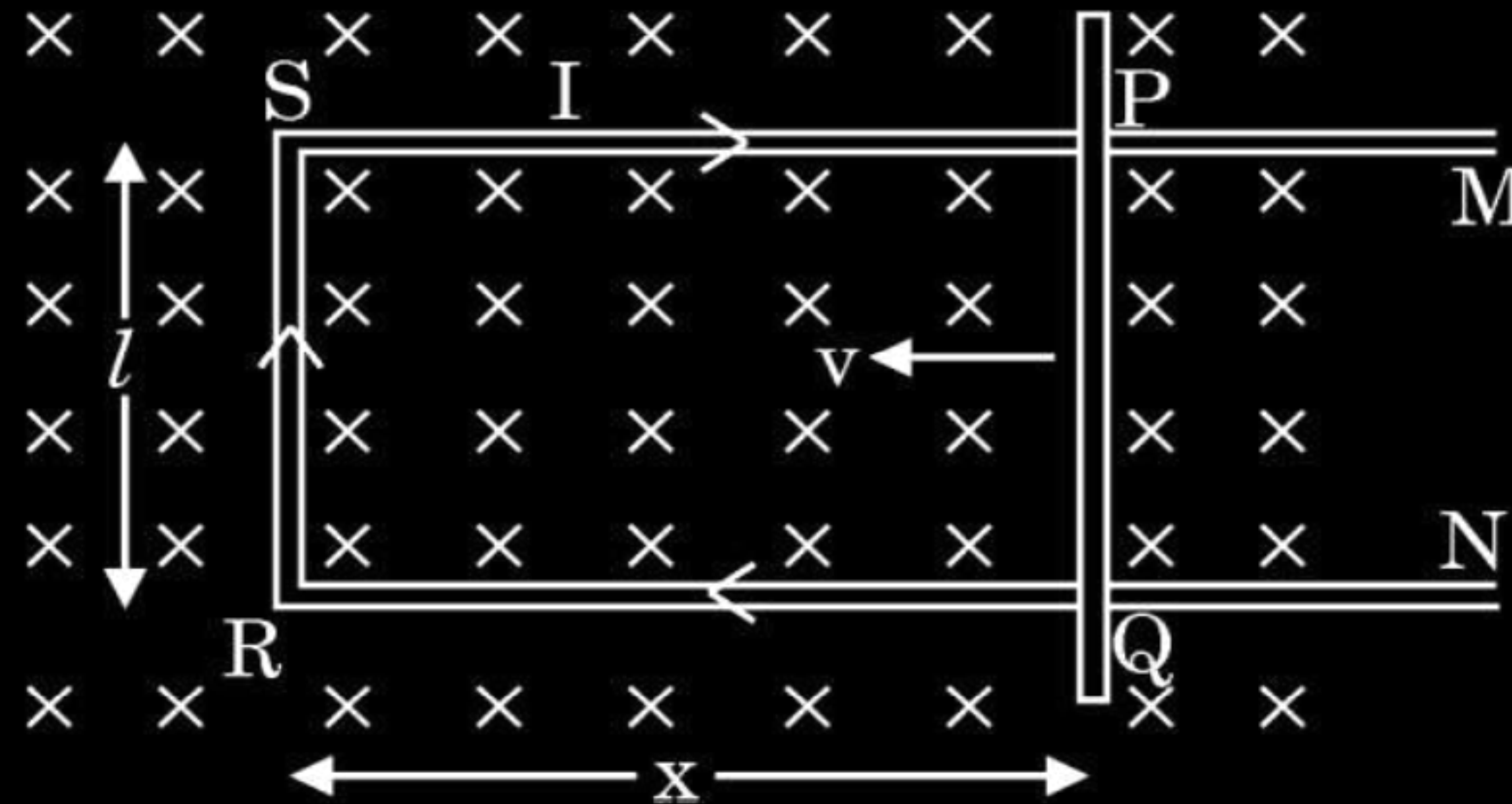
(b) 0.15 V

(c) 0.19 V

(d) 0.22 V

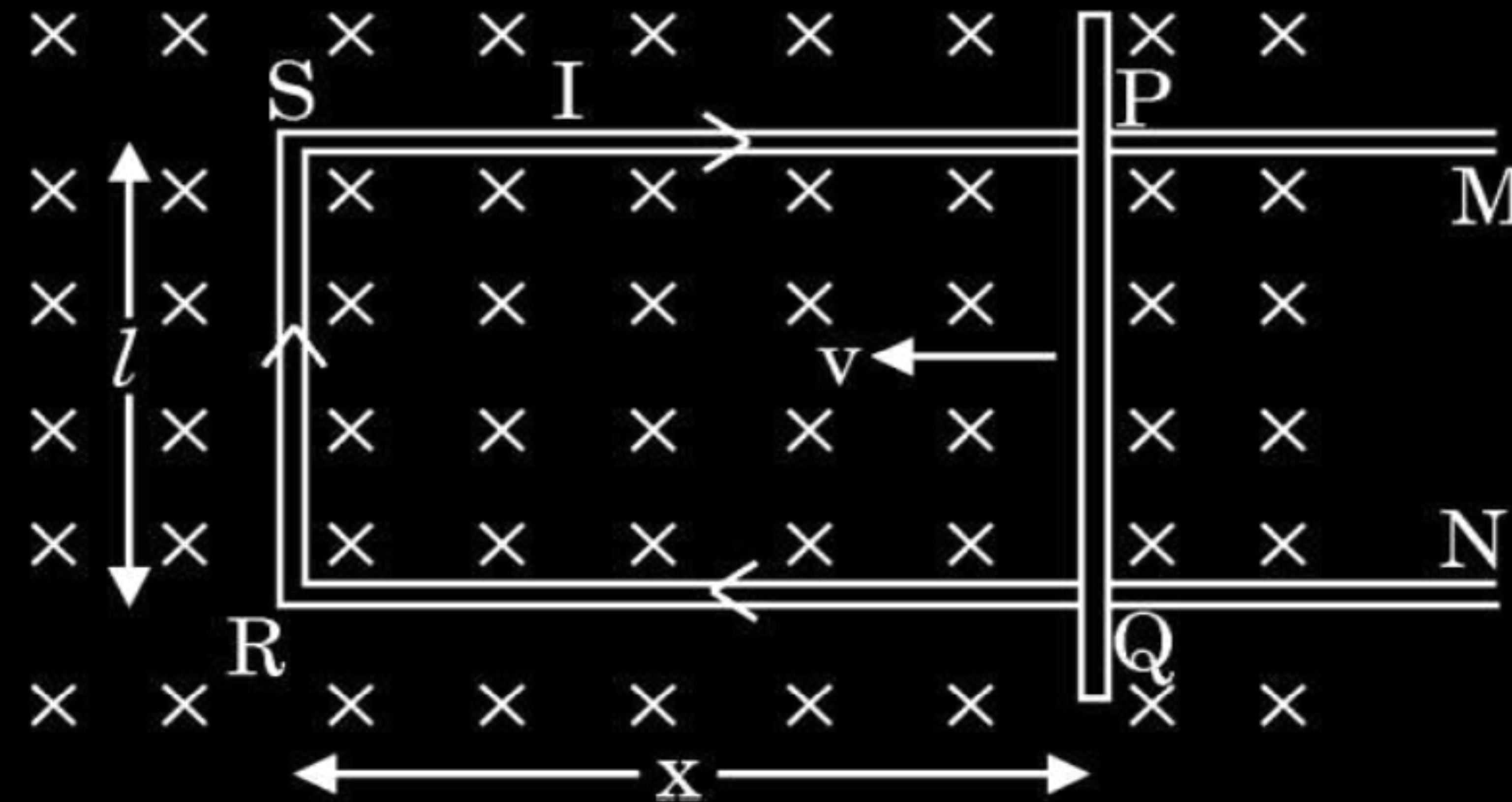
(c) 0.19 V

7. Figure shows a rectangular conductor PSRQ in which movable arm PQ has a resistance 'r' and resistance of PSRQ is negligible. The magnitude of emf induced when PQ is moved with a velocity $\vec{v}$ does **not** depend on :



- | | |
|----------------------------------|----------------------------|
| (a) magnetic field ($\vec{B}$) | (b) velocity ($\vec{v}$) |
| (c) resistance (r) | (d) length of PQ |

7. Figure shows a rectangular conductor PSRQ in which movable arm PQ has a resistance 'r' and resistance of PSRQ is negligible. The magnitude of emf induced when PQ is moved with a velocity $\vec{v}$ does **not** depend on :



- | | |
|----------------------------------|----------------------------|
| (a) magnetic field ($\vec{B}$) | (b) velocity ($\vec{v}$) |
| (c) resistance (r) | (d) length of PQ |

(c) Resistance (r)

- 28.** A long solenoid of radius r consists of n turns per unit length. A current $I = I_0 \sin \omega t$ flows in the solenoid. A coil of N turns is wound tightly around it near its centre. What is :
- (a) the induced emf in the coil ?
 - (b) the mutual inductance between the solenoid and the coil ?

a) magnetic field produced in the solenoid near the center	
$B = \mu_0 n I$	$\frac{1}{2}$
Flux linked with the coil wound over solenoid	
$\phi = NBA = N \pi r^2 B$	
$= N \pi r^2 \mu_0 n I$	$\frac{1}{2}$
Induced emf $e = \frac{-d\phi}{dt} = -\pi r^2 N n \mu_0 \frac{dI}{dt}$	$\frac{1}{2}$
$= -\mu_0 \pi r^2 n N I_0 \omega \cos \omega t$	$\frac{1}{2}$
b) comparing Eq (i) with $e = -M \frac{dI}{dt}$	$\frac{1}{2}$
$M = \mu_0 \pi r^2 n N$	$\frac{1}{2}$

31. (a) (i) Define coefficient of self-induction. Obtain an expression for self-inductance of a long solenoid of length l , area of cross-section A having N turns.
- (ii) Calculate the self-inductance of a coil using the following data obtained when an AC source of frequency $\left(\frac{200}{\pi}\right)$ Hz and a DC source is applied across the coil.

AC Source		
S.No.	V (Volts)	I (A)
1	3.0	0.5
2	6.0	1.0
3	9.0	1.5

DC Source		
S.No.	V (Volts)	I (A)
1	4.0	1.0
2	6.0	1.5
3	8.0	2.0

(i) Coefficient of self induction is defined as the amount of magnetic flux associated with a coil when unit current flows through it. Alternatively It is defined as the magnitude of emf induced in a coil when current changes at the rate of 1 A/s through it.	1
(ii) The magnetic field due to a current I flowing in solenoid is $B = \frac{\mu_0 NI}{l}$ The total magnetic flux linked with solenoid $N\phi_B = (N) \left(\frac{\mu_0 NI}{l} \right) (A)$ $= \frac{\mu_0 N^2 IA}{l}$ The self inductance is $L = \frac{N\phi_B}{I}$ $L = \frac{\mu_0 N^2 A}{l}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

(iii) From the table, $Z=6\ \Omega$, $R=4\Omega$

$$Z^2 = R^2 + X_L^2$$

$$X_L^2 = Z^2 - R^2 = 36 - 16 = 20$$

$$X_L = 2\sqrt{5} \approx 4.5\ \Omega$$

$$2\pi\nu L = 4.5$$

$$L = \frac{4.5}{2 \times \pi \times \frac{200}{\pi}}$$

$$L = 1.1 \times 10^{-2}\ H = 11\text{mH}$$

Note : Please do not deduct marks if a student writes answer as

$$0.5\sqrt{5} \times 10^{-2}\ \text{H}$$

1/2

1/2

1/2

1/2

- (b) (i) With the help of a labelled diagram, describe the principle and working of an ac generator. Hence, obtain an expression for the instantaneous value of the emf generated.
- (ii) The coil of an ac generator consists of 100 turns of wire, each of area 0.5 m^2 . The resistance of the wire is 100Ω . The coil is rotating in a magnetic field of 0.8 T perpendicular to its axis of rotation, at a constant angular speed of $60 \text{ radian per second}$. Calculate the maximum emf generated and power dissipated in the coil.

Principle – It is based on the principle of electromagnetic induction.

Whenever there is a change in magnetic flux linked with a coil, an emf is induced in the coil.

Alternatively

Give full credit if a candidate writes, it is based on the principle of electromagnetic induction.

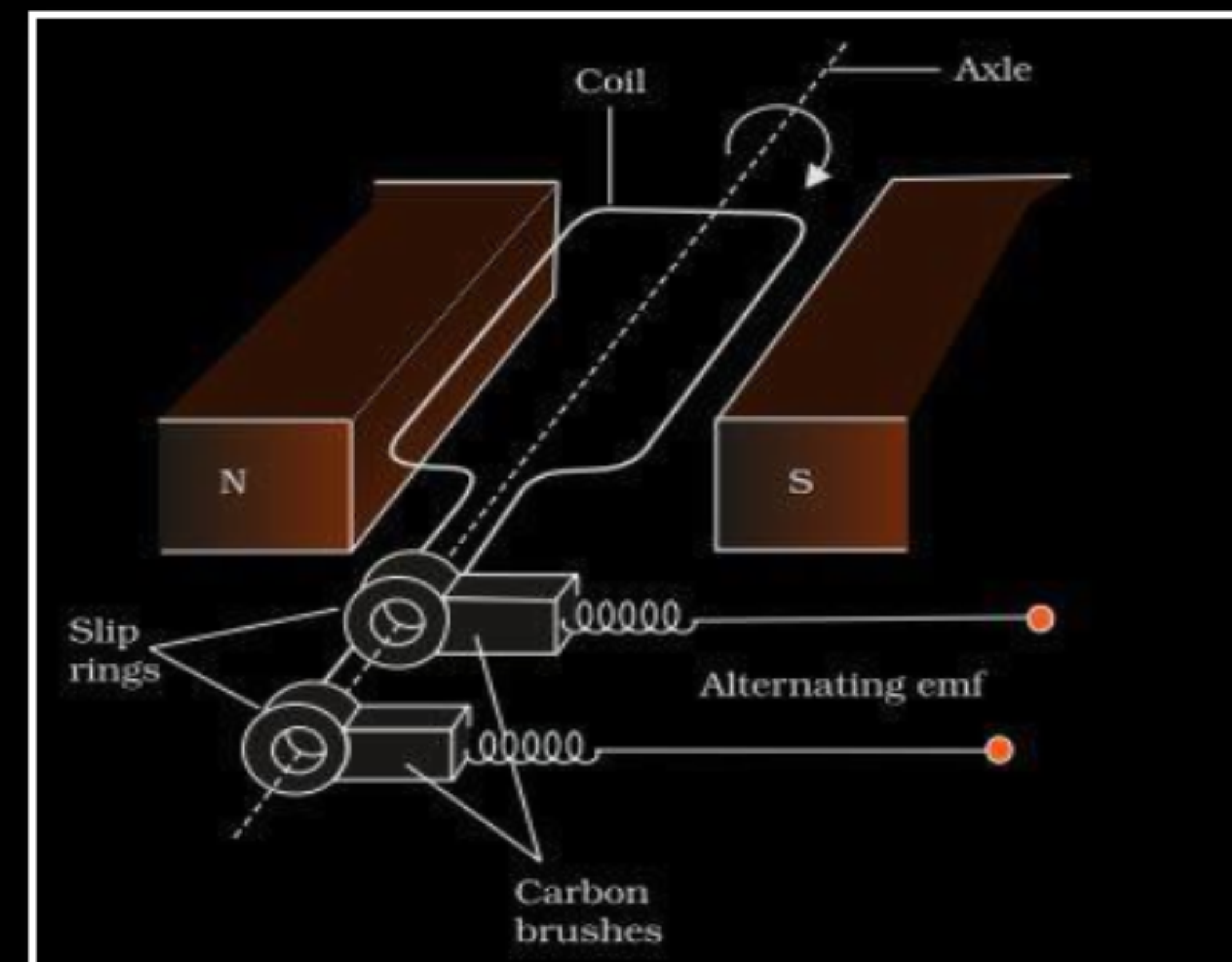
Working - When a rectangular coil is rotated in a magnetic field, the magnetic flux changes continuously which induces an emf and the direction of current changes periodically.

$$\begin{aligned}\varepsilon &= \frac{-Nd\phi}{dt} \\ &= -NBA \frac{d}{dt}(\cos \omega t)\end{aligned}$$

$$\varepsilon = NBA\omega \sin \omega t$$

$$\begin{aligned}\text{(ii)} \quad \varepsilon_0 &= NBA\omega \\ &= 100 \times 0.8 \times 0.5 \times 60 \\ &= 2400 \text{ V}\end{aligned}$$

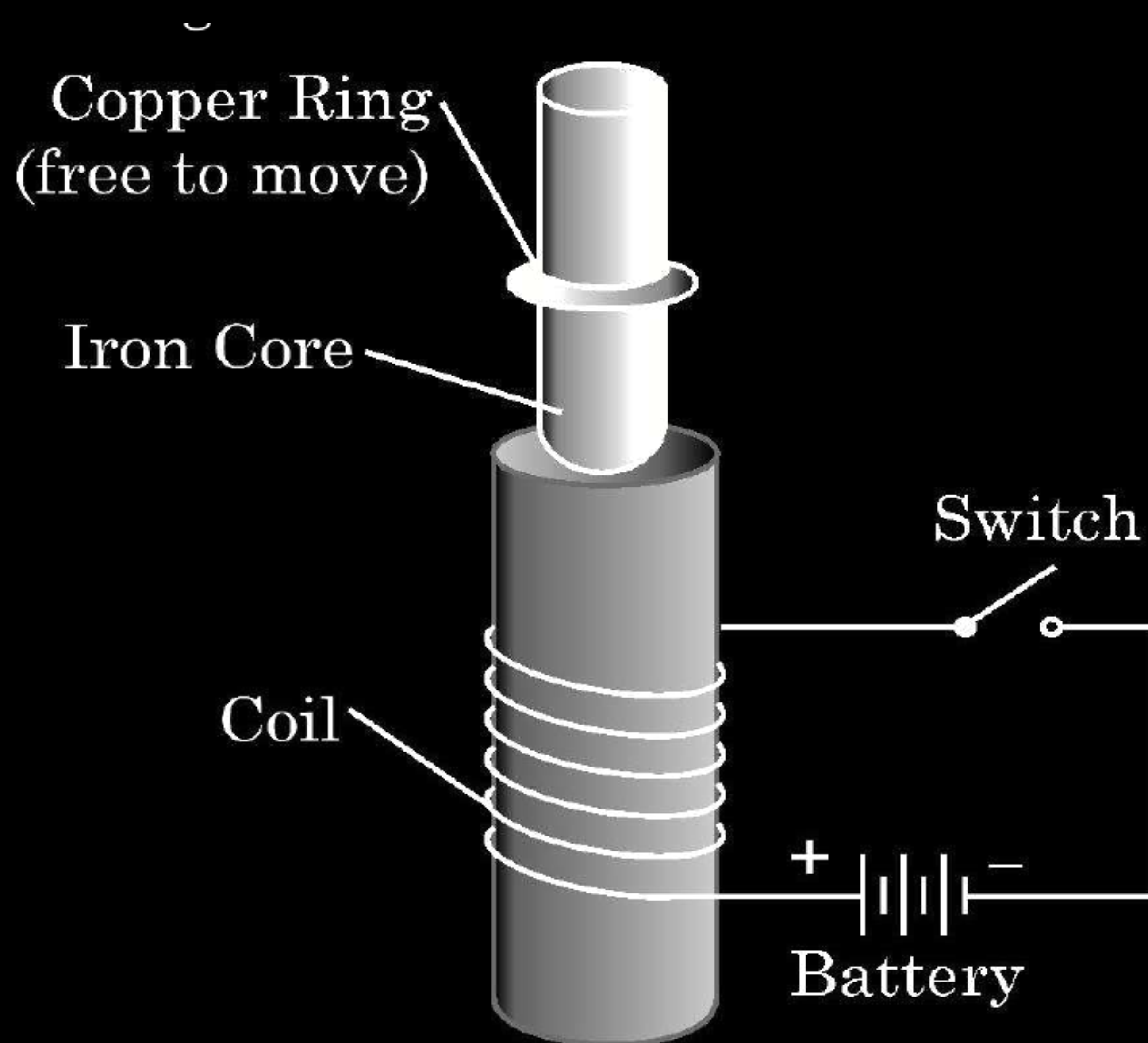
$$\begin{aligned}\text{Power dissipated, } P &= \frac{\varepsilon_{rms}^2}{R} \\ &= \frac{\left(\frac{2400}{\sqrt{2}}\right)^2}{100} \\ &= 28.8 \text{ kW}\end{aligned}$$



34. (a) Consider the experimental set up shown in the figure. This jumping ring experiment is an outstanding demonstration of some simple laws of Physics. A conducting non-magnetic ring is placed over the vertical core of a solenoid. When current is passed through the solenoid, the ring is thrown off.

Answer the following questions :

- (i) Explain the reason of jumping of the ring when the switch is closed in the circuit.
- (ii) What will happen if the terminals of the battery are reversed and the switch is closed ? Explain.
- (iii) Explain the two laws that help us understand this phenomenon.



(i) The direction of induced current in the ring is such that the polarity developed in the ring is same as that of the polarity on the face of the coil , hence it will jump up due to repulsive force.	1
(ii) The polarity of the induced current in the ring will get reversed on changing the terminals of the battery, so the ring will jump again.	1
(iii) <u>Lenz's law</u> It states that the polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produces it.	
<u>Faraday's law of EMI</u>	1
Whenever there is change in magnetic flux through a coil, an emf is induced. The magnitude of the induced emf in a coil is equal to the time rate of change of magnetic flux through the coil.	1

4. A square shaped coil of side 10 cm, having 100 turns is placed perpendicular to a magnetic field which is increasing at 1 T/s. The induced emf in the coil is

(A) 0.1 V

(B) 0.5 V

(C) 0.75 V

(D) 1.0 V

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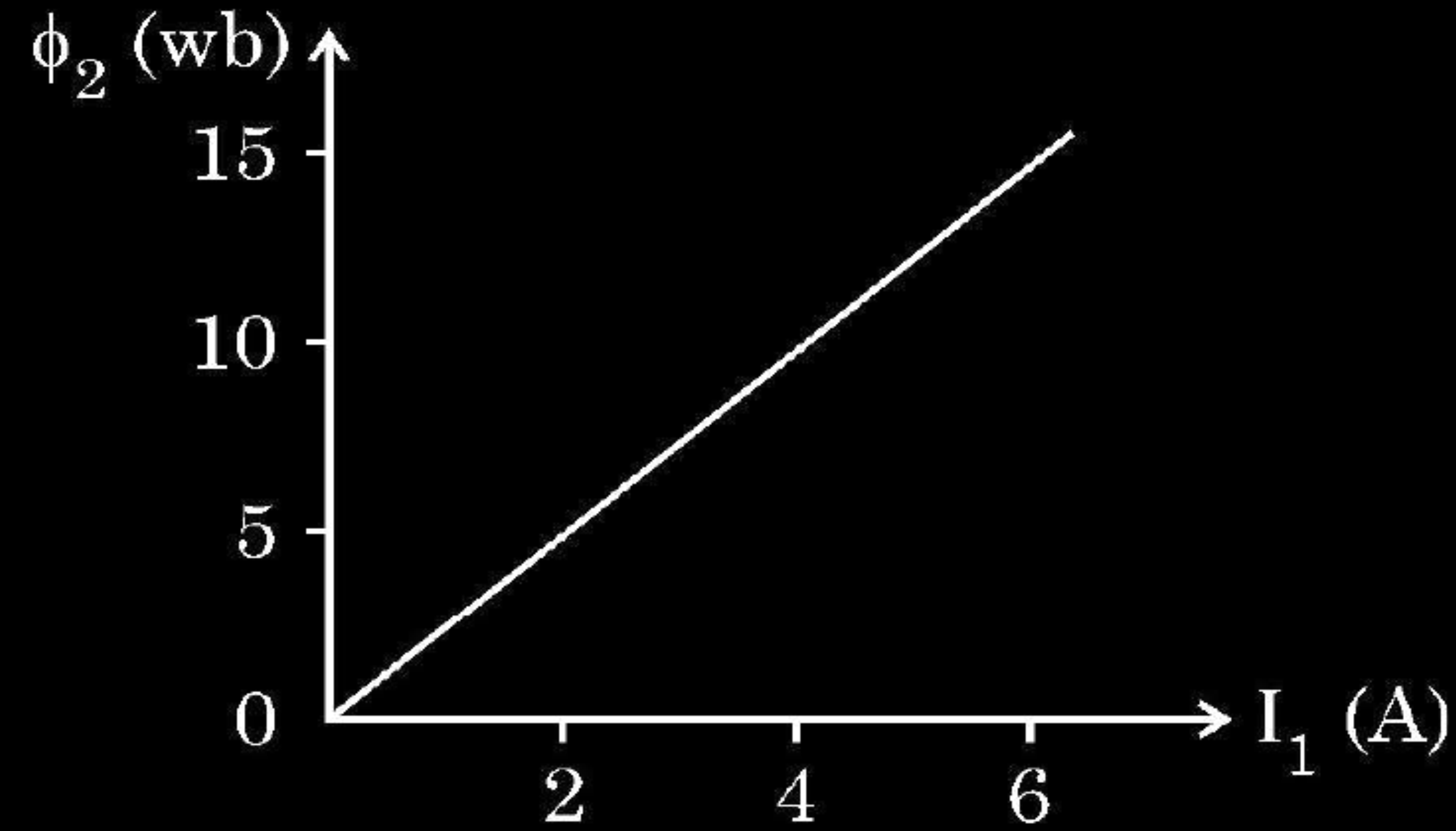
(C) 0.75 V

(D) 1.0 V

(D) 1.0 V

25. Two coils C_1 and C_2 are placed close to each other. The magnetic flux ϕ_2 linked with the coil C_2 varies with the current I_1 flowing in coil C_1 , as shown in the figure. Find

2



- (i) the mutual inductance of the arrangement, and
- (ii) the rate of change of current $\left(\frac{dI_1}{dt}\right)$ that will induce an emf of 100 V in coil C_2 .

(i) $\phi_2 = MI_1$ Slope of the graph , $M = \frac{\phi_2}{I_1}$ $M = 2.5 \text{ H}$	$\frac{1}{2}$
(ii) $\epsilon_2 = M \frac{dI_1}{dt}$ $\frac{dI_1}{dt} = \frac{ \epsilon_2 }{M} = \frac{100}{2.5} = 40 \text{ As}^{-1}$	$\frac{1}{2}$

28. State the basic principle behind the working of an ac generator. Briefly describe its working and obtain the expression for the instantaneous value of emf induced.

28. State the basic principle behind the working of an ac generator. Briefly describe its working and obtain the expression for the instantaneous value of emf induced.

3

Principle: Whenever magnetic flux linked through a coil changes an emf is induced.	1
Alternatively: ac generator is based on the phenomenon of electromagnetic induction.	
Working: The coil (armature) is mechanically rotated in the uniform magnetic field by some external means. The rotation of the coil causes the magnetic flux through it to change. So an emf is induced in the coil.	1
Expression for induced emf	
Flux linked with the coil at any instant of time is-	
$\phi_B = BA \cos \omega t$	
$\epsilon = -N \frac{d\phi_B}{dt}$	$\frac{1}{2}$
$\epsilon = -NBA \frac{d}{dt}(\cos \omega t)$	
Thus, instantaneous value of emf is	
$\epsilon = NBA\omega \sin \omega t$	$\frac{1}{2}$

- 19.** A closely wound coil having 100 turns carries a current of 2 A. If the magnetic flux linked with coil is $\pi^2 \times 10^{-5}$ Wb, find the radius of the coil.

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2

$\phi = BNA$	$\frac{1}{2}$
$B = \frac{\mu_o IN}{2r}$	$\frac{1}{2}$
$\phi = N \times \frac{\mu_o NI}{2r} \times \pi r^2$	
$\Rightarrow \pi^2 \times 10^{-5} = \frac{4\pi \times 10^{-7} \times (100)^2 \times 2 \times r^2}{2r}$	$\frac{1}{2}$
$r = 0.25 \text{ cm}$	$\frac{1}{2}$

28. State Lenz's law. Show that it is consistent with the principle of conservation of energy.

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3

Lenz's Law	
The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it.	1
The e.m.f. induced opposes the causes of its change. So in order to change the magnetic flux work is always needed to be done. This work done gets converted to induced e.m.f. Hence this is consistent with conservation of energy.	2

13. A planar loop is rotated in a magnetic field about an axis perpendicular to the field. The polarity of induced emf changes once in each :

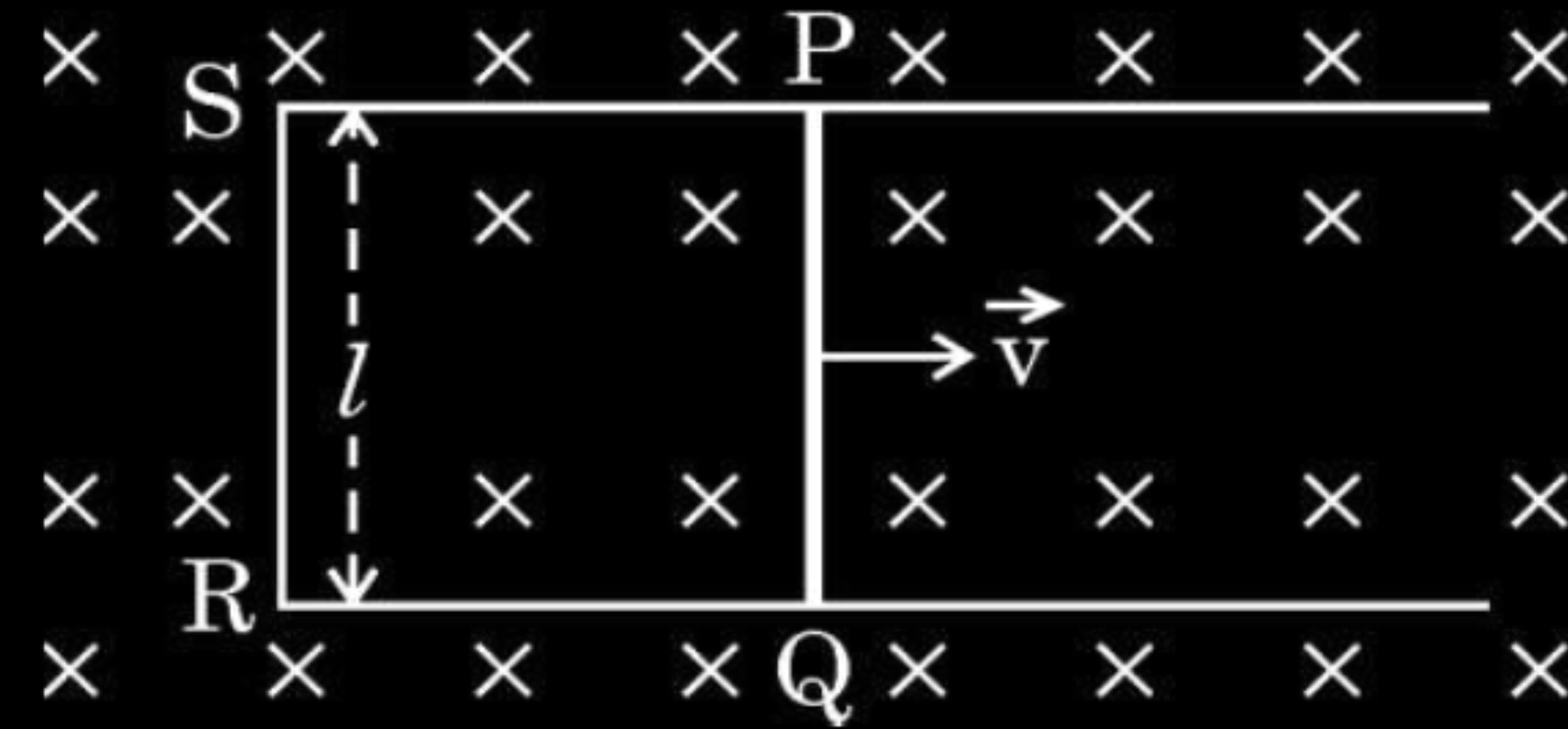
- | | |
|---|---|
| (a) 1 revolution | (b) $\left(\frac{1}{2}\right)$ revolution |
| (c) $\left(\frac{1}{4}\right)$ revolution | (d) $\left(\frac{3}{4}\right)$ revolution |

13. A planar loop is rotated in a magnetic field about an axis perpendicular to the field. The polarity of induced emf changes once in each :

- | | |
|---|---|
| (a) 1 revolution | (b) $\left(\frac{1}{2}\right)$ revolution |
| (c) $\left(\frac{1}{4}\right)$ revolution | (d) $\left(\frac{3}{4}\right)$ revolution |

(b) $\left(\frac{1}{2}\right)$ revolution

- 26.** The figure shows a rectangular conductor PQRS with a movable arm PQ, kept in a uniform magnetic field $\vec{B}$ pointing into the page.



- (i) PQ is moved towards the right with a velocity $\vec{v}$. Obtain the expression for the emf developed across PQ.
- (ii) If r is the resistance of PQRS, find the force required to move PQ with constant velocity $\vec{v}$.

(i)	$\phi_B = B l x$	$\frac{1}{2}$
	$\mathcal{E} = - \frac{d\phi_B}{dt}$	$\frac{1}{2}$
	$= - \frac{d}{dt} (B l x)$	
	$= - B l \frac{dx}{dt}$	
	$\mathcal{E} = - B l v$	$\frac{1}{2}$
(ii)	$i = \frac{\mathcal{E}}{r}$	$\frac{1}{2}$
	$\vec{F} = i (\vec{l} \times \vec{B})$	$\frac{1}{2}$
	$F = \frac{\mathcal{E}}{r} i B = \frac{B l v}{r} (l B)$	
	$F = \frac{B^2 l^2 v}{r}$	$\frac{1}{2}$

30. Answer the following, giving reasons :

3

- (a) Induced emf does not always produce induced current.
- (b) The motion of a copper plate is damped when it is allowed to oscillate between pole pieces of a strong magnet.
- (c) No power is consumed in an ac circuit containing a pure inductor.

30. Answer the following, giving reasons :

3

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- (b) The motion of a copper plate is damped when it is allowed to oscillate between pole pieces of a strong magnet.
- (c) No power is consumed in an ac circuit containing a pure inductor.

- | | |
|--|---|
| (a) Even in the open loop e.m.f. can be induced but since circuit is open no current can flow. | 1 |
| (b) When copper plate oscillates between poles of a strong magnet eddy currents are generated in the plate opposing the change in flux. | 1 |
| (c) Power consumed by a pure inductor oscillates. For first half of the cycle it is positive and for the next half it is negative. So for a complete cycle it is zero. | 1 |

26. A horizontal straight metallic rod of length 4 m is held at some height above the surface of Earth, in east-west direction. If it is allowed to fall from rest, find the :

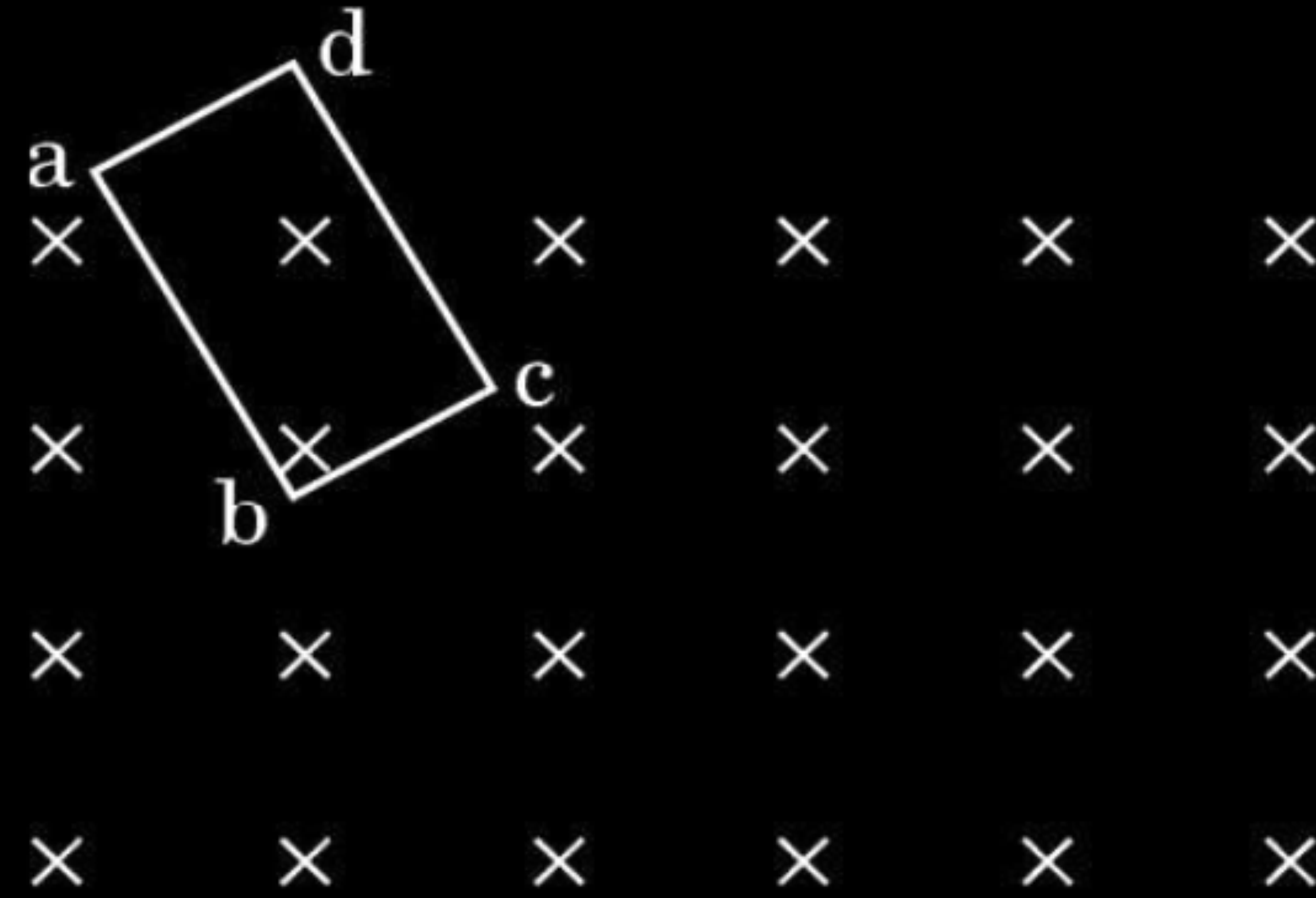
- (a) emf induced in the rod 2 s after it starts falling,
- (b) polarity of the emf induced, and
- (c) the end of the rod which is at the higher potential.

The horizontal component of the Earth's magnetic field at the place is $0.3 \times 10^{-4} \text{ Wb/m}^2$ and take $g = 10 \text{ m/s}^2$.

(a) $\mathcal{E} = Blv$ $B = 0.3 \times 10^{-4} \text{ Wbm}^{-2}$ $v = 20 \text{ ms}^{-1}$ $l = 4 \text{ m}$ $v = 0 + 10 \times 2$ $v = 20 \text{ m / s}$ $\mathcal{E} = 0.3 \times 10^{-4} \times 20 \times 4$ $= 24 \times 10^{-4} \text{ V} = 2.4 \text{ mV}$ (b) West to east (c) East end of the rod is at higher potential.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
--	--

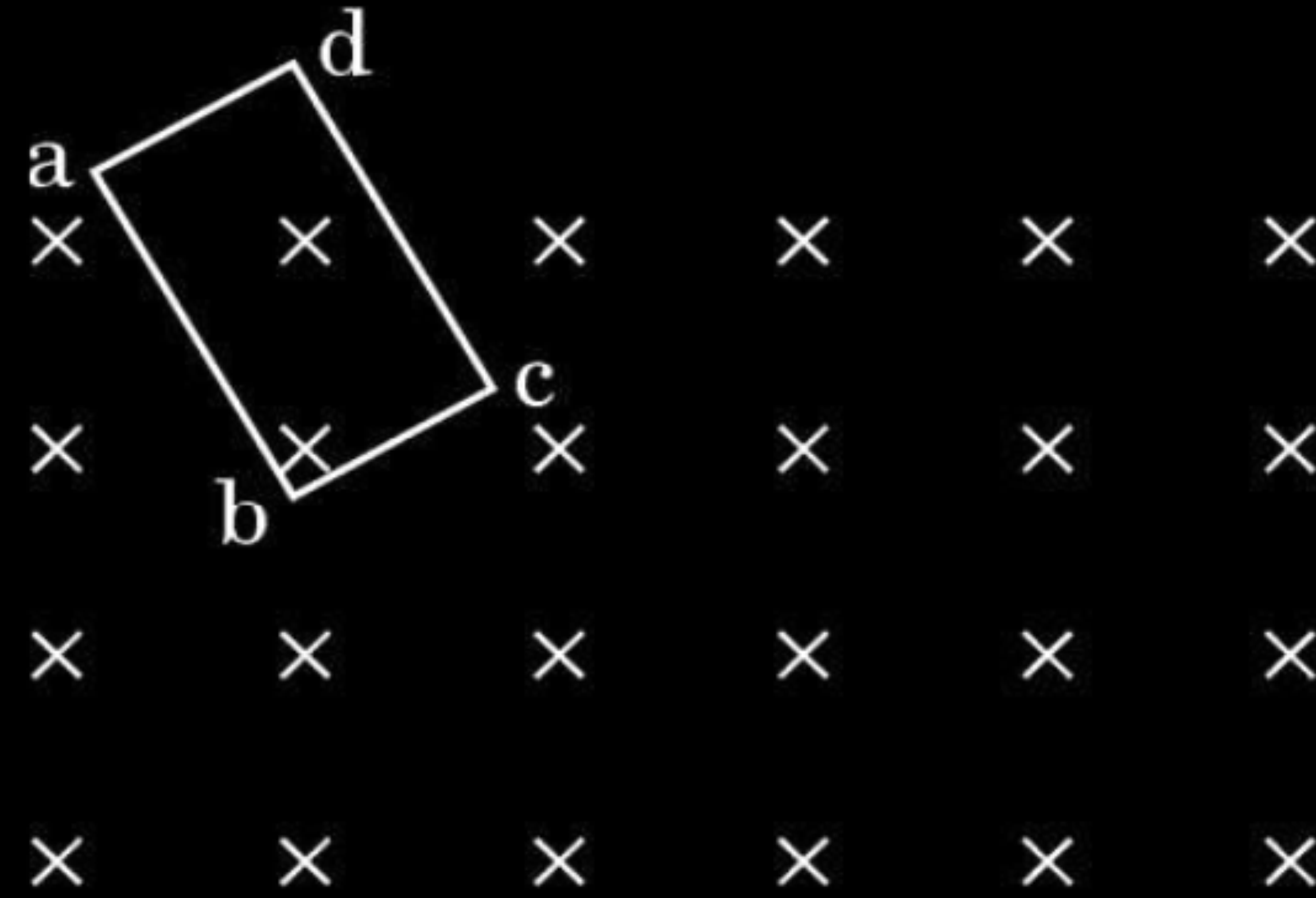
- 26.** (a) State Lenz's law. Determine the direction of the induced current when a rectangular conducting loop abcd in xy-plane is moved into a region of magnetic field which is directed along z-axis.

3



- 26.** (a) State Lenz's law. Determine the direction of the induced current when a rectangular conducting loop abcd in xy-plane is moved into a region of magnetic field which is directed along z-axis.

3



(a) Lenz's law “ the polarity of induced e.m.f is such that it tends to produce a current which opposes the change in magnetic flux that produced it”



Direction of induced current is abcd/anticlockwise

2

1

- (b) Two identical circular loops, one of copper and the other of aluminium are rotated about their diameters with the same angular speed in a magnetic field directed perpendicular to their axes of rotation. Compare (i) the emf induced, and (ii) the current in the two loops. Justify your answers.

- (b) Two identical circular loops, one of copper and the other of aluminium are rotated about their diameters with the same angular speed in a magnetic field directed perpendicular to their axes of rotation. Compare (i) the emf induced, and (ii) the current in the two loops. Justify your answers.

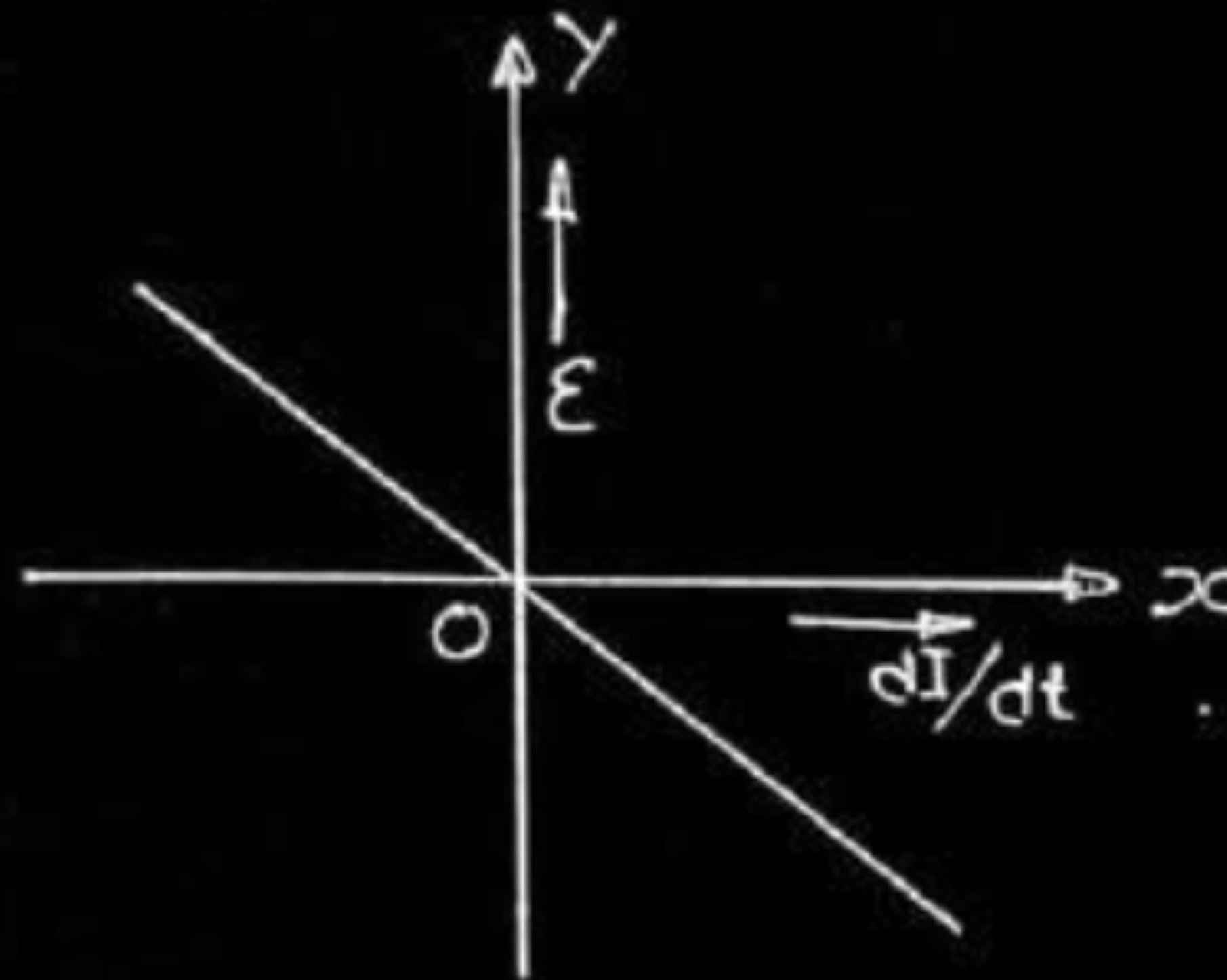
3

(b) (i) Same e.m.f induced in both cases because they are identical and moving with same angular speed in same magnetic field.	$1\frac{1}{2}$
(ii) Induced current is more in copper loops, as its resistance is lesser than that of aluminum.	$1\frac{1}{2}$

19. Plot a graph showing variation of induced e.m.f. with the rate of change of current flowing through a coil.

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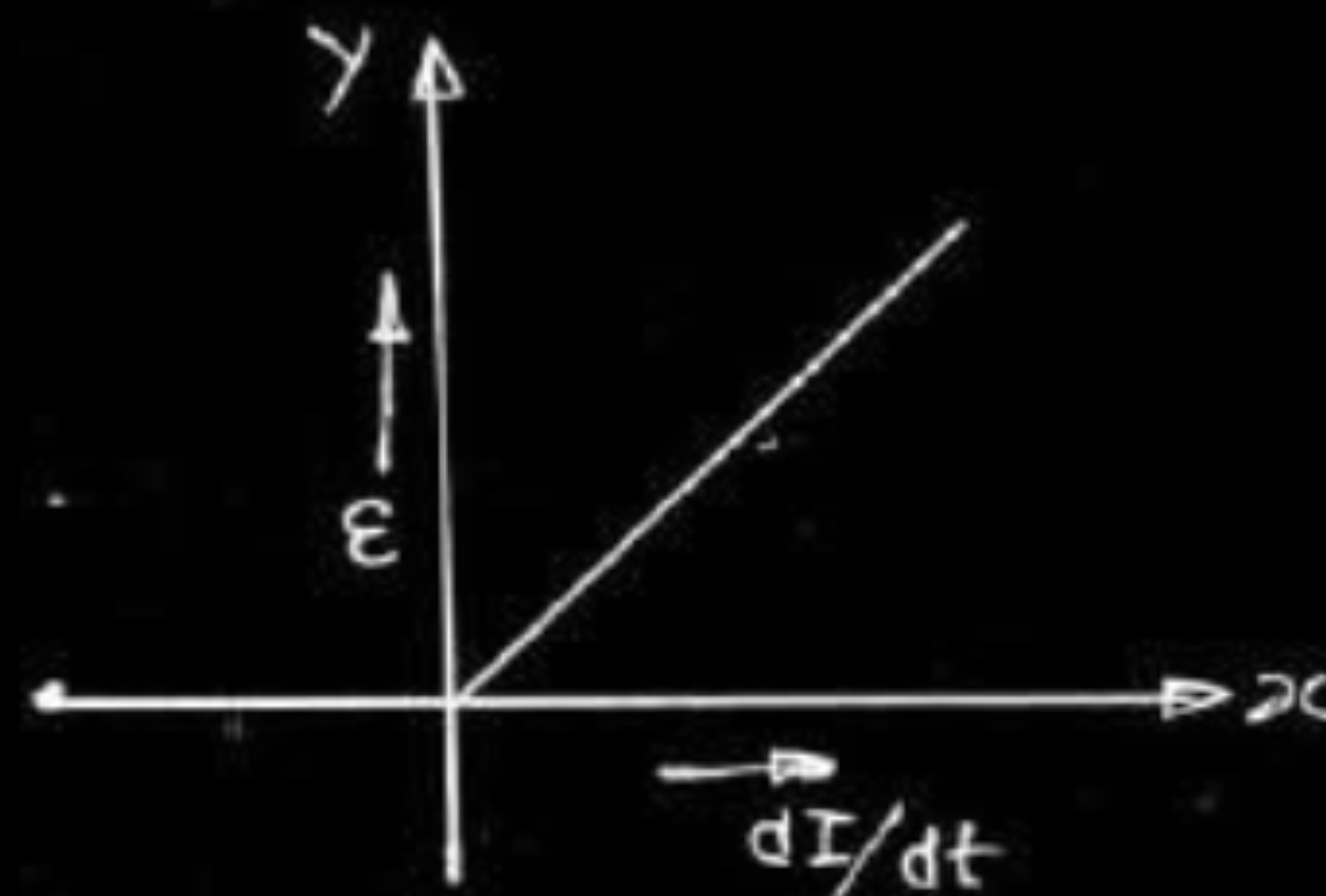
1



Induced e.m.f. in a coil, $\varepsilon = -L \frac{dI}{dt}$

[Award one full mark even if the student just draws the graph without writing the expression of induced emf]

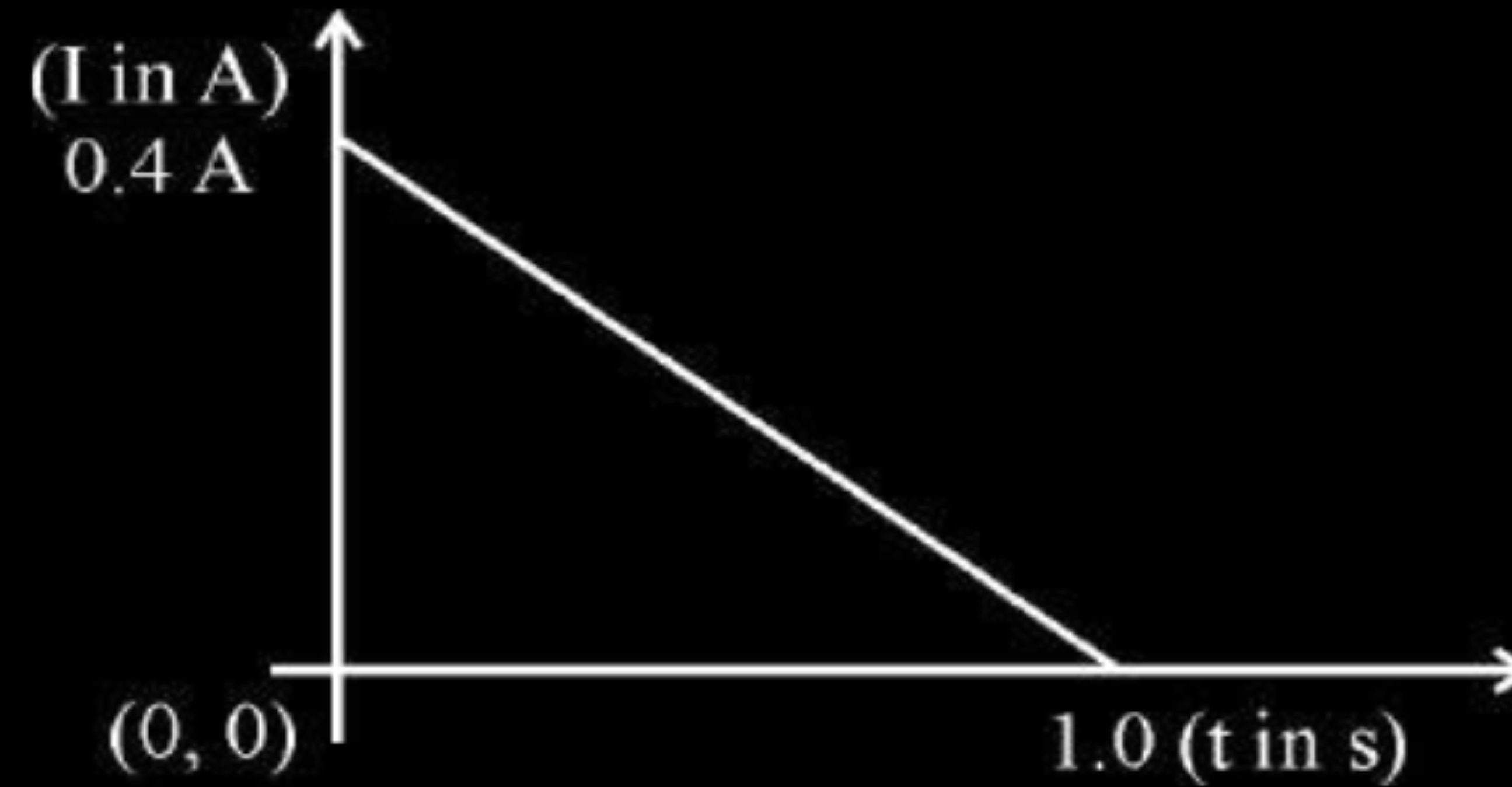
(Note: Award this one mark if a student draws the graph in first quadrant as shown below.)



1/2

1/2

34. When a conducting loop of resistance $10\ \Omega$ and area $10\ \text{cm}^2$ is removed from an external magnetic field acting normally, the variation of induced current in the loop with time is shown in the figure.



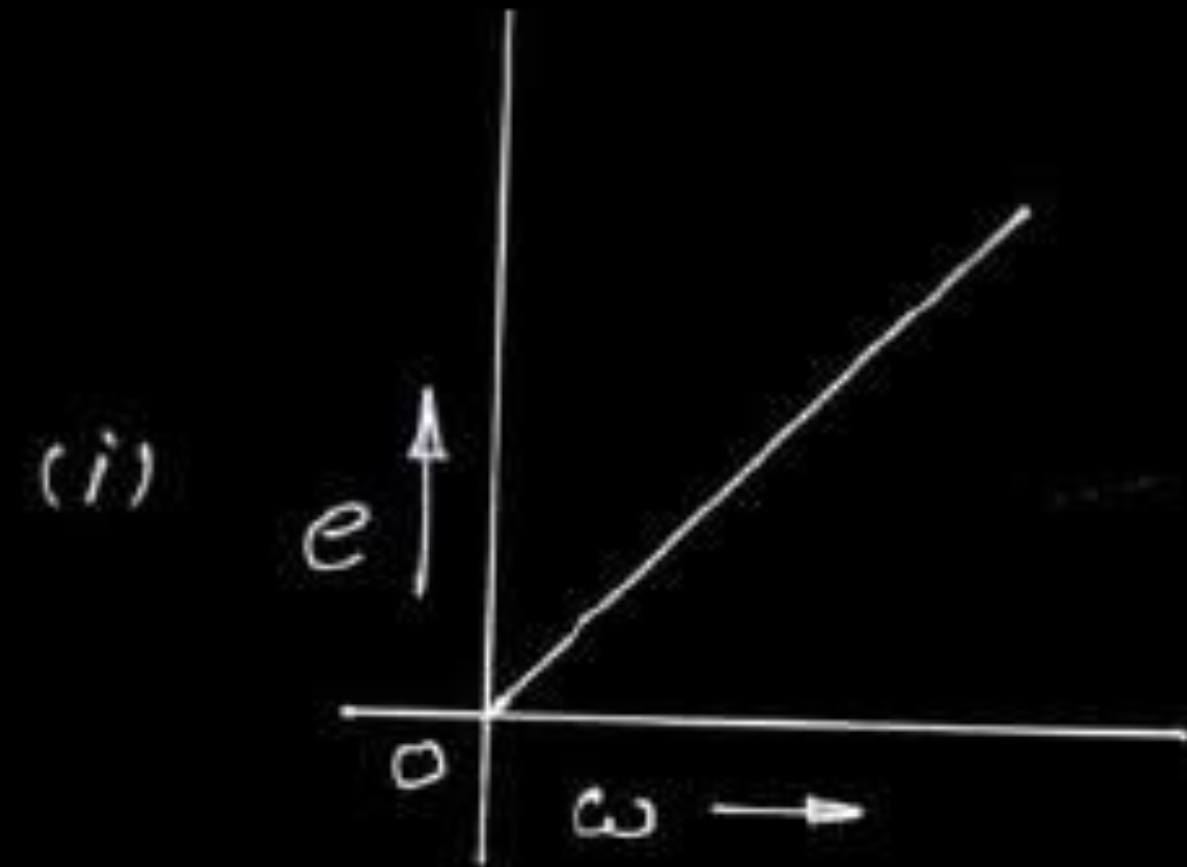
Find the

- (i) total charge passed through the loop.
- (ii) change in magnetic flux through the loop.
- (iii) magnitude of the magnetic field applied.

- (a) A conductor of length ' l ' is rotated about one of its ends at a constant angular speed ' ω ' in a plane perpendicular to a uniform magnetic field B . Plot graphs to show variations of the emf induced across the ends of the conductor with (i) angular speed ω and (ii) length of the conductor l .
- (b) Two concentric circular loops of radius 1 cm and 20 cm are placed coaxially.
- (i) Find mutual inductance of the arrangement.
 - (ii) If the current passed through the outer loop is changed at a rate of 5 A/ms, find the emf induced in the inner loop. Assume the magnetic field on the inner loop to be uniform.

$$\text{e.m.f. induced in the rod} = \left| -\frac{d\phi}{dt} \right|$$

$$\therefore e = \frac{1}{2} Bl^2 \omega$$



(i) Imagine a current I to flow through the larger coil.

Magnetic flux linked with the smaller coil = MI

$$= B_{\text{centre}} \times \pi \times 10^{-4} \text{ Wb}$$

$$= \frac{\mu_0 I}{2 \times 20 \times 10^{-2}} \times \pi \times 10^{-4} \text{ Wb}$$

$$\Rightarrow M = \frac{\mu_0 \pi}{4} \times 10^{-3} \text{ H}$$

$$= \frac{4\pi \times 10^{-7} \pi \times 10^{-3}}{4} \text{ H}$$

$$= 9.9 \times 10^{-10} \text{ H}$$

$$= 10^{-9} \text{ H}$$

1/2

1/2

1/2+1/2

1/2

(ii)

$$\text{e.m.f. induced} = -M \frac{dI}{dt}$$

$$= -10^{-9} \times \frac{5}{10^{-3}} \text{ V}$$

$$= -5 \times 10^{-6} \text{ V}$$

Alternatively

$$\text{e.m.f. induced} = -\frac{d\phi}{dt}$$

$$= -\frac{d}{dt} (B_c A_i) = -A_i \frac{dB_c}{dt}$$

$$= -A_i \frac{d}{dt} \left(\frac{\mu_0 I}{2R} \right)$$

$$= -\frac{A_i \mu_0}{2R} \frac{dI}{dt}$$

$$= \frac{-\pi \times 10^{-4} \times 4\pi \times 10^{-7} \times 5}{2 \times (20 \times 10^{-2}) \times 10^{-3}} \text{ V}$$

$$= -\frac{20\pi^2 \times 10^{-6}}{2 \times 20} \text{ V}$$

$$= -5 \times 10^{-6} \text{ V}$$

1/2

1/2

1/2

1/2

1/2

1/2

4. A circular loop of radius r , carrying a current I lies in y - z plane with its centre at the origin. The net magnetic flux through the loop is :
- (a) directly proportional to r (b) zero
(c) inversely proportional to r (d) directly proportional to I

4. A circular loop of radius r , carrying a current I lies in y - z plane with its centre at the origin. The net magnetic flux through the loop is :

- (a) directly proportional to r (b) zero
(c) inversely proportional to r (d) directly proportional to I

1

(b) Zero

30. (a) Differentiate between self inductance and mutual inductance.
- (b) The mutual inductance of two coaxial coils is 2H . The current in one coil is changed uniformly from zero to 0.5A in 100 ms . Find the :
- (i) change in magnetic flux through the other coil.
 - (ii) emf induced in the other coil during the change.

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 (i) change in magnetic flux through the other coil.
 (ii) emf induced in the other coil during the change.

3

(a) Self inductance is the response of the coil/ solenoid to the change in current in the coil/ solenoid itself (or definition of self inductance)	$\frac{1}{2}$
Mutual inductance is the response of a coil to the change of current in a neighbouring coil (or definition of mutual inductance)	$\frac{1}{2}$
Alternatively	
Self-inductance is the property of given coil/solenoid	
Mutual inductance is the property of given pair of coils /solenoids	
(b)	
(i) $\Delta\phi = M\Delta I = 2 \times 0.5 = 1Wb$	$\frac{1}{2} + \frac{1}{2}$
(ii) $e = -\frac{d\phi}{dt} = \frac{1}{100 \times 10^{-3}} = 10V$	$\frac{1}{2} + \frac{1}{2}$

17. Two identical coils, one of copper and the other of aluminium are rotated with the same angular speed in an external magnetic field. In which of the two coils will the induced current be more ?

17. Two identical coils, one of copper and the other of aluminium are rotated with the same angular speed in an external magnetic field. In which of the two coils will the induced current be more ?

1

Copper

What happens when a block of metal is kept in a varying magnetic field ? 1

What happens when a block of metal is kept in a varying magnetic field ? 1

Eddy currents are produced in metal block / block gets heated

- 23.** Two coplanar and concentric coils 1 and 2 have respectively the number of turns N_1 and N_2 and radii r_1 and r_2 ($r_2 \gg r_1$). Deduce the expression for mutual inductance of this system.

The given system has the shape shown here
Let a current I flow through the larger coil.

Magnetic field, due to the current at the centre of coil is

$$B_c = \frac{\mu_0 I N_2}{2r_2}$$

We can consider this to be the value of the magnetic field over the whole area of the smaller coil (as $r_1 \ll r_2$)

$\therefore$ Magnetic flux through the smaller coil

$$\begin{aligned} &= B_c (\pi r_1^2) N_1 = \frac{\mu_0 I N_2}{2r_2} \pi r_1^2 N_1 \\ &= \left(\frac{\mu_0 \pi r_1^2}{2r_2} N_1 N_2 \right) I \end{aligned}$$

But Magnetic flux = MI

Where M = mutual Inductance of the system

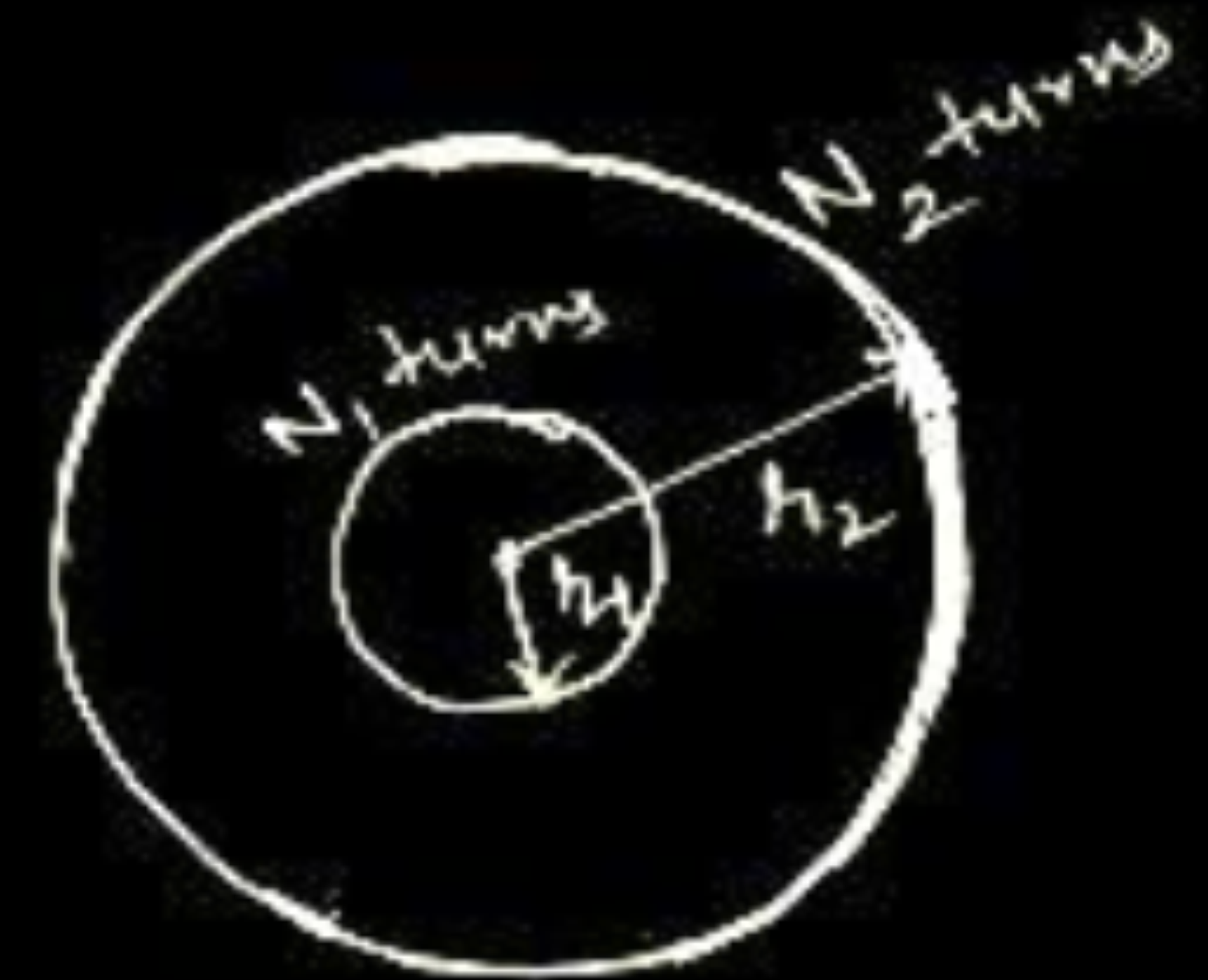
$$\therefore MI = \frac{\mu_0 \pi r_1^2 N_1 N_2}{2r_2} I$$

$$M = \frac{\mu_0 \pi r_1^2 N_1 N_2}{2r_2}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



- 17.** A conducting rod of length l is kept parallel to a uniform magnetic field $\vec{B}$. It is moved along the magnetic field with a velocity $\vec{v}$. What is the value of emf induced in the conductor ?

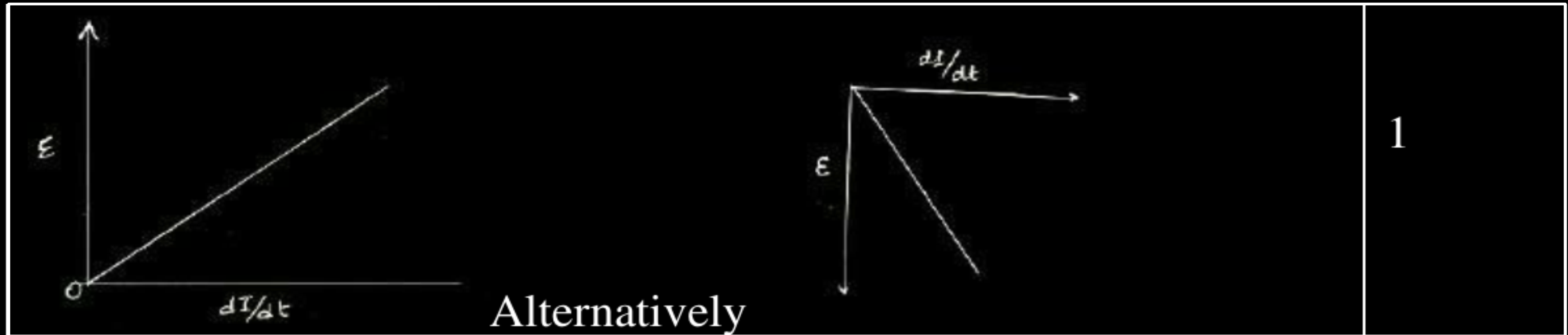
17. A conducting rod of length l is kept parallel to a uniform magnetic field $\vec{B}$. It is moved along the magnetic field with a velocity $\vec{v}$. What is the value of emf induced in the conductor ?

1

| Zero

18. Draw the graph showing variation of the value of the induced emf as a function of rate of change of current flowing through an ideal inductor. 1

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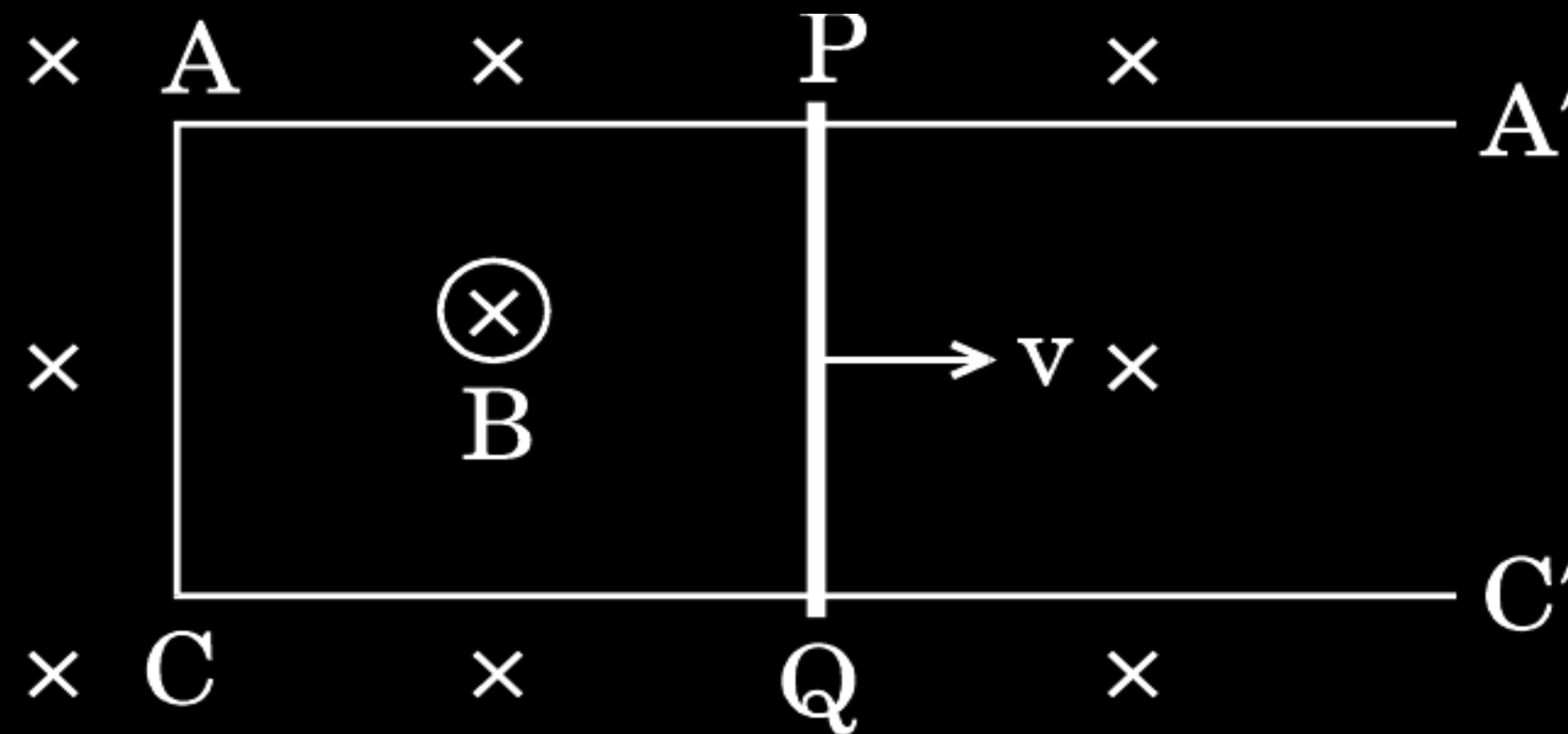
- 13.** The number of turns of a solenoid are doubled without changing its length and area of cross-section. The self-inductance of the solenoid will become _____ times.

13. The number of turns of a solenoid are doubled without changing its length and area of cross-section. The self-inductance of the solenoid will become _____ times.

1

Four times

- (b) A conducting rod PQ of length 20 cm and resistance $0.1\ \Omega$ rests on two smooth parallel rails of negligible resistance AA' and CC'. It can slide on the rails and the arrangement is positioned between the poles of a permanent magnet producing uniform magnetic field $B = 0.4\ \text{T}$. The rails, the rod and the magnetic field are in three mutually perpendicular directions as shown in the figure. If the ends A and C of the rails are short circuited, find the
- (i) external force required to move the rod with uniform velocity $v = 10\ \text{cm/s}$, and
 - (ii) power required to do so.



(b)

(i)

$$\begin{aligned} F &= BIl \\ I &= \frac{E}{R} = \frac{Bvl}{R} \\ F &= \frac{B^2 vl^2}{R} \\ &= \frac{0.4 \times 0.4 \times 0.1 \times 0.2 \times 0.2}{0.1} \\ &= 6.4 \times 10^{-3} \text{ N} \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

$$\begin{aligned} P &= F.v = 6.4 \times 10^{-3} \times 0.1 \\ &= .64 \times 10^{-3} \text{ W} \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

- (a) A metallic rod of length ' l ' and resistance ' R ' is rotated with a frequency ' ν ' with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius ' l ', about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field ' B ' parallel to the axis is present everywhere.
- (i) Derive the expression for the induced emf and the current in the rod.
 - (ii) Due to the presence of current in the rod and of the magnetic field, find the expression for the magnitude and direction of the force acting on this rod.
 - (iii) Hence, obtain an expression for the power required to rotate the rod.
- (b) A copper coil is taken out of a magnetic field with a fixed velocity. Will it be easy to remove it from the same field if its ohmic resistance is increased ?

(i)

$$\varepsilon' = Blv$$

$$\varepsilon = Bl \frac{v}{2}$$

$$\varepsilon = Bl \left(\frac{l\omega}{2} \right)$$

$$\varepsilon = \frac{bl^2\omega}{2}$$

[Student may use any method to arrive at this result]

$$I = \frac{\varepsilon}{R} = \frac{bl^2\omega}{2R}$$

(ii) $F = I (l \times b)$

$$F = \frac{bl^3\omega B}{2R}$$

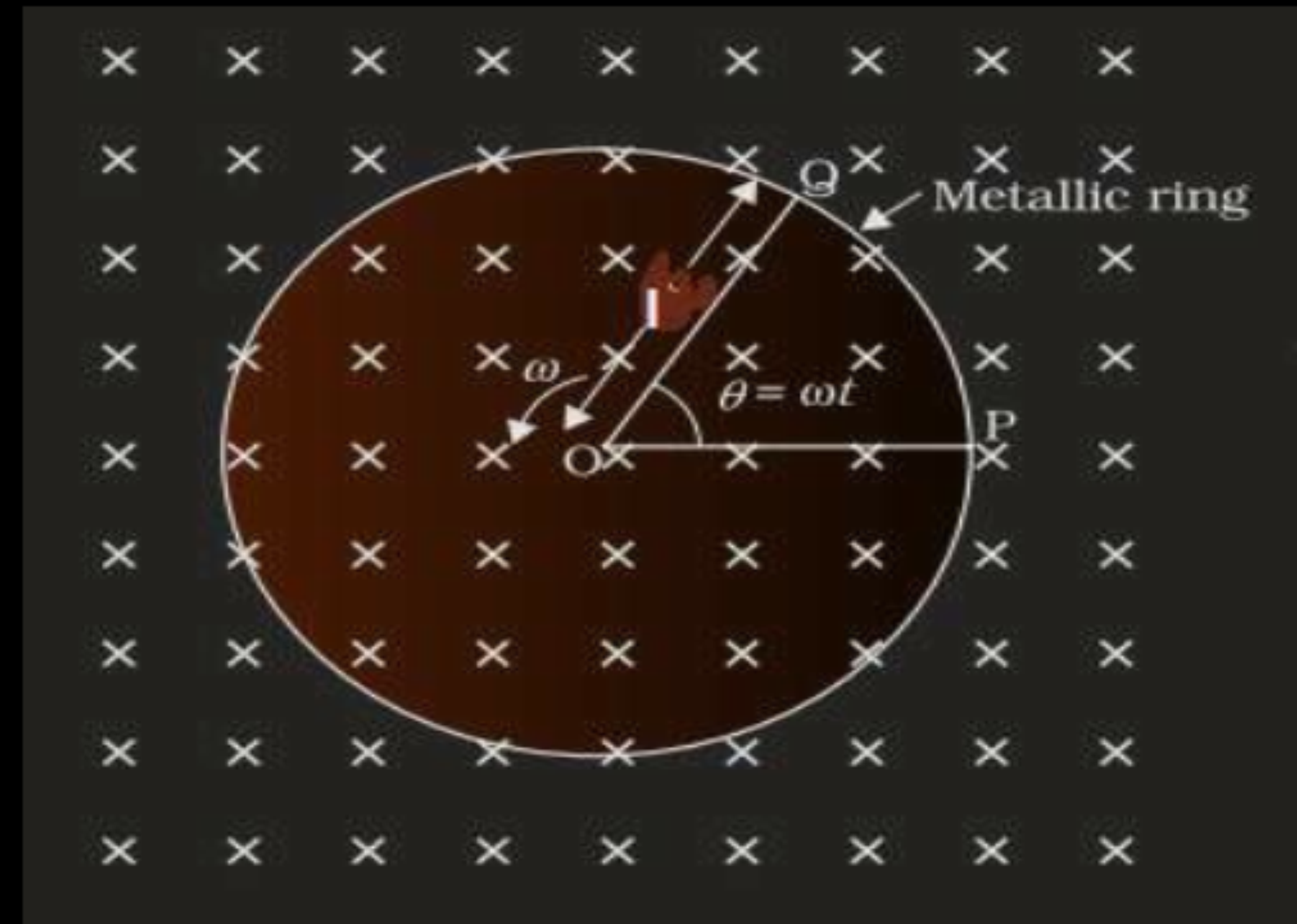
Direction of force is perpendicular to both $\vec{l}$ and $\vec{B}$

iii)

$$Power = i^2 R = \left(\frac{bl^2\omega}{2R} \right)^2 R$$

$$= \frac{b^2 l^4 \omega^2}{4R}$$

Since induced current will reduce, it will be a little easier to remove the coil
[Even if student writes induced current decreases award 1/2 mark]



1

1

1/2

1/2

1/2

1/2

1/2

1/2

- (a) A rectangular coil rotates in a uniform magnetic field. Obtain an expression for induced emf and current at any instant. Also find their peak values. Show the variation of induced emf versus angle of rotation (ωt) on a graph.
- (b) An iron bar falling through the hollow region of a thick cylindrical shell made of copper experiences a retarding force. What can you conclude about the nature of the iron bar ? Explain.

(a)

$$\phi = N \vec{B} \cdot \vec{A}$$

$$\phi = NBA \cos \omega t$$

$$\varepsilon = \frac{-d\phi}{dt}$$

$$\varepsilon = NBA\omega \sin \omega t$$

$$\text{here } \varepsilon_0 = NBA\omega$$

$$i = \frac{\varepsilon}{R} = \frac{NBA\omega}{R} \sin \omega t$$

$$i_0 = \frac{NBA\omega}{R}$$

1/2

1/2

1/2

1/2

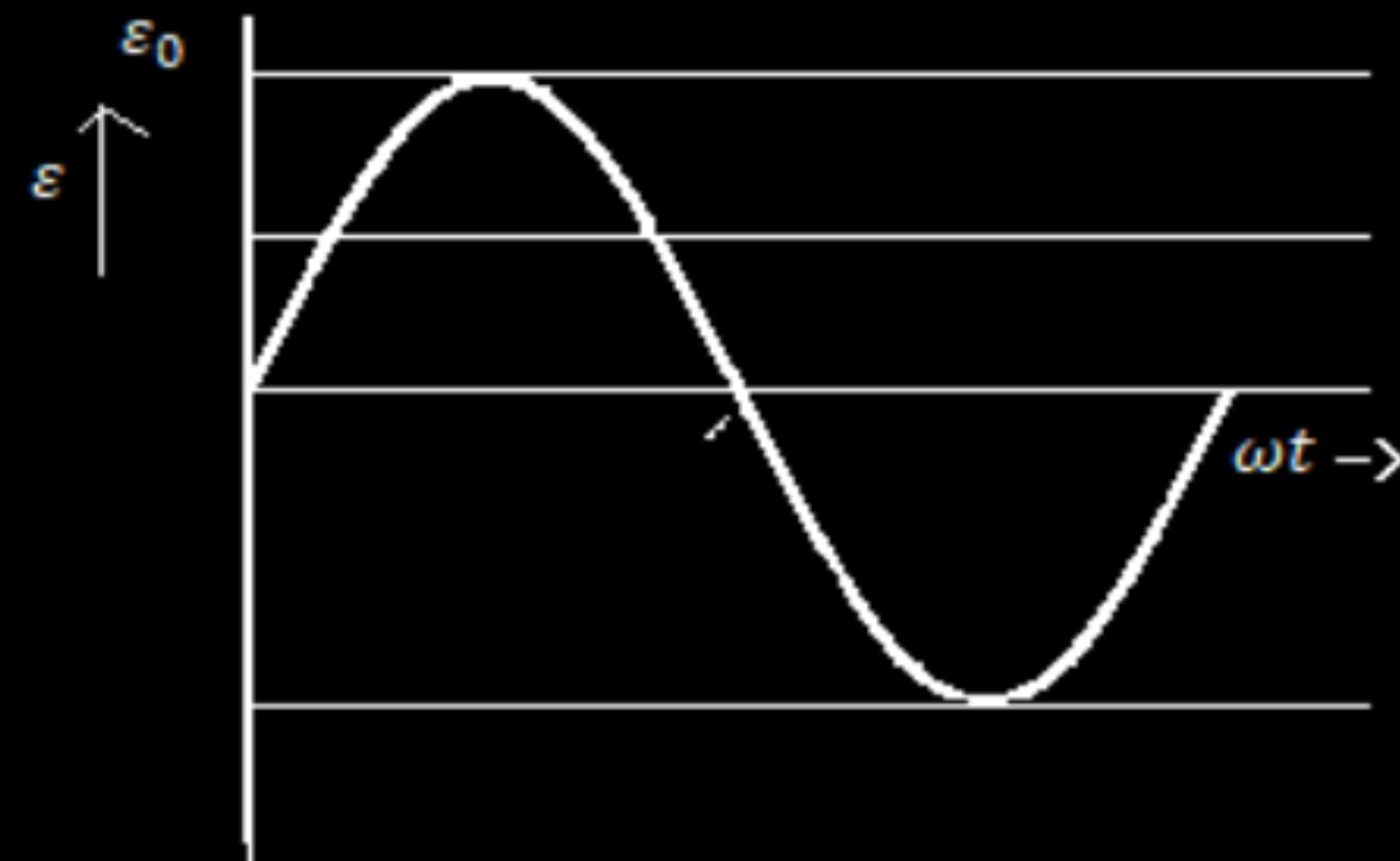
1/2

1/2

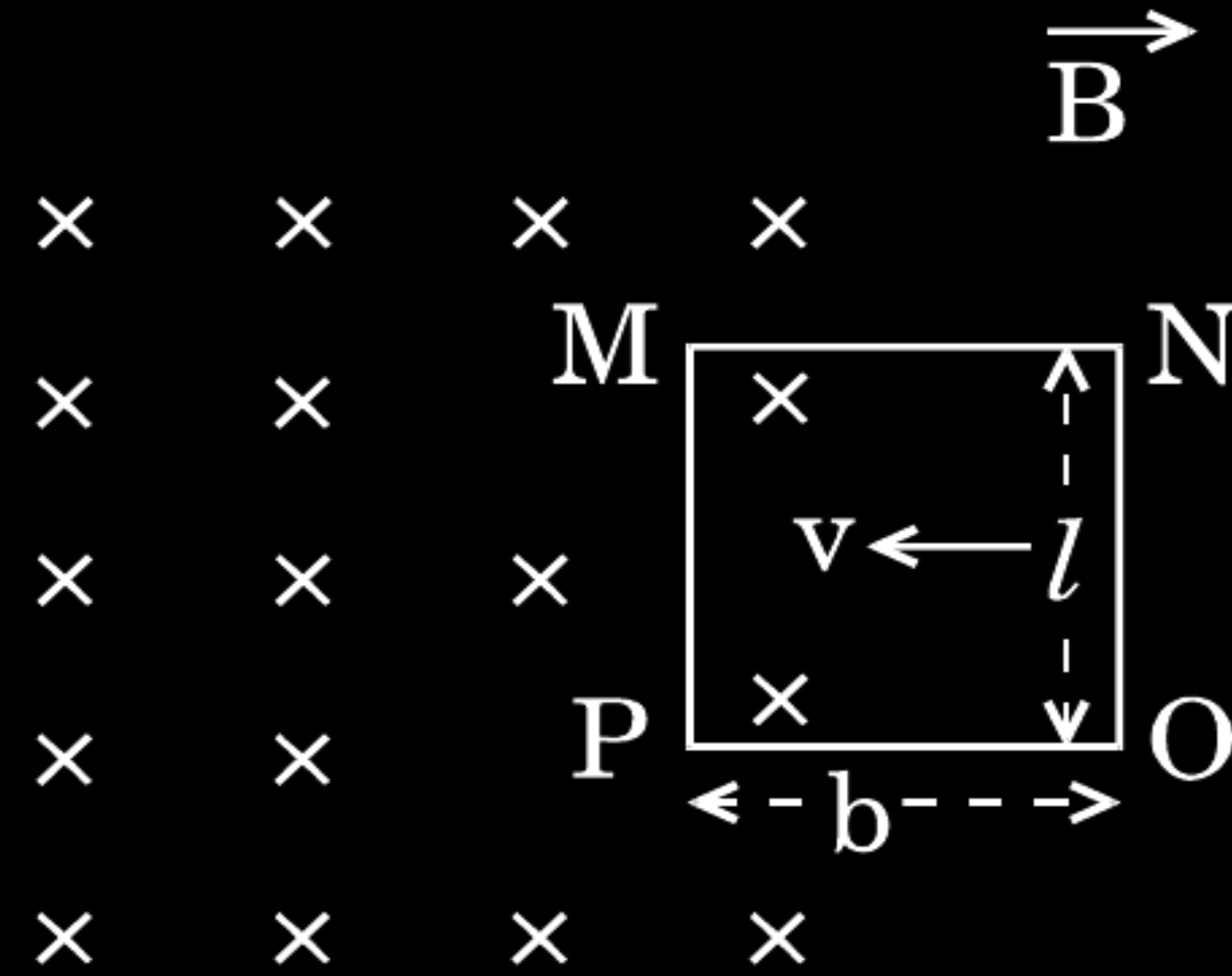
(b) Bar is magnetic

Reason: Lenz's law/(Induced emf/current opposes its cause)

1/2



The figure shows a rectangular conducting frame MNOP of resistance R placed partly in a perpendicular magnetic field $\vec{B}$ and moved with velocity $\vec{v}$ as shown in the figure.



Obtain the expressions for the

- force acting on the arm 'ON' and its direction, and
- power required to move the frame to get a steady emf induced between the arms MN and PO.

a) The induced emf in the moving conductor MNOP	$\frac{1}{2}$
$e = Blv$	
The induced current, $i = \frac{e}{R} = \frac{Blv}{R}$	
Force on the arm 'ON', $F = Bil$	$\frac{1}{2}$
$= \frac{B^2 l^2 v}{R}$	$\frac{1}{2}$
The force is directed in the direction opposite to velocity of rod (v)	$\frac{1}{2}$
Note : Award the last half mark if the student write $F = 0$ as $B = 0$ in the position shown	
b) Power $P = F \times v$	$\frac{1}{2}$
$= \frac{B^2 l^2 v}{R}$	$\frac{1}{2}$
Note : Award the last half mark if the student write $P = 0$ as $B = 0$ in the position shown	

- (a) Draw a schematic diagram of an ac generator. Explain its working and obtain the expression for the instantaneous value of the emf in terms of the magnetic field B , number of turns N of the coil of area A rotating with angular frequency ω . Show how an alternating emf is generated by a loop of wire rotating in a magnetic field.
- (b) A circular coil of radius 10 cm and 20 turns is rotated about its vertical diameter with angular speed of 50 rad s^{-1} in a uniform horizontal magnetic field of $3.0 \times 10^{-2} \text{ T}$.
- (i) Calculate the maximum and average emf induced in the coil.
 - (ii) If the coil forms a closed loop of resistance 10Ω , calculate the maximum current in the coil and the average power loss due to Joule heating.

Working of AC Generator -

Whenever coil placed in uniform magnetic field is rotated, flux linked with it changes, and an emf induces in the coil. The ends of the coil are connected to an external circuit by means of slip rings and brushes.

Flux linked with the coil of Area a , placed in uniform magnetic field 'B'

$$\phi_B = BA \cos \theta$$

or $\phi_B = BA \cos \omega t$ (eqⁿ1)

∴ From Faraday's law

e.m.f induced in the coil

$$\varepsilon = -N \frac{d\phi_B}{dt}$$
$$= -NBA \frac{d}{dt} \cos \omega t$$

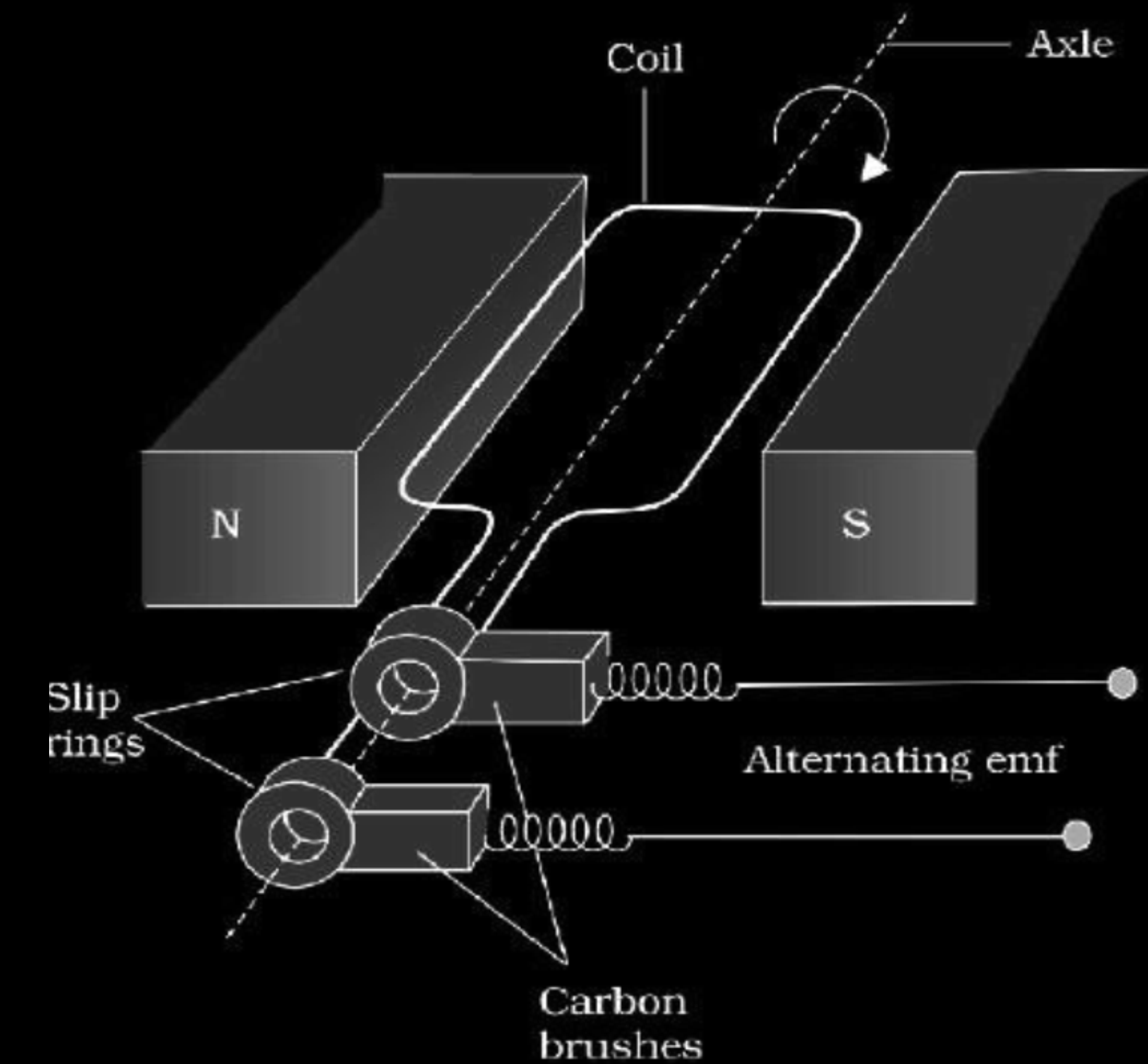
$$\varepsilon = NBA \omega \sin \omega t$$

or $\varepsilon = \varepsilon_0 \sin \omega t$ where $\varepsilon_0 = NBA \omega$

1/2

1/2

1/2



$$\text{b) i) } r = 10 \text{ cm, } N = 20 \text{ turns, } \omega = 50 \text{ rad s}^{-1}$$

$$B = 3.0 \times 10^{-2} \text{ T}$$

$$\varepsilon_0 = NBA\omega$$

$$= 20 \times 3 \times 10^{-2} \times \pi (10 \times 10^{-2})^2 \times 50$$

$$= 0.942 \text{ volt}$$

$$\varepsilon_{AV} = 0, \text{ over a cycle}$$

$$\text{ii) } i_0 = \frac{e_0}{R} = \frac{0.942}{10}$$

$$= 0.094 \text{ A}$$

$$P = \frac{1}{2} \varepsilon_0 \times I_0$$

$$= \frac{1}{2} \times 0.942 \times 0.094$$

$$= 0.045 \text{ watt.}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

A 0.5 m long solenoid of 10 turns/cm has area of cross-section 1 cm^2 . Calculate the voltage induced across its ends if the current in the solenoid is changed from 1 A to 2 A in 0.1 s.

A 0.5 m long solenoid of 10 turns/cm has area of cross-section 1 cm^2 . Calculate the voltage induced across its ends if the current in the solenoid is changed from 1 A to 2 A in 0.1 s.

Induced voltage	
$ V = L \frac{dI}{dt}$	$\frac{1}{2}$
$\therefore V = \mu_0 n^2 l a \frac{dI}{dt}$	$\frac{1}{2}$
$= 4\pi \times 10^{-7} \times \left(\frac{10}{10^{-2}}\right)^2 \times 0.5 \times 1 \times 10^{-4} \times \frac{(2-1)}{0.1}$	$\frac{1}{2}$
$= 6.28 \times 10^{-4} \text{ V or } 0.628 \text{ mV}$	$\frac{1}{2}$

A small flat search coil of area 5 cm^2 with 140 closely wound turns is placed between the poles of a powerful magnet producing magnetic field 0.09 T and then quickly removed out of the field region. Calculate

- (a) change of magnetic flux through the coil, and
- (b) emf induced in the coil.

A small flat search coil of area 5 cm^2 with 140 closely wound turns is placed between the poles of a powerful magnet producing magnetic field 0.09 T and then quickly removed out of the field region. Calculate

- (a) change of magnetic flux through the coil, and
- (b) emf induced in the coil.

$$\begin{aligned} \text{i) } \Delta\phi &= \phi_2 - \phi_1 = 0 - NBA \cos\theta \\ &= 140 \times 0.09 \times 5 \times 10^{-4} \cos 0 = 63 \times 10^{-4} \text{ Wb} \end{aligned}$$

$\frac{1}{2}$

Alternatively

If student assumes that the coil was initially kept with its plane parallel to the field, i.e. $\phi = 90^\circ$, $\Delta\phi = (0 - 0) = 0 \text{ Wb}$

1

award 1 mark

$$\text{ii) } e = \frac{-\Delta\phi}{\Delta t} = \frac{-63 \times 10^{-4}}{\Delta t} \text{ V}$$

$\frac{1}{2} + \frac{1}{2}$

Alternatively, if the student takes $\Delta\phi = 0$, then $e = 0$,

[Note: Award this 1 mark, If a student writes that induced emf cannot be calculated as value of time interval Δt it is not given.]

- (a) Derive an expression for the induced emf developed when a coil of N turns, and area of cross-section A , is rotated at a constant angular speed ω in a uniform magnetic field B .
- (b) A wheel with 100 metallic spokes each 0.5 m long is rotated with a speed of 120 rev/min in a plane normal to the horizontal component of the Earth's magnetic field. If the resultant magnetic field at that place is 4×10^{-4} T and the angle of dip at the place is 30° , find the emf induced between the axle and the rim of the wheel.

Flux linked with the coil at any instant of time is:

$$\phi = NBA \cos \omega t$$

$$\frac{d\phi}{dt} = NBA\omega(-\sin \omega t) S$$

$$\varepsilon = -\frac{d\phi}{dt}$$

$$\varepsilon = NBA\omega \sin \omega t$$

$$\varepsilon = \varepsilon_0 \sin \omega t \quad (\text{Here } \varepsilon_0 = NBA\omega)$$

(b) $l = 0.5m, \nu = 120rpm = 2rps$

$$\omega = 2\pi\nu = 4\pi \text{ rad} / s, \quad B = 4 \times 10^{-4} T, \quad \delta = 30^\circ$$

$$B_H = 4 \times 10^{-4} \times \frac{\sqrt{3}}{2}$$

$$B_H = 2\sqrt{3} \times 10^{-4} T$$

$$\varepsilon = \frac{1}{2} B \omega l^2$$

$$\varepsilon = \frac{1}{2} \times 2\sqrt{3} \times 10^{-4} \times 4\pi \times (0.5)^2$$

$$\varepsilon = 5.4 \times 10^{-4} \text{ volt}$$

$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$

$$1$$

- (a) Derive the expression for the magnetic energy stored in an inductor when a current I develops in it. Hence, obtain the expression for the magnetic energy density.

- (a) Derive the expression for the magnetic energy stored in an inductor when a current I develops in it. Hence, obtain the expression for the magnetic energy density.

(a) When external source supplies current to the inductor, e.m.f. is induced in it due to self induction. So the external supply has to do work to establish current. The amount of work done is:

$$dw = |\varepsilon| Idt \quad \because \varepsilon = L \frac{dI}{dt}$$

$$dw = LIdt$$

$$W = \frac{1}{2} LI^2$$

$$\text{Energy density} = \frac{\text{Energy}}{\text{Volume}}$$

$$u = (1/2 LI^2) / \text{volume}$$

$1/2$

1

$1/2$

(a) Define mutual inductance and write its S.I. unit.

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(a) Mutual inductance equals the magnetic flux associated with a coil when unit current flows in its neighbouring coil.

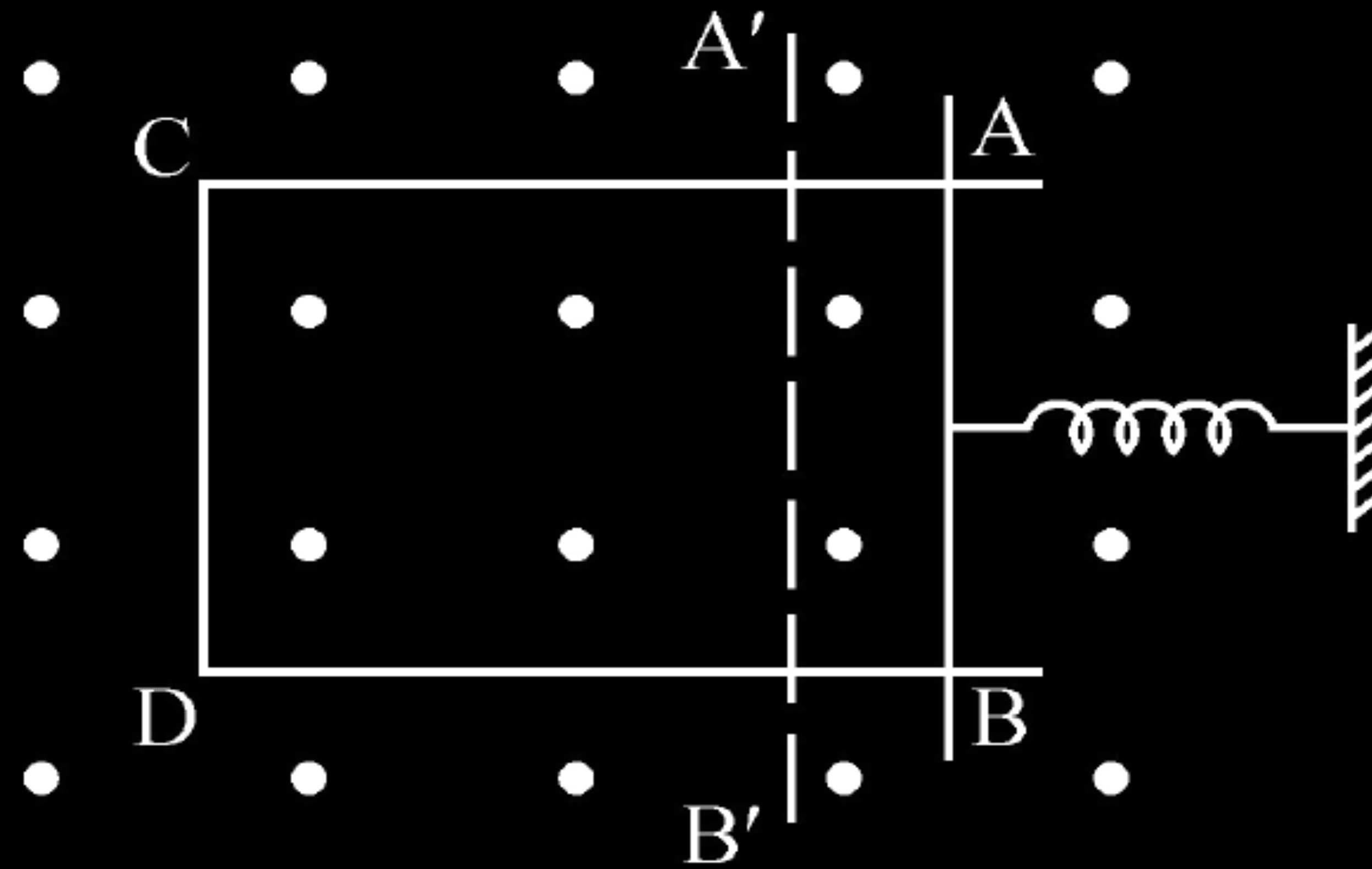
Alternatively: Mutual inductance equals the induced emf in a coil when the rate of change of current in its neighbouring coil is one ampere/ second.

S.I unit : henry (H) or weber/ampere (or any other correct SI unit)

1

$\frac{1}{2}$

A rectangular frame of wire is placed in a uniform magnetic field directed outwards, normal to the paper. AB is connected to a spring which is stretched to A'B' and then released at time $t = 0$. Explain qualitatively how induced e.m.f. in the coil would vary with time. (Neglect damping of oscillations of spring)



A long straight current carrying wire passes normally through the centre of circular loop. If the current through the wire increases, will there be an induced emf in the loop ? Justify.

A long straight current carrying wire passes normally through the centre of circular loop. If the current through the wire increases, will there be an induced emf in the loop ? Justify.

No,

As the magnetic field due to current carrying wire will be in the plane of the circular loop, so magnetic flux will remain zero.

Alternatively

[Magnetic flux does not change with the change of current.]

$\frac{1}{2}$

$\frac{1}{2}$

- (a) Define the term 'self-inductance' and write its S.I. unit.
- (b) Obtain the expression for the mutual inductance of two long co-axial solenoids S_1 and S_2 wound one over the other, each of length L and radii r_1 and r_2 and n_1 and n_2 number of turns per unit length, when a current I is set up in the outer solenoid S_2 .

a) Definition of self inductance and its SI unit b) Derivation of expression for mutual inductance	1 + ½ 1 ½
Self inductance of a coil equals, the magnitude of the magnetic flux, linked with it, when a unit current flows through it. Alternatively Self inductance, of a coil, equals the magnitude of the emf induced in it, when the current in the coil, is changing at a unit rate.	1
SI unit : henry / (weber/ampere) / (ohm second.)	½

- (a) Draw a labelled diagram of AC generator. Derive the expression for the instantaneous value of the emf induced in the coil.
- (b) A circular coil of cross-sectional area 200 cm^2 and 20 turns is rotated about the vertical diameter with angular speed of 50 rad s^{-1} in a uniform magnetic field of magnitude $3.0 \times 10^{-2} \text{ T}$. Calculate the maximum value of the current in the coil.

[Deduct ½ mark, If diagram is not labeled]

When the coil is rotated with constant angular speed ω , the angle θ between the magnetic field and area vector of the coil, at instant t , is given by $\theta = \omega t$,

Therefore, magnetic flux, (ϕ_B), at this instant, is

$$= BA \cos \omega t$$

∴ Induced emf $e = -N \frac{d\phi}{dt}$

$$e = NBA \omega \sin \omega t$$

$$e = e_0 \sin \omega t$$

$$\text{where } e_0 = NBA$$

b) Maximum value of emf

$$= NBA \omega$$

$$= 20 \times 200 \times 3 \times 50 \text{ V}$$

$$= 600 \text{ mV}$$

Maximum induced current $= \frac{e_0}{R} = \frac{600}{1000} \text{ mA}$

[Note 1: If the student calculates the value of the maximum induced emf and says that “since R is not given, the value of maximum induced current cannot be calculated”, the ½ mark, for the last part, of the question, can be given.]

[Note 2: The direction of magnetic field has not been given. If the student takes this direction along the axis of rotation and hence obtains the value of induced emf and, therefore, maximum current, as zero, award full marks for this part.]

½

½

½

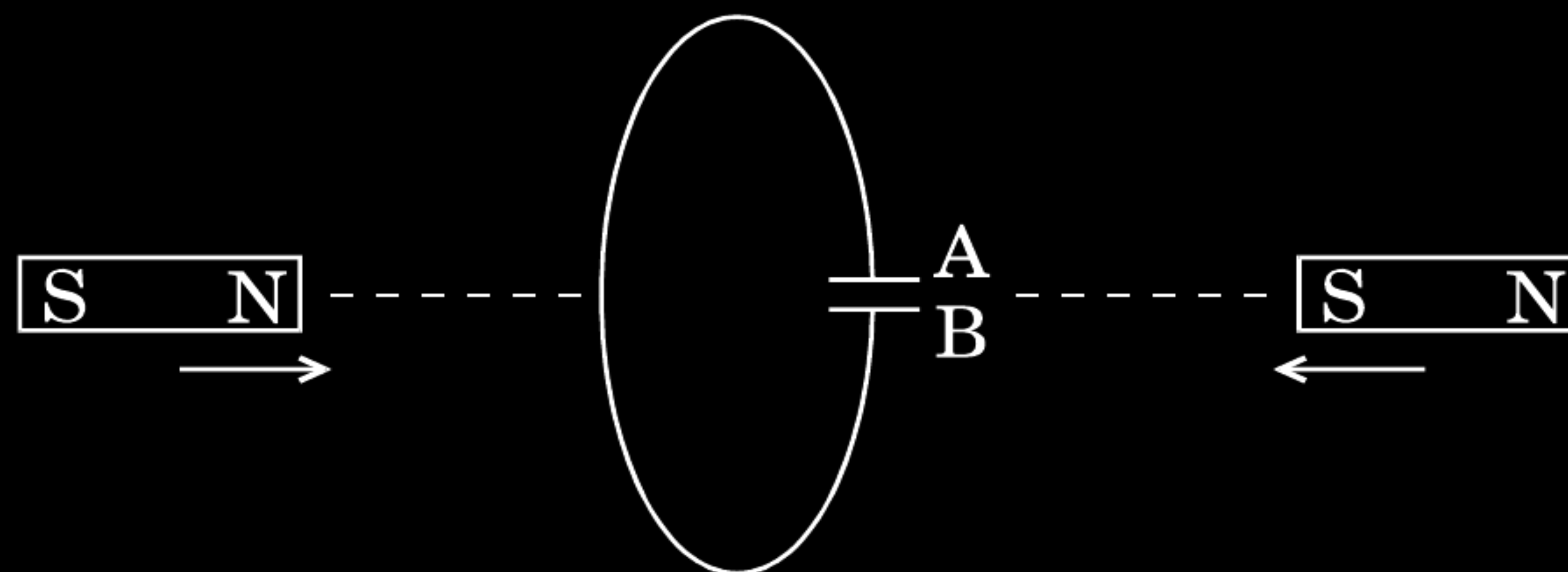
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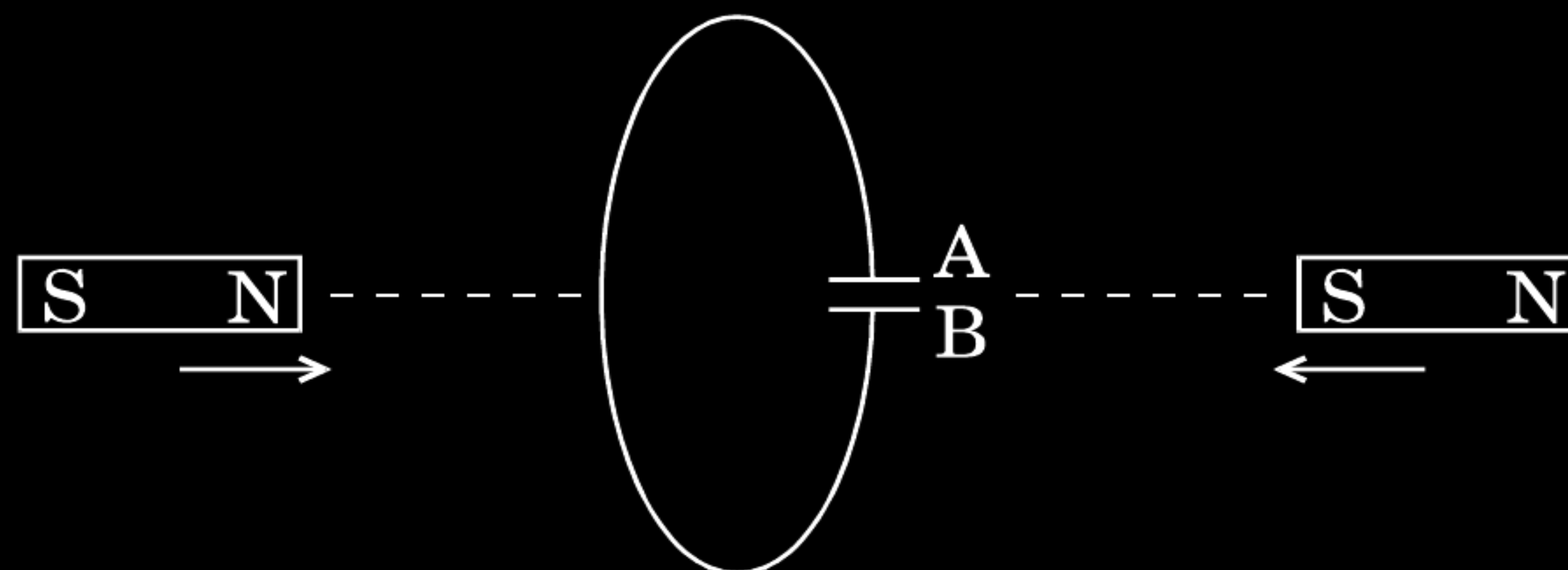
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Predict the polarity of the capacitor in the situation described below :



Predict the polarity of the capacitor in the situation described below :



A is positive and

B is negative

(Also accept: A is negative and B is positive)

$\frac{1}{2}$

$\frac{1}{2}$

Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other.

Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other.

- (i) Mutual inductance is numerically equal to the induced emf in the secondary coil when the current in the primary coil changes by unity.

Alternatively: Mutual inductance is numerically equal to the magnetic flux linked with one coil/secondary coil when unit current flows through the other coil /primary coil.

Let a current, i_2 , flow in the secondary coil

$$\therefore B_2 = \frac{\mu_0 N_2 i_2}{l}$$

$\therefore$ Flux linked with the primary coil

$$= N_1 A_1 B_2 = \frac{\mu_0 N_2 N_1 A_1 i_2}{l} = M_{12} i_2$$

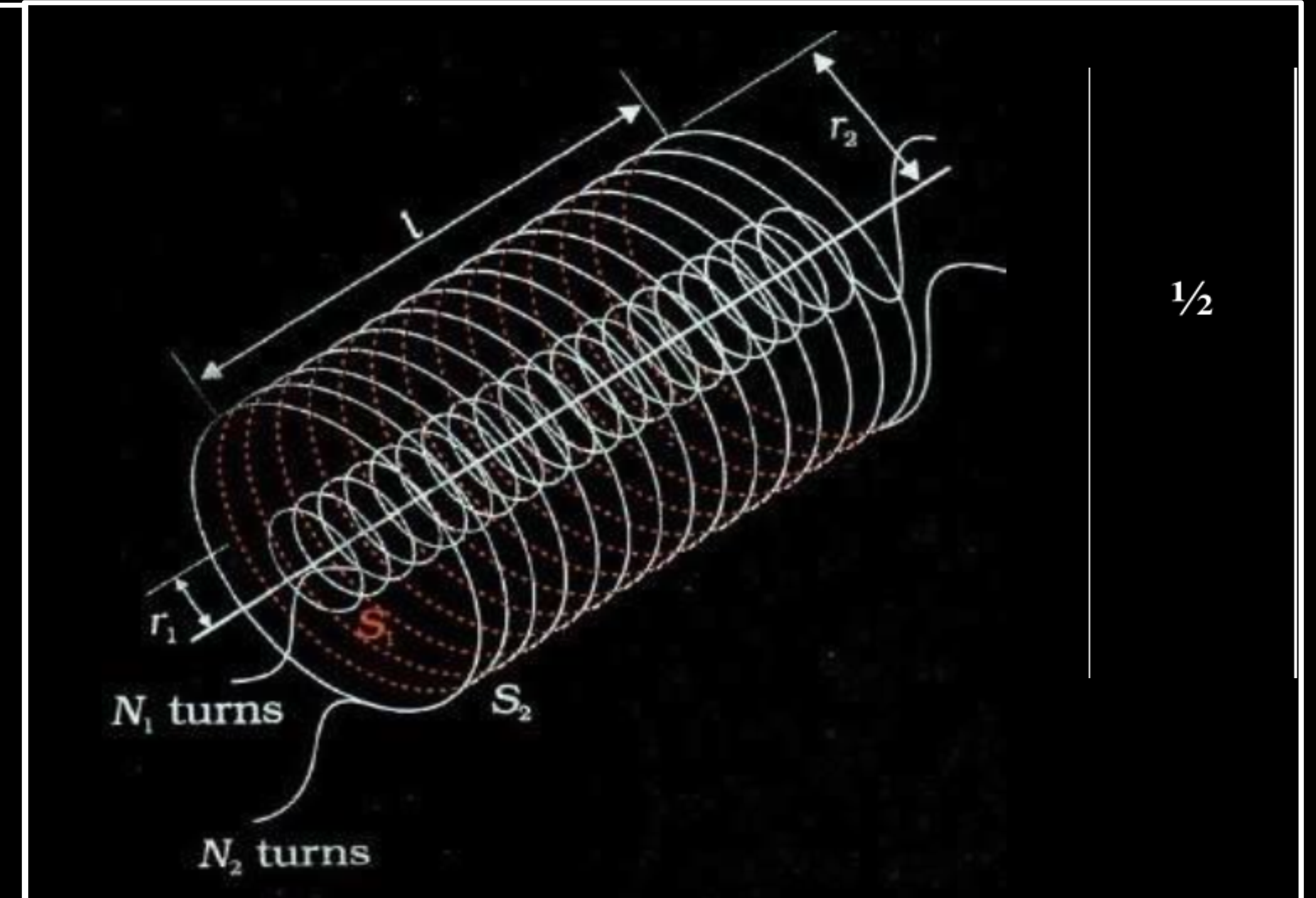
$$\text{Hence, } M_{12} = \frac{\mu_0 N_2 N_1 A_2}{l} = \mu_0 n_2 n_1 A_1 l \left(n_1 = \frac{N_1}{l}; n_2 = \frac{N_2}{l} \right)$$

1

1/2

1/2

1/2



Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor L connected across a source of emf.

Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor L connected across a source of emf.

(i)	Self inductance, of a coil, is numerically equal to the emf induced in that coil when the current in it changes at a unit rate. (Alternatively: The self inductance of a coil equals the flux linked with it when a unit current flows through it.)	1
(ii)	The work done against back /induced emf is stored as magnetic potential energy.	$\frac{1}{2}$
	The rate of work done, when a current i is passing through the coil, is	$\frac{1}{2}$
	$\frac{dW}{dt} = \varepsilon i = \left(L \frac{di}{dt} \right) i$	$\frac{1}{2}$
	$\therefore W = \int dW = \int_0^I L i di$	$\frac{1}{2}$
	$= \frac{1}{2} Li^2$	

- (i) Define mutual inductance.
- (ii) A pair of adjacent coils has a mutual inductance of 1.5 H. If the current in one coil changes from 0 to 20 A in 0.5 s, what is the change of flux linkage with the other coil ?

(i) Magnetic flux, linked with the secondary coil due to the unit current flowing in the primary coil, $\phi_2 = MI_1$

[Alternatively

Induced emf associated with the secondary coil, for a unit rate of change of current in the primary coil. $e_2 = -M \frac{dI_1}{dt}$]

[Also accept the Definition of Mutual Induction, as per the Hindi translation of the question]

[i.e. the phenomenon of production of induced emf in one coil due to change in current in neighbouring coil]

(ii) Change of flux linkage

$$\begin{aligned}d\phi &= M dI \\&= 1.5 \times (20-0)W \\&= 30 \text{ weber}\end{aligned}$$

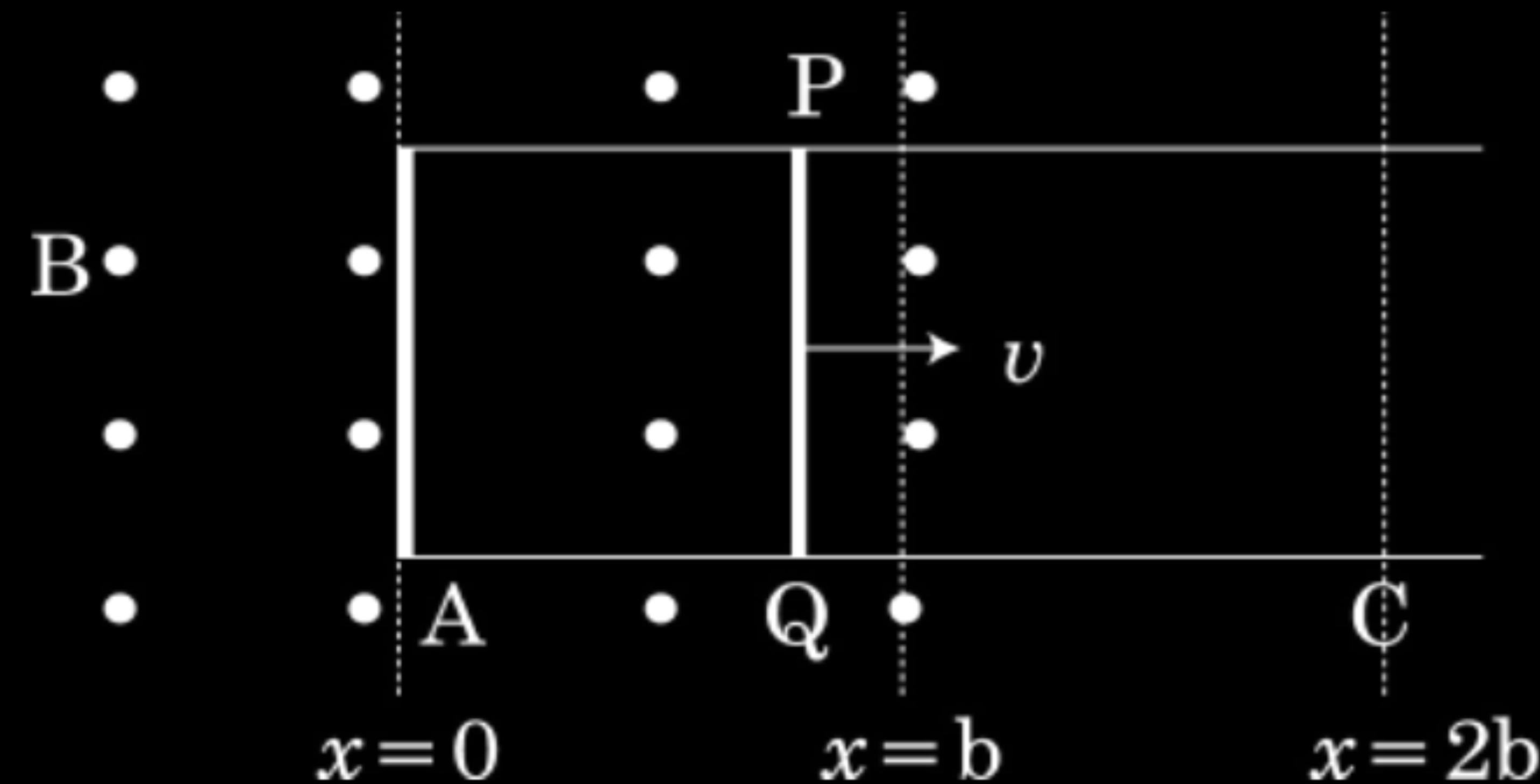
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$\frac{1}{2}$

$\frac{1}{2}$

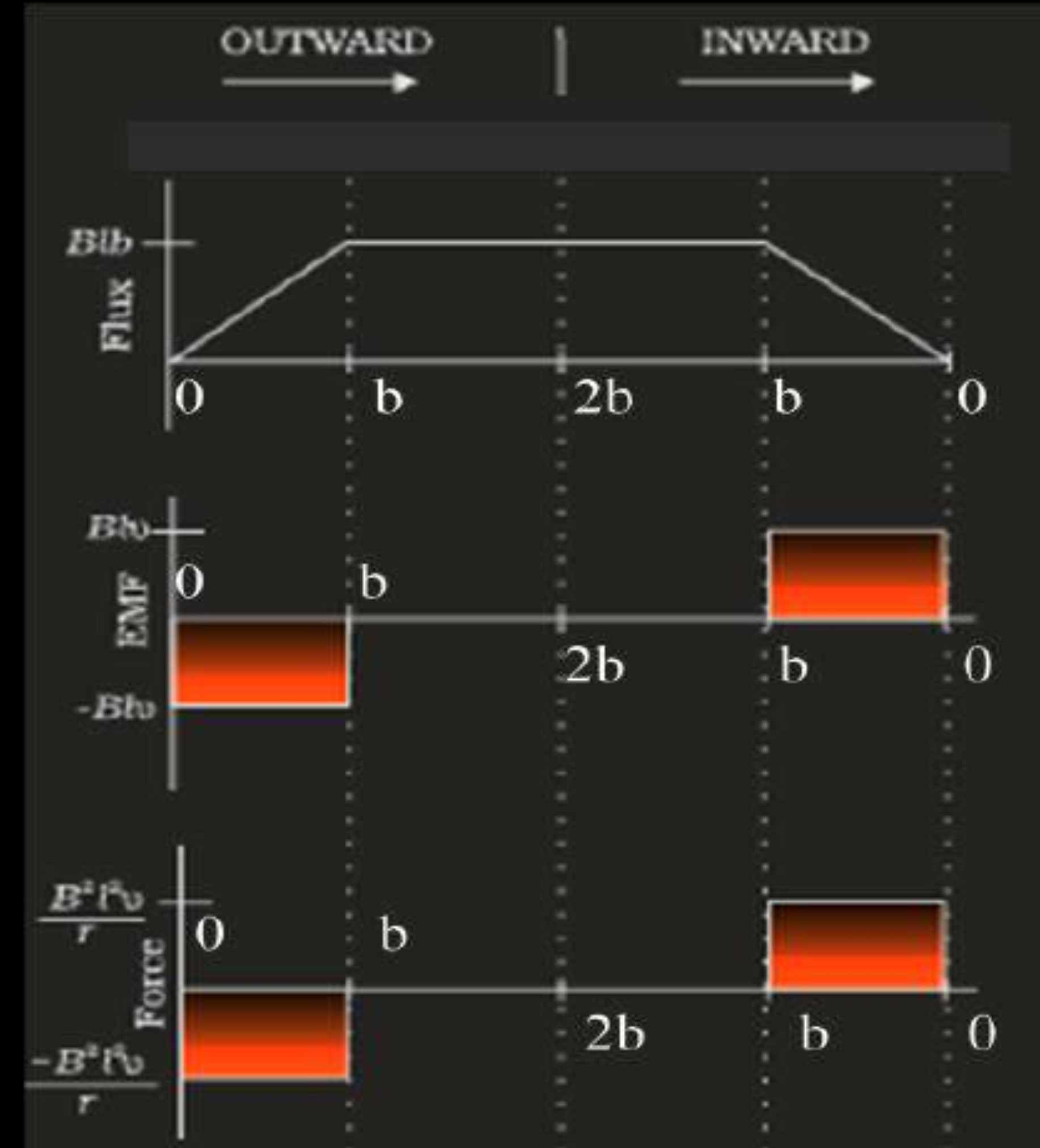
- (a) When a bar magnet is pushed towards (or away) from the coil connected to a galvanometer, the pointer in the galvanometer deflects. Identify the phenomenon causing this deflection and write the factors on which the amount and direction of the deflection depends. State the laws describing this phenomenon.
- (b) Sketch the change in flux, emf and force when a conducting rod PQ of resistance R and length l moves freely to and fro between A and C with speed v on a rectangular conductor placed in uniform magnetic field as shown in the figure.



- (a) The phenomenon involved is electromagnetic induction (EMI) 1/2
 For the deflection:
 Amount depends upon the speed of movement of the magnet. 1/2
 Direction depends on the sense (towards, or away) of the movement of the magnet. 1/2
 The law describing the phenomenon is :
 The magnitude of the induced emf, in a circuit, is equal to the time rate of change of the magnetic flux through the circuit. 1/2
- (Note: Also accept if a student writes: whenever magnetic flux linked with a conductor changes, an induced emf is setup in the conductor.)

(Alternatively, $\epsilon = -\frac{d\phi_B}{dt}$)

(b)



1

1

1

- (a) Explain the meaning of the term mutual inductance. Consider two concentric circular coils, one of radius r_1 and the other of radius r_2 ($r_1 < r_2$) placed coaxially with centres coinciding with each other. Obtain the expression for the mutual inductance of the arrangement.
- (b) A rectangular coil of area A , having number of turns N is rotated at ' f ' revolutions per second in a uniform magnetic field B , the field being perpendicular to the coil. Prove that the maximum emf induced in the coil is $2 \pi f NBA$.

- a. Mutual Inductance is the property of a pair of coils due to which an emf induced in one of the coils due to the change in the current in the other coil.

$$\text{Mathematically } e_2 = - \frac{M di_1}{dt}$$

$$\therefore M = - \frac{e_2}{di_1/dt}$$

Let a current I_2 flow through the outer circular coil. Then

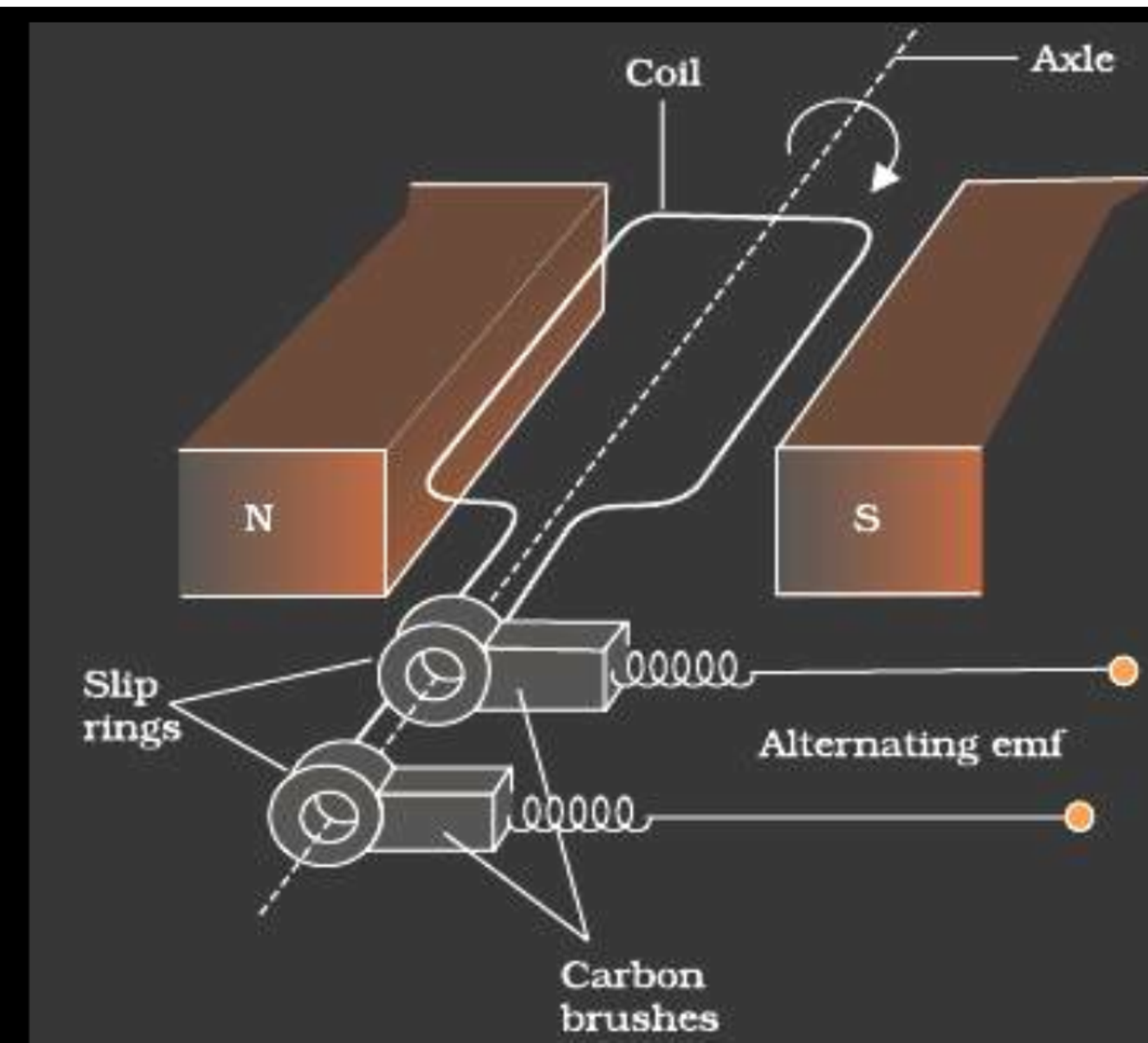
$$B_2 = \mu I_2 / 2r_2$$

$$\therefore \Phi_1 = \pi r^2 B_2 = \frac{\mu \pi r_1^2}{2r_2} I_2 = M_{12} I_2$$

$$\text{Thus } M_{12} = \frac{\mu \pi r_1^2}{2r_2} = M_{21}$$

1/2

1/2



1/2

1/2

1/2

1/2

Flux at any time 't'.

$$\Phi_B = BA \cos \theta = BA \cos \omega t$$

From Faraday's Law, induced emf

$$e = -N \frac{d\Phi_B}{dt} = NBA \frac{d}{dt} (\cos \omega t)$$

Thus the instantaneous value of emf is

$$E = NBA \omega \sin \omega t$$

For maximum value of emf $\sin \omega t = \pm 1$

$$\text{i.e., } e_0 = NBA \omega = 2\pi f NBA$$